THE HODGE CONJECTURE FOR FERMAT FOURFOLDS OF DEGREE NOT DIVISIBLE BY 6 OR 110

DEEP BHATTACHARJEE† AND USHASHI BHATTACHARYA

Abstract. We prove the Hodge conjecture for the Fermat fourfold of every degree divisible neither by 6 nor by 110. Each Hodge class is carried, through the cycles of Shioda and Aoki, to classes of lower degree or lower order, except for a finite number of classes, which come from uniruled quotients of Fermat threefolds.

1. Introduction

A rational cohomology class on a smooth complex projective variety is a *Hodge class* if it has type (p,p), and it is *algebraic* if it is a rational combination of classes of subvarieties. Algebraic classes are Hodge classes, and Hodge asked whether the converse holds [Hod52]. For divisors it does, by the theorem of Lefschetz on classes of type $(1,1)$ [Lef24]; in general it is open.

Let $X_M^n \subseteq \mathbb{P}^{n+1}$ be the Fermat variety $x_0^M + \cdots + x_{n+1}^M = 0$. The group μ_M^{n+2} acts on it, its primitive cohomology splits into lines indexed by characters of this group, and the Hodge classes are spanned by the lines whose characters satisfy a balancing condition [Shi79]. Fermat varieties are the varieties on which the Hodge conjecture has been tested most closely. It is known for X_M^n in every dimension when M is prime or $M = 4$ [Ran80, Shi79], when $M \leq 20$ [Shi79], and when M is a power of a prime [Aok87]. Aoki reduced the conjecture for all Fermat varieties of degree M to finitely many classes, one for each divisor of M that is a product of an even number of distinct elements of $\{4, 3, 5, 7, 11, \dots\}$, and proved it for the degrees $2^a 3^b 5^c 7^d$ with $c = 0$ or $d = 0$ and for $2p^e$ [Aok83, Aok00]. Peterson added the degrees $2^a 3^b 5^c 7^d$ not divisible by 420 [Pet26], and the first author the degrees $2^a 3^b 11^c$, $2^a 3^b 13^c$, and $2^a 3^b 17^c$, $2^a 3^b 19^c$ with $a \leq 1$ [Bha26b]. For fourfolds more is known: da Silva treated the degrees up to 100 [dSJ21],[1] Jumagulov the odd degrees up to 199 [Jum26b], a computer census of Jumagulov the

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† Corresponding author: Deep Bhattacharjee. ORCID iDs: Deep Bhattacharjee 0000-0003-0466-750X; Ushashi Bhattacharya 0009-0002-2254-3914.

An earlier preprint of the authors, *Relative secant cycles and Hodge classes*, Preprints.org (2026), doi:10.20944/preprints202602.0462.v5, claimed a proof of the Hodge conjecture that does not hold, and the present paper replaces it. The secant variety $\mathrm{Sec}^n(A^\vee)$ contains $A^\vee$, so the cycle $\Sigma \cap A^\vee$ of its Definition 3 is $A^\vee$ itself, whose Fourier–Mukai transform is a skyscraper sheaf with no component in degree $2n$, contrary to its Lemma 4. The Weil classes span a space of dimension two on which $K^\times$ acts by $\tau \mapsto \tau^{2n}$, while η^n has the character $N(\tau)^n$, contrary to its Propositions 1 and 2. Its general case, Theorem 12, applies the Kuga–Satake construction to Hodge structures that are not of K3 type and assumes that the Kuga–Satake correspondence is algebraic, which is open.

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[1] The second arXiv version of [dSJ21] states, as its Theorem 3.3, the Hodge conjecture for X_M^4 for every M prime to 6, but its proof treats only the classes in tensor products $V(\beta) \otimes V(\gamma)$ of two Hodge lines of lower dimension; the published version keeps the bound $M \leq 100$.

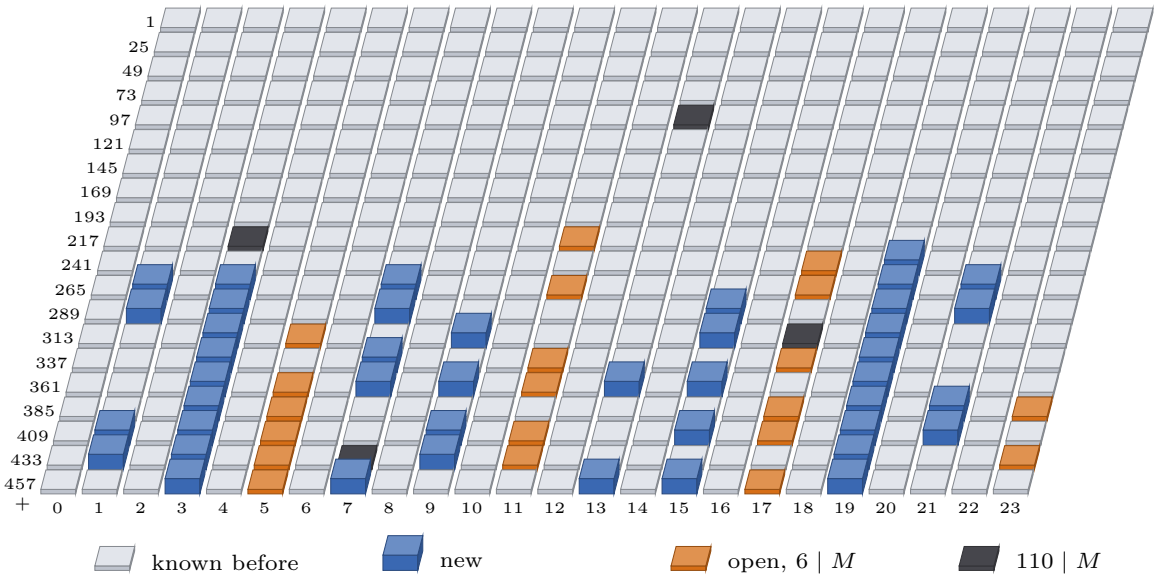


FIGURE 1. The degrees $M \leq 480$, in rows of 24: the row label plus the column number is M. Light tiles are degrees covered by the results quoted above, raised tiles are the degrees added by Theorem A, orange tiles are the degrees divisible by 6 that remain open, and the dark tiles are the multiples of 110.

even degrees up to 250 other than 110, 114, 220 and 228 [Jum26a], and [Bha26b, Theorem B] every odd degree.

Theorem A. *Let $M \geq 1$ be an integer divisible neither by 6 nor by 110. Then every Hodge class on the Fermat fourfold X_M^4 is algebraic.*[2]

The degrees that the results above leave open begin with $228 = 4 \cdot 3 \cdot 19$, $258 = 2 \cdot 3 \cdot 43$, $260 = 4 \cdot 5 \cdot 13$ and $266 = 2 \cdot 7 \cdot 19$. Theorem A settles 260, 266 and every other degree divisible neither by 6 nor by 110. For the degrees divisible by 6, such as 228 and 258, Section 10 reduces the conjecture to a statement about configurations of at most six residues, Hypothesis 10.4, and proves the part of it that concerns large powers of 2 and 3; the rest of that statement is open. The degree 110 is an exception of the method. At that level Aoki's reduction leaves one class, the image of his class at level 55, and the induction leaves sextuples that it cannot lower; we do not know whether their lines are algebraic, and the theorem says nothing about X_M^4 when $110 \mid M$ (Remark 9.1).

The method. A Hodge line of X_M^4 is given by a *sextuple*: six nonzero residues modulo M, up to order, that are balanced in the sense of Section 2. Three kinds of sextuples are known to give algebraic lines: those that contain a pair $\{y, -y\}$, through the inductive structure of

[2]Kang claimed this for all degrees [Kan16, Corollary 3.2], but his proof has a gap. It bounds a refined motivic dimension of $X_M^1 \times X_M^3$ through [Kan15, Proposition 2.2(iv)], whose proof writes the level filtration of a tensor product as $F^m(H \otimes H') = \bigoplus_{t+s=m} F^t H \otimes F^s H'$, where F^p is the largest rational sub-Hodge structure contained in the p-th step of the Hodge filtration. This already fails for $H = H' = H^1(E, \mathbb{Q})$, E an elliptic curve, and $m = 1$: the Künneth component of the diagonal of $E \times E$ is a nonzero rational class of type $(1,1)$ in $H^1(E) \otimes H^1(E)$, so it lies in $F^1(H \otimes H')$, while $F^1 H^1(E) = 0$ and the right-hand side vanishes. The preprint [MMRV26] relies on the same statement for the degrees 33, 39 and 57.

Shioda and Katsura [SK79]; the sextuples that split into two triples with sum zero, through a product of two Fermat curves; and the standard multisets of Aoki, through his explicit cycles [Aok87]. A balanced multiset that meets a standard multiset can be traded for the rest of it, and we call such a trade a *move*. The proof is an induction on the level and on the orders of the entries: every sextuple without a pair has a chain of moves to at most four elements, to a sextuple of smaller orders or to a split sextuple, or it lies in one of finitely many explicit families, whose lines we show to be algebraic directly.

The engine of the induction is a linear relation. If the largest order of an entry of a sextuple A is e, the balancing condition, read through Dirichlet characters of conductor dividing e, says that the function counting the entries of order e in each residue class is a sum of functions that are invariant under the subgroups K_p of units congruent to 1 modulo e/p, one for each prime $p \mid e$ (Section 3). Such *split* functions are rigid: a split function of small total mass is supported on a single coset of one K_p, and a coset of K_p is exactly what a move needs. The only input from analytic number theory is the nonvanishing of the generalised Bernoulli numbers $B_{1,\chi}$ for odd characters χ.

Section 2 recalls the description of the Hodge lines and the algebraic tools, and Section 3 proves the rigidity of split functions. Section 4 treats odd degrees. Section 5 sets up Aoki's reduction, and Section 6 supplies the classes it leaves at the levels 33, 35, 39, 44, 52, 102 and 114, by showing that the relevant parts of the cohomology of Fermat threefolds are supported on divisors. Sections 7 and 8 treat even degrees prime to 3, according to the power of 2 in the largest order of an entry. Section 9 assembles the proof, and Section 10 carries the method to the degrees divisible by 6 as far as it goes.

2. Fermat varieties and their Hodge characters

2.1. **Correspondences.** All varieties are smooth and projective over $\mathbb{C}$, possibly disconnected, and cohomology has rational coefficients unless we say otherwise. Let B and X be varieties, with B of pure dimension b. An algebraic cycle Γ of codimension c on $B \times X$ gives the map

$$\Gamma_*\colon H^k(B) \to H^{k+2c-2b}(X), \qquad y \mapsto \mathrm{pr}_{X*}\bigl(\mathrm{cl}(\Gamma) \cup \mathrm{pr}_B^* y\bigr),$$

a morphism of Hodge structures $H^k(B)(b-c) \to H^{k+2c-2b}(X)$. It carries algebraic classes to algebraic classes, the composite of two such maps is the map of the composite correspondence, and for a morphism $f\colon B \to X$ the graph of f and its transpose give f_* and f^* [Ful98, Chapter 16].

A pure rational Hodge structure V of weight w is *polarizable* if it carries a bilinear form Q with $Q(Cx, \bar{x}) > 0$ for $x \neq 0$, where C acts on $V^{a,b}$ by i^{a-b}. The cohomology of a variety and its Tate twists are polarizable [Voi02, Chapter 6], and polarizable Hodge structures are semisimple.[3]

Lemma 2.1. *Let $\varphi\colon V \to W$ be a surjective morphism of polarizable Hodge structures of weight $2p$. Then φ maps the Hodge classes of V, the rational classes of type (p,p), onto those of W.*

Proof. By semisimplicity the kernel of φ has a complement V' that is a sub-Hodge structure, and φ restricts to an isomorphism $V' \to W$ of Hodge structures. Its inverse carries a Hodge class of W to a Hodge class of $V' \subseteq V$. □

Theorem 2.2 (Lefschetz [Lef24], Kodaira–Spencer [KS53])**.** *Every Hodge class of degree two is algebraic.*

[3] If $W \subseteq V$ is a sub-Hodge structure, the restriction of Q to W is again a polarization, so it is nondegenerate and $V = W \oplus W^\perp$. The orthogonal $W^\perp$ is a sub-Hodge structure, because $V_{\mathbb{C}} = \bigoplus V^{a,b}$ is an orthogonal decomposition for the hermitian form $Q(Cx, \bar{y})$, and $W_{\mathbb{C}}$ is the sum of its intersections with the $V^{a,b}$.

Proof. In the exact sequence $H^1(X,\mathcal{O}_X^\times)\to H^2(X,\mathbb{Z})\to H^2(X,\mathcal{O}_X)$ of the exponential sequence, the second map is the composite of $H^2(X,\mathbb{Z})\to H^2(X,\mathbb{C})$ with the projection to $H^{0,2}(X)$. So an integral class of type $(1,1)$ is the first Chern class of a holomorphic line bundle, which is algebraic by GAGA, and its Chern class is the class of a divisor. A rational Hodge class has an integral multiple. □

2.2. **The Fermat varieties.** For $m\ge 1$ and $n\ge 0$ let $X_m^n\subseteq\mathbb{P}^{n+1}$ be the Fermat variety $x_0^m+\cdots+x_{n+1}^m=0$; thus X_m^0 is a set of m points and X_m^1 the Fermat curve. The group $G_n=\mu_m^{n+2}/\mu_m$ acts by multiplying the coordinates, and its characters are the vectors $a=(a_0,\dots,a_{n+1})\in(\mathbb{Z}/m)^{n+2}$ with $\sum_i a_i=0$. Let $\mathfrak{A}_m^n$ be the set of characters with every $a_i\neq 0$. For $y\in\mathbb{Z}/m$ let $\bar y\in\{0,\dots,m-1\}$ be its representative and $\langle y\rangle=\bar y/m$, and for $a\in\mathfrak{A}_m^n$ put $\langle a\rangle=\sum_i\langle a_i\rangle$, an integer between 1 and $n+1$.

Proposition 2.3. (i) *The primitive cohomology $H^n_{\mathrm{prim}}(X_m^n,\mathbb{C})$ is the direct sum of lines $V(a)$, $a\in\mathfrak{A}_m^n$, on which G_n acts by pull-back through the character a, and $V(a)\subseteq H^{n-q,q}$ with $q=\langle a\rangle-1$.*

(ii) *Let n be even. The space of primitive Hodge classes, tensored with $\mathbb{C}$, is the sum of the lines $V(a)$ with $\langle ta\rangle=n/2+1$ for every $t\in(\mathbb{Z}/m)^\times$.*

Proof. (i) By Griffiths' description of the cohomology of a hypersurface by residues [Gri69, §8],[4] $H^{n-q,q}_{\mathrm{prim}}(X_m^n)$ is isomorphic to the graded piece of degree $(q+1)m-(n+2)$ of the Jacobian ring $\mathbb{C}[x_0,\dots,x_{n+1}]/(x_0^{m-1},\dots,x_{n+1}^{m-1})$; the monomial x^b corresponds to the residue of $x^b\Omega/F^{q+1}$, where $F=\sum x_i^m$ and $\Omega=\sum_i(-1)^i x_i\,dx_0\wedge\dots\widehat{dx_i}\cdots\wedge dx_{n+1}$. That residue is an eigenvector of G_n with character $a=b+(1,\dots,1)$, as Ω has character $(1,\dots,1)$ and F is invariant. As b runs over the exponents with $0\le b_i\le m-2$, a runs once over the characters with all entries nonzero, and the degree condition reads $\langle a\rangle=q+1$.

(ii) The decomposition is defined over $\mathbb{Q}(\mu_m)$, and the automorphism $\zeta\mapsto\zeta^t$ of $\mathbb{Q}(\mu_m)$ carries $V(a)$ to $V(ta)$. So for each orbit O of $(\mathbb{Z}/m)^\times$ on $\mathfrak{A}_m^n$ the sum $\bigoplus_{a\in O}V(a)$ is defined over $\mathbb{Q}$, and these rational subspaces decompose $H^n_{\mathrm{prim}}(X_m^n,\mathbb{Q})$. A rational class with a nonzero component on some $V(a)$ has nonzero components on the whole orbit of a; so it is of type $(n/2,n/2)$ only if every $V(ta)$ is, that is, only if $\langle ta\rangle=n/2+1$ for every unit t. □

2.3. **The inductive structure.** Fix ε with $\varepsilon^m=-1$. For $r,s\ge 1$, let $Y\subseteq X_m^r\times X_m^s$ be the subvariety $x_{r+1}=y_{s+1}=0$, which is isomorphic to $X_m^{r-1}\times X_m^{s-1}$, and let Z be the blow-up of $X_m^r\times X_m^s$ along Y. Shioda and Katsura show that the rational map

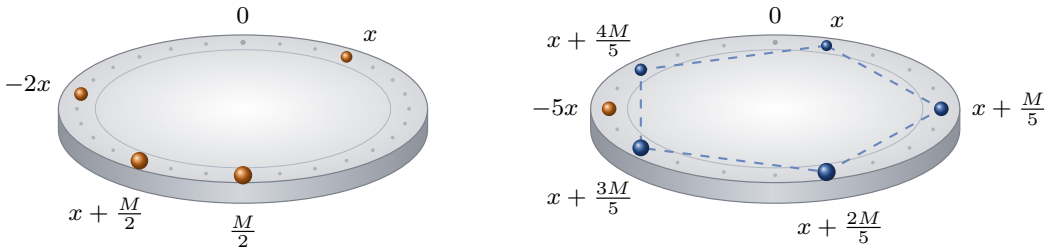
$$(x,y)\mapsto\left[x_0y_{s+1}:\cdots:x_ry_{s+1}:\varepsilon x_{r+1}y_0:\cdots:\varepsilon x_{r+1}y_s\right]$$

extends to a surjective morphism $\psi\colon Z\to X_m^{r+s}$ [SK79, Lemma 1.2].[5] Let $E\subseteq Z$ be the exceptional divisor, $j\colon E\to Z$ the inclusion and $\varpi\colon E\to Y$ the projection. The map

$$f=\psi_*j_*\varpi^*\colon H^{r+s-2}(Y)\to H^{r+s}(X_m^{r+s})$$

[4] For $m=n+2$ the canonical bundle $\mathcal{O}(m-n-2)$ of X_m^n is trivial, and this description gives the Hodge numbers of the Calabi–Yau hypersurface. For the sextic fourfold X_6^4 the graded pieces of degrees 0, 6 and 12 of $\mathbb{C}[x_0,\dots,x_5]/(x_0^5,\dots,x_5^5)$ have dimensions 1, 426 and 1751, so $h^{4,0}=1$, $h^{3,1}=426$, $h^{2,2}=1752$ and $\chi(X_6^4)=2610$. The Hodge numbers of the Calabi–Yau hypersurfaces of degree $n+2$ are derived from the Jacobian ring in [Bha26a], and characteristic classes of Calabi–Yau fourfolds are studied in [BSB+26].

[5] The image lies in X_m^{r+s}, because $y_{s+1}^m\sum_{i\le r}x_i^m+\varepsilon^m x_{r+1}^m\sum_{j\le s}y_j^m=-y_{s+1}^m x_{r+1}^m+x_{r+1}^m y_{s+1}^m=0$. All coordinates vanish exactly on Y, and on the chart of the blow-up where $y_{s+1}=t\,x_{r+1}$ the map is $[tx_0:\cdots:tx_r:\varepsilon y_0:\cdots:\varepsilon y_s]$, where $y_0,\dots,y_s$ do not all vanish.



(a) $T(x)$ for $M = 28$, $x = 3$ (b) $\sigma_{5,x} = \Phi_5(x) + \{-5x\}$ for $M = 20$, $x = 1$

FIGURE 2. Standard multisets, with the residues modulo M drawn on a circle and 0 at the back. (a) $T(3) = \{3, 17, 22, 14\}$ at level 28. (b) At level 20, the coset $\Phi_5(1) = \{1, 5, 9, 13, 17\}$ is a regular pentagon, and $\sigma_{5,1}$ adds the element $-5 = 15$.

carries $V(a') \otimes V(b')$ onto $V(a' * b')$ for $a' \in \mathfrak{A}_m^{r-1}$ and $b' \in \mathfrak{A}_m^{s-1}$, where $a' * b'$ is the juxtaposed character [Shi79, Theorem II], [Aok87, §1]. As a composite of pull-backs and push-forwards along morphisms, it carries algebraic classes to algebraic classes.

2.4. **Balanced multisets.** In the rest of the paper $M \geq 2$ is an integer, and we work with multisets of elements of $\mathbb{Z}/M$; $A + B$ is the sum of multisets, $A - B$ the difference when B is contained in A, tA and $-A$ the images under multiplication.

Definition 2.4. Let A be a multiset of nonzero elements of $\mathbb{Z}/M$ with sum 0. It is *balanced* if $h_A(t) = \sum_{a \in A} \langle ta \rangle$ is the same integer for every $t \in (\mathbb{Z}/M)^\times$. A *pair* is a submultiset $\{a, -a\}$; when M is even, $\{M/2, M/2\}$ is a pair. A multiset is *split* if it is the sum of two multisets of three elements, each with sum 0.

Comparing t with $-t$ shows that a balanced multiset of n elements has $h_A = n/2$, so n is even. Ordering the elements of a balanced multiset of $n \geq 2$ elements gives a character $\alpha \in \mathfrak{A}_M^{n-2}$, and by Proposition 2.3(ii) the line $V(\alpha)$ consists of Hodge classes (for $n = 2$, X_M^0 is a set of M points). Permuting the coordinates permutes these lines, so the following notion does not depend on the order: A is *algebraic* if $V(\alpha) \subseteq \mathrm{Alg}^{(n-2)/2}(X_M^{n-2}) \otimes \mathbb{C}$. Conversely every Hodge class of X_M^4 is, over $\mathbb{C}$, a sum of a multiple of the square of the hyperplane class and of vectors of lines $V(\alpha)$ with α balanced. Since $\mathrm{Alg}^2 \subseteq \mathrm{Hdg}^2$ are rational subspaces, this gives:

Lemma 2.5. *The Hodge conjecture holds for X_M^4 if and only if every balanced multiset of six nonzero elements of $\mathbb{Z}/M$ is algebraic.*

A balanced multiset of six nonzero elements is a *sextuple at level M*.

For an odd prime $\ell \mid M$ and $x \in \mathbb{Z}/M$ put $\Phi_\ell(x) = \{x + jM/\ell : \ 0 \leq j < \ell\}$, the solutions y of $\ell y = \ell x$, and

$$\sigma_{\ell,x} = \Phi_\ell(x) + \{-\ell x\}.$$

For even M and $x \in \mathbb{Z}/M$ with $2x \neq 0$ put

$$T(x) = \{x,\ x + M/2,\ -2x,\ M/2\}.$$

These are the *standard multisets* of Aoki [Aok87, §1].

Lemma 2.6. *Let A and B be balanced multisets of nonzero elements of $\mathbb{Z}/M$.*

(a) *If $|A| \leq 4$, then A is algebraic.*

(b) *If A and B are algebraic, so is $A+B$. If P is a sum of pairs and $A+P$ is algebraic, then A is algebraic.*
(c) *For a unit t, tA is balanced, and it is algebraic if and only if A is.*
(d) *Let $g \mid M$, $M' = M/g$, let A' be a multiset of nonzero elements of $\mathbb{Z}/M'$ with sum 0, and let $gA' \subseteq \mathbb{Z}/M$ be its image under $y \mapsto gy$. Then gA' is balanced if and only if A' is, and in that case gA' is algebraic if and only if A' is.*
(e) *If $\ell x \neq 0$, then $\sigma_{\ell,x}$ is balanced and algebraic. If M is even and $2x \neq 0$, then $T(x)$ is balanced and algebraic.*
(f) *A split sextuple is algebraic.*

Proof. (a) For $|A| = 2$, $A = \{a, -a\}$ and $H^0(X_M^0)$ is spanned by classes of points. For $|A| = 4$, $V(\alpha)$ consists of Hodge classes of degree two on the surface X_M^2, which are algebraic by Theorem 2.2.

(b) The line of $A+B$ is the image of $V(\alpha) \otimes V(\beta)$ under the map f of the inductive structure, which preserves algebraic classes; this is [Aok87, Theorem 1-4], which also gives the converse for a pair $\{a, -a\}$, whose line on X_M^0 is spanned by a class of points.

(c) The automorphism $\zeta \mapsto \zeta^t$ of $\mathbb{Q}(\mu_M)$, acting on $H^*(X_M^{n-2}, \mathbb{Q}) \otimes \mathbb{Q}(\mu_M)$ through the second factor, carries $V(\alpha)$, which is defined over $\mathbb{Q}(\mu_M)$, onto $V(t\alpha)$, and preserves $\operatorname{Alg} \otimes \mathbb{Q}(\mu_M)$.

(d) Every unit modulo M' lifts to a unit modulo M, and $\langle tgy \rangle = \langle t'y \rangle$ for $y \in \mathbb{Z}/M'$ and a unit t with image t', the first bracket taken modulo M and the second modulo M'; this gives the first assertion. Let $\pi\colon X_M^{n-2} \to X_{M'}^{n-2}$ be the finite morphism $[x_i] \mapsto [x_i^g]$, of degree g^{n-1}. It is equivariant for the homomorphism $G \to G'$, $\zeta \mapsto \zeta^g$, so for $\xi \in V(\alpha')$ and $\zeta \in G$, $\zeta^* \pi^* \xi = \pi^*(\zeta^g)^* \xi = \alpha'(\zeta^g) \pi^* \xi$, and $\pi^* V(\alpha') \subseteq V(g\alpha')$. Since $\pi_* \pi^*$ is multiplication by $\deg \pi$, π^* is injective, and as both are lines, $\pi^* V(\alpha') = V(g\alpha')$ and $V(\alpha') = \pi_* V(g\alpha')$. Both π^* and π_* preserve algebraic classes.

(e) The elements of $\Phi_\ell(x)$ are nonzero since $\ell x \neq 0$, and the sum of $\sigma_{\ell,x}$ is $\ell x + \binom{\ell}{2} M/\ell - \ell x = 0$, as ℓ is odd. For a unit t, $t\Phi_\ell(x) = \Phi_\ell(tx)$, and writing $\overline{tx} = qM/\ell + r$ with $0 \le r < M/\ell$,

$$\sum_{y \in \Phi_\ell(tx)} \bar{y} = \sum_{j=0}^{\ell-1} \Big(((q+j) \bmod \ell) \frac{M}{\ell} + r \Big) = \frac{(\ell-1)M}{2} + \overline{\ell t x};$$

adding $\overline{-\ell t x} = M - \overline{\ell t x}$ gives $(\ell+1)M/2$, so $\sigma_{\ell,x}$ is balanced. For $\ell = 3$ it is algebraic by (a). Let $\ell \ge 5$, $d = M/\ell$, and let i be the representative of x in $\{1, \dots, d-1\}$ modulo d (it is not 0, as $\ell x \neq 0$); then $\Phi_\ell(x) = \Phi_\ell(i)$ and $\ell x = \ell i$. Put $g = \gcd(i, d)$, $M' = M/g$, $d' = d/g$, $i' = i/g$. Then $\sigma_{\ell,x} = g\, \sigma_{\ell,i'}$ with $\sigma_{\ell,i'}$ formed at level M', and $\gcd(i', d') = 1$, $d' \ge 2$. If $d' \ge 3$, Aoki constructs a subvariety of $X_{M'}^{\ell-1}$ of codimension $(\ell-1)/2$ that represents $\sigma_{\ell,i'}$ [Aok87, Theorem 2-1], so $\sigma_{\ell,i'}$ is algebraic. If $d' = 2$, then $i' = 1$, $M' = 2\ell$, and $\sigma_{\ell,1} = \{1, 3, \dots, 2\ell-1\} + \{\ell\}$ is the sum of the pairs $\{y, 2\ell - y\}$, $y < \ell$ odd, and $\{\ell, \ell\}$, algebraic by (a) and (b). In both cases (d) gives the claim.

For $T(x)$: let t be a unit, which is odd, and $y = \overline{tx}$; then $y \neq M/2$ and $tM/2 = M/2$. The representatives of the elements of $tT(x)$ are $y,\ y + M/2,\ M - 2y,\ M/2$ if $y < M/2$, and $y,\ y - M/2,\ 2M - 2y,\ M/2$ if $y > M/2$; both add up to $2M$. So $T(x)$ is balanced, and it is algebraic by (a).

(f) Let $A = A_1 + A_2$ with $|A_i| = 3$ and sums 0. Their elements are nonzero, and the entries of A_1 and A_2 are characters a' and b' of the Fermat curve X_M^1, with $\langle ta' \rangle, \langle tb' \rangle \in \{1, 2\}$ and $\langle ta' \rangle + \langle tb' \rangle = 3$ for every unit t. So $V(ta') \otimes V(tb')$ has type $(1,1)$ in $H^2(X_M^1 \times X_M^1)$, and the sum L of these lines over the units is stable under $\operatorname{Gal}(\mathbb{Q}(\mu_M)/\mathbb{Q})$, so $L = L_{\mathbb{Q}} \otimes \mathbb{C}$ for a space $L_{\mathbb{Q}}$ of Hodge classes of degree two on the surface $X_M^1 \times X_M^1$. They are algebraic by Theorem 2.2, and f with $r = s = 2$ carries $V(a') \otimes V(b')$ onto the line of A. □

2.5. **Moves.** The proofs below replace a sextuple by another multiset in a way that keeps track of algebraicity.

Definition 2.7. Let A be a balanced multiset of nonzero elements, σ an algebraic balanced multiset, and $S \subseteq A$ a submultiset that is also contained in σ. The multiset

$$A'' = (A - S) + (-(\sigma - S))$$

is the result of the *move* (σ, S). We usually take σ standard, S consisting of the elements of A that lie in σ, and we remove from A'' every pair it contains.

Lemma 2.8. *In Definition 2.7, A'' is a balanced multiset of nonzero elements and $A''+\sigma = A+P$ for a sum of pairs P. If A'' is algebraic, so is A. The same holds for a chain of moves.*

Proof. With $P = (\sigma-S)+(-(\sigma-S))$ we have $A''+\sigma = (A-S)+S+(\sigma-S)+(-(\sigma-S)) = A+P$. The sum and the function h are additive, and A, P, σ are balanced; so A'' is balanced. If A'' is algebraic, then $A + P = A'' + \sigma$ is algebraic by Lemma 2.6(b), and so is A. □

The *profile* $o(A)$ of a sextuple is the sequence of the additive orders of its elements in nonincreasing order, and profiles are compared lexicographically. For two sextuples A and A'', $o(A'') < o(A)$ exactly when, at the largest integer f at which A and A'' have different numbers of elements of order f, A has more. A chain of moves that turns a sextuple A into a multiset $A^\flat$ with at most four elements is *direct*; it is *lowering* if $A^\flat$ is a sextuple with $o(A^\flat) < o(A)$; it *succeeds* if it is direct or lowering, or $A^\flat$ is a split sextuple, or a sextuple that lies in a proper subgroup of $\mathbb{Z}/M$.

Proposition 2.9 (Induction)**.** *Let $M \geq 2$, and suppose that the Hodge conjecture holds for $X^4_{M'}$ for every proper divisor M' of M. Suppose that every sextuple at level M that contains no pair, is not split and generates $\mathbb{Z}/M$ has a chain of moves that succeeds, or is algebraic for another reason. Then the Hodge conjecture holds for X^4_M.*

Proof. By Lemma 2.5 we show that every sextuple A at level M is algebraic, by induction on the profile, which takes finitely many values. If A contains a pair, then removing it leaves a balanced multiset of four elements, and A is algebraic by Lemma 2.6(a),(b). If A is split, Lemma 2.6(f) applies. If A lies in the subgroup $g\mathbb{Z}/M$, $g > 1$, then $A = gA'$ for a sextuple A' at level M/g, which is algebraic by Lemma 2.5 and the hypothesis, and so is A by Lemma 2.6(d). Otherwise A has a chain of moves that succeeds, or is algebraic by assumption. In the first case the last multiset $A^\flat$ is algebraic by Lemma 2.6(a), by induction on the profile, or by the cases already treated, and A is algebraic by Lemma 2.8. □

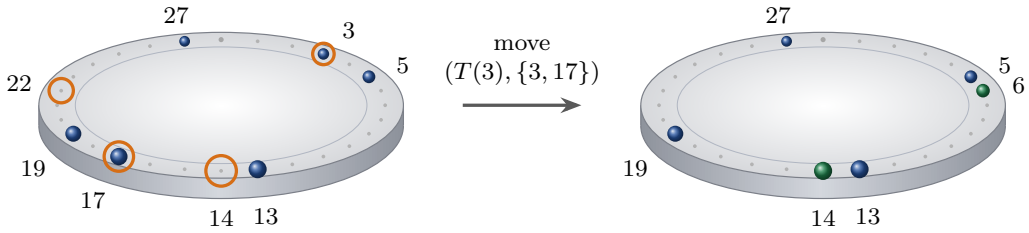
3. Degrees prime to six

In this section M is prime to 6. For $x \in \mathbb{Z}/M$ with $5 \mid M$ we write $\Phi(x) = \Phi_5(x)$, so that $\sigma_{5,x} = \Phi(x) + \{-5x\}$.

Theorem 3.1. *Let M be prime to 6.*

(i) *Every balanced multiset of four nonzero elements of $\mathbb{Z}/M$ contains a pair.*
(ii) *Every balanced multiset of six nonzero elements of $\mathbb{Z}/M$ contains a pair or is $\sigma_{5,x}$ for some x with $5x \neq 0$.*

Part (i) is Aoki's theorem that the Hodge classes of a Fermat surface of degree prime to 6 are spanned by classes of lines [Aok83, Theorem A]; we prove it again because the proof of (ii) uses it and runs on the same lines. When $5 \nmid M$, every prime factor of M exceeds 6, and (ii) is also contained in [Aok83, Theorem A].



$A = \{3, 5, 13, 17, 19, 27\}$, $T(3) = \{3, 17, 22, 14\}$
orders $28, 28, 28, 28, 28, 28$

$A'' = \{5, 13, 19, 27\} + \{6, 14\}$
orders $28, 28, 28, 28, 14, 2$

FIGURE 3. A lowering move at level 28. The sextuple $A = \{3, 5, 13, 17, 19, 27\}$ meets $T(3)$ in $S = \{3, 17\}$; the move replaces S by $-\{22, 14\} = \{6, 14\}$, and two elements of order 28 give way to elements of orders 14 and 2.

Corollary 3.2. *The Hodge conjecture holds for X_M^4 when M is prime to 6.*

Proof. By Lemma 2.5 it suffices to show that every sextuple A at level M is algebraic. If $A = A' + \{a, -a\}$, then A' is balanced, because $\langle ta\rangle + \langle -ta\rangle = 1$ for every unit t; it has four elements and is algebraic by Lemma 2.6(a), and A is algebraic by Lemma 2.6(b). Otherwise $A = \sigma_{5,x}$ with $5x \neq 0$ by Theorem 3.1(ii), and Lemma 2.6(e) applies. □

The rest of this section proves Theorem 3.1. We fix a balanced multiset A of $n \in \{4, 6\}$ nonzero elements of $\mathbb{Z}/M$ that contains no pair, and show that $n = 6$ and that A is a multiset $\sigma_{5,x}$. Since A is balanced, $\sum_{a\in A} \overline{ta} = nM/2$ for every unit t.

3.1. **The counting function.** Let e be the largest additive order of an element of A. It divides M, so it is prime to 6, and $e > 1$. The map $\kappa\colon \mathbb{Z}/e \to \mathbb{Z}/M$, $y \mapsto (M/e)y$, is an injective homomorphism, and the elements of $\mathbb{Z}/M$ of order exactly e are the $\kappa(z)$ with $z \in U = (\mathbb{Z}/e)^\times$. For $z \in U$ put

$$F(z) = \mathrm{mult}_A(\kappa z) - \mathrm{mult}_A(-\kappa z),$$

the difference of the multiplicities in A. Then $F(-z) = -F(z)$. If $a_0 = \kappa(z_0) \in A$ has order e, then $-a_0 \notin A$, so $F(z_0) \geq 1$. We call $\|F\| = \sum_{z\in U} |F(z)|$ the *mass* of F.

Lemma 3.3. *$\|F\| \leq 2n$. If $\|F\| = 2n$, then every element of A has order e and $\sum_{z\in U} F(z)\, z = 0$ in $\mathbb{Z}/e$.*

Proof. $\|F\| \leq \sum_z (\mathrm{mult}_A(\kappa z) + \mathrm{mult}_A(-\kappa z)) = 2N$, where $N \leq n$ is the number of elements of A of order e. If $\|F\| = 2n$, then $N = n$, and writing $a = \kappa(z_a)$ for $a \in A$,

$$\sum_z F(z)z = \sum_{a\in A} z_a - \sum_{a\in A} (-z_a) = 2\sum_{a\in A} z_a,$$

while $\kappa(\sum_a z_a) = \sum_a a = 0$; as κ is injective, $\sum_a z_a = 0$. □

3.2. **The characters.** For a Dirichlet character χ modulo e let $B_{1,\chi} = e^{-1}\sum_{s=1}^{e} \chi(s)s$. If χ is odd and primitive, then $B_{1,\chi} \neq 0$: it is a nonzero multiple of $L(1, \bar\chi)$ [Was97, Theorem 4.9], and $L(1, \bar\chi) \neq 0$.[6]

Lemma 3.4. *$\sum_{z\in U} F(z)\chi(z)^{-1} = 0$ for every primitive Dirichlet character χ modulo e.*

[6]The relative class number formula [Was97, Theorem 4.17] expresses the relative class number of $\mathbb{Q}(\mu_e)$, a positive integer, as a nonzero rational multiple of the product of the numbers $-B_{1,\chi}/2$ over the odd characters χ; so no factor vanishes. This is the only analytic input of the proof.

Proof. If χ is even, the sum vanishes because F is odd. Let χ be odd, and regard it as a character of $(\mathbb{Z}/M)^\times$ through reduction modulo e; it is not trivial. Since A is balanced,

$$\sum_{a\in A}\sum_{t\in(\mathbb{Z}/M)^\times}\chi(t)\,\overline{ta}=\frac{nM}{2}\sum_t\chi(t)=0.$$

Let $a\in A$ have order f. If $f\neq e$, then $e\nmid f$, since $f\le e$. For a divisor f of M let $W_f\subseteq(\mathbb{Z}/M)^\times$ be the units congruent to 1 modulo f. If χ were trivial on W_f, it would be trivial on $W_fW_e=W_{\gcd(f,e)}$ by the Chinese remainder theorem, so it would be induced from a character modulo the proper divisor $\gcd(f,e)$ of e, against primitivity. So χ is not trivial on W_f; as $\overline{ta}$ depends only on t modulo f, the inner sum vanishes. If $f=e$, write $a=\kappa(z)$. For a unit t with image $s\in U$ we have $ta=\kappa(sz)$ and $\overline{ta}=(M/e)\,\overline{sz}$, where now $\overline{sz}\in\{0,\dots,e-1\}$. Reduction $(\mathbb{Z}/M)^\times\to U$ is onto, with fibres of size $\varphi(M)/\varphi(e)$, so

$$\sum_t\chi(t)\,\overline{ta}=\frac{M\,\varphi(M)}{e\,\varphi(e)}\sum_{s\in U}\chi(s)\,\overline{sz}=\frac{M\,\varphi(M)}{\varphi(e)}\,\chi(z)^{-1}B_{1,\chi},$$

after the substitution $s\mapsto sz^{-1}$. Adding over A and dividing by $M\varphi(M)B_{1,\chi}/\varphi(e)\neq0$ gives $\sum_z\operatorname{mult}_A(\kappa z)\chi(z)^{-1}=0$. The substitution $z\mapsto-z$ turns $\sum_z\operatorname{mult}_A(-\kappa z)\chi(z)^{-1}$ into $\chi(-1)^{-1}$ times the same sum, so it vanishes too, and so does their difference. □

3.3. **Splitting along the subgroups K_p.** For a prime $p\mid e$ let $K_p\subseteq U$ be the subgroup of units congruent to 1 modulo e/p, the kernel of $U\to(\mathbb{Z}/(e/p))^\times$.

Lemma 3.5. (a) *$|K_p|=p$ if $p^2\mid e$, and $|K_p|=p-1$ if $p\parallel e$. So $|K_p|\ge4$, with equality only for $p=5\parallel e$.*

(b) *Let $p^v\parallel e$. The product of the K_q with $q\neq p$ consists of units congruent to 1 modulo p^v. Hence K_p meets it trivially, the product H of all the K_q is direct, and -1 does not lie in the product of the K_q with $q\neq p$.*

(c) *A Dirichlet character modulo e that is not primitive is trivial on some K_p.*

Proof. (a) $|K_p|=\varphi(e)/\varphi(e/p)$. (b) For $q\neq p$, p^v divides e/q. An element of K_p congruent to 1 modulo p^v is congruent to 1 modulo $\operatorname{lcm}(e/p,p^v)=e$, and $-1\not\equiv1\pmod p$ as p is odd. (c) A character induced from a proper divisor f of e is trivial on the units congruent to 1 modulo f, and f divides e/p for some $p\mid e$. □

Definition 3.6. Let $K_1,\dots,K_r$ be subgroups of a finite abelian group, H their product and C a coset of H. A function $f\colon C\to\mathbb{Z}$ is *split along* $(K_1,\dots,K_r)$ if $f=f_1+\cdots+f_r$ with functions $f_i\colon C\to\mathbb{C}$ such that $f_i(zk)=f_i(z)$ for $z\in C$ and $k\in K_i$; for $r=0$ this means $f=0$. A *line* along K_i is a coset wK_i.

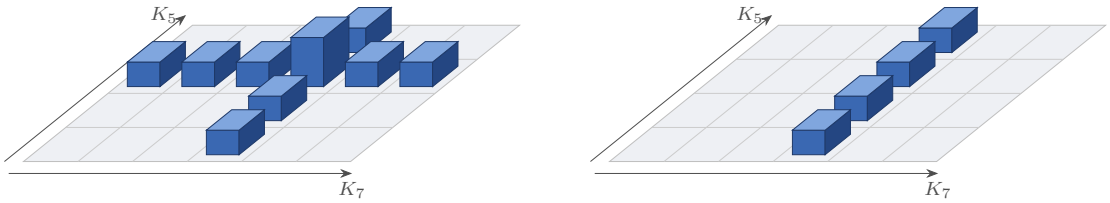
Lemma 3.7. *On every coset of H in U, the function F is split along the family $(K_p)_{p\mid e}$.*

Proof. By Fourier inversion on U, $F=\varphi(e)^{-1}\sum_\chi\hat F(\chi)\chi$ with $\hat F(\chi)=\sum_z F(z)\chi(z)^{-1}$. By Lemma 3.4 only characters that are not primitive contribute, and by Lemma 3.5(c) each of them is trivial on some K_p. Collect the terms according to such a p. □

Four properties of split functions are used below, for f split along $(K_1,\dots,K_r)$ on C and $H'=K_2\cdots K_r$.

(S1) For $k\in K_1$ and every coset $S\subseteq C$ of H', the function $z\mapsto f(z)-f(zk)$ on S is split along $(K_2,\dots,K_r)$: the terms f_1 cancel.

(S2) If f vanishes on one coset $S_0\subseteq C$ of H', then f is split along $(K_2,\dots,K_r)$ on every coset $S\subseteq C$ of H': choose $k\in K_1$ with $Sk=S_0$ and apply (S1).



(a) $\mathbf{1}_{wK_5} + \mathbf{1}_{w'K_7}$, split, mass 10 (b) $\mathbf{1}_{wK_5}$, a line, mass 4

FIGURE 4. Functions on $(\mathbb{Z}/35)^\times = K_5 \times K_7$, a grid of 4×6 cells, drawn as bars. (a) The sum of the indicators of a line along K_5 and a line along K_7 is split and has mass 10. (b) A split function of mass at most 6 is $\pm$ the indicator of a single line (Lemma 3.8).

(S3) For $r = 2$ and $C = xK_1K_2$: $f(xij) = f(xi) + f(xj) - f(x)$ for $i \in K_1$ and $j \in K_2$.
(S4) For $r = 1$, f is invariant under K_1.

3.4. **Three lemmas on split functions.** In the next three lemmas, $K_1, \dots, K_r$ are subgroups of a finite abelian group whose product H is direct (the first one is sometimes called K), and C is a coset of H. Fix $x \in C$ and put $H' = K_2 \cdots K_r$. For c in the first subgroup, the *slices* of C are the cosets $S_c = xcH'$; they are pairwise distinct, and for a function f on C we write $\mu(c) = \sum_{z \in S_c} |f(z)|$.

Lemma 3.8. *Suppose $|K_i| \geq 4$ for every i, with equality for at most one i. Let $f \colon C \to \mathbb{Z}$ be split along $(K_1, \dots, K_r)$, not identically zero, with $\sum_C |f| \leq 6$. Then for some i with $|K_i| \leq 6$ there are a line $L = wK_i \subseteq C$ and $\sigma \in \{\pm 1\}$ such that $f = \sigma$ on L and $f = 0$ on $C \setminus L$.*

Proof. Induction on r. For $r = 0$ the function vanishes, which is excluded. Let $r \geq 1$ and $K = K_1$. First an observation.

(P) *Let $c, c' \in K$ with $\mu(c) + \mu(c') \leq 6$, and suppose $g(z) = f(z) - f(zc^{-1}c')$ is not identically zero on S_c. Then $\mu(c) + \mu(c') \geq |K_i|$ for some $i \geq 2$.* Indeed, g is split along $(K_2, \dots, K_r)$ on S_c by (S1), and $\sum_{S_c} |g| \leq \mu(c) + \mu(c')$ because $z \mapsto zc^{-1}c'$ maps S_c onto $S_{c'}$. By induction $|g|$ is 1 on a line along some K_i, $i \geq 2$, and 0 elsewhere, so $\sum_{S_c} |g| = |K_i|$.

Case A: $f(zk) = f(z)$ for all $z \in C$, $k \in K$. Let $f(z_0) \neq 0$. Then f equals $f(z_0)$ on the $|K| \geq 4$ points of z_0K, so $|f(z_0)| = 1$ and $|K| \leq 6$, and a nonzero value off z_0K would add at least $|K| \geq 4$ more. So $f = f(z_0)$ on z_0K and 0 elsewhere.

Case B1: $\mu(c_0) = 0$ for some c_0. By (S2), f is split along $(K_2, \dots, K_r)$ on every slice. On a slice where it does not vanish, by induction it is ± 1 on a line along some K_i with $i \geq 2$ and $|K_i| \leq 6$ and 0 elsewhere, which uses mass $|K_i| \geq 4$; a second such slice would bring the mass to 8. This is the claim.

Case B2: f is not K-invariant, and $\mu \geq 1$ on every slice. As $\sum \mu \leq 6 < 2|K|$, some c_s has $\mu(c_s) = 1$. Choose $c, k \in K$ such that $f(z) \neq f(zk)$ for some $z \in S_c$. By (P), applied to c and ck, or because $\mu(c) + \mu(ck) > 6$, one of them, c_1, has $\mu(c_1) \geq 2$. Then $f(z) - f(zc_1^{-1}c_s)$ does not vanish identically on S_{c_1}, since translation carries S_{c_1} onto S_{c_s} and $\mu(c_1) \neq \mu(c_s)$. By (P), $\mu(c_1) + 1 \geq |K_i| \geq 4$ for some $i \geq 2$, so

$$6 \geq \sum_{c \in K} \mu(c) \geq \mu(c_1) + (|K| - 1) \geq 3 + 3.$$

All these are equalities: $|K| = 4$ and $|K_i| = \mu(c_1) + 1 = 4$, two subgroups of order four, which is excluded. □

Lemma 3.9. *Let $|K| \geq 10$ and $|K_i| \geq 4$ for $i \geq 2$, with equality for at most one i. Let $f\colon C \to \mathbb{Z}$ be split along $(K, K_2, \dots, K_r)$, and suppose that for some $\varepsilon \in K \setminus \{1\}$ and $\varepsilon' \in H'$ we have $f(z\varepsilon\varepsilon') = -f(z)$ for all $z \in C$. If $\sum_C |f| \leq 12$ and $f(x) \geq 1$, then $f = 1$ on a line along some K_i, $i \geq 2$, with $|K_i| \leq 6$.*

Proof. Multiplication by $\varepsilon\varepsilon'$ carries S_c onto $S_{c\varepsilon}$ and preserves $|f|$, so $\mu(c\varepsilon) = \mu(c)$. If some slice has $\mu = 0$, then f is split along $(K_2, \dots, K_r)$ on $S_1 \ni x$ by (S2), and $2\mu(1) = \mu(1) + \mu(\varepsilon) \leq 12$; by Lemma 3.8, f is σ on a line along some K_i in S_1 and 0 elsewhere on S_1, and $\sigma = 1$ because the line passes through x. Otherwise $\mu \geq 1$ everywhere, and as $\sum \mu \leq 12 < 2|K|$, some c has $\mu(c) = 1$, hence $\mu(c\varepsilon) = 1$. The function $f(z) - f(z\varepsilon)$ on S_c is split along $(K_2, \dots, K_r)$ by (S1) and has mass at most 2, so it vanishes by Lemma 3.8. Let z_0 be the point of S_c with $f(z_0) = \pm 1$. The points $z_0\varepsilon$ and $z_0\varepsilon\varepsilon'$ of $S_{c\varepsilon}$ carry the values $f(z_0)$ and $-f(z_0)$, both nonzero, so they coincide because $\mu(c\varepsilon) = 1$; then $f(z_0) = -f(z_0)$, a contradiction. □

Lemma 3.10. *Let $K \cap L = 1$, $|K| = 4$, $|L| = 6$, and let $\varepsilon \in K$ and $\varepsilon' \in L$ be different from 1, with $\varepsilon^2 = 1$. Let $f\colon xKL \to \mathbb{Z}$ be split along (K, L), with $f(z\varepsilon\varepsilon') = -f(z)$, $\sum |f| \leq 12$ and $f(x) \geq 1$. Then*

(a) *$f(xi) = 1$ for all $i \in K$; or*
(b) *$f(xj) = 1$ for all $j \in L$; or*
(c) *$\sum |f| = 12$, and there are $u\colon K \to \{\pm 1\}$ and $v\colon L \to \{\pm 1\}$ with $u(i\varepsilon) = -u(i)$, $v(j\varepsilon') = -v(j)$ and $2f(xij) = u(i) + v(j)$.*

Proof. By (S3), $f(xij) = f(xi) + f(xj) - f(x)$. Put $\delta = f(x) - f(x\varepsilon')$, $u(i) = 2f(xi) - \delta$ and $v(j) = 2f(xj) - 2f(x) + \delta$, so that $2f(xij) = u(i) + v(j)$. Comparing $f(xi\varepsilon \cdot \varepsilon') = -f(xi)$ with (S3) gives $f(xi\varepsilon) + f(xi) = \delta$, that is $u(i\varepsilon) = -u(i)$; for $i = 1$ it gives $f(x\varepsilon) = -f(x\varepsilon')$, and then the same comparison for $x\varepsilon \cdot j\varepsilon'$ gives $v(j\varepsilon') = -v(j)$. Since $j \mapsto j\varepsilon'$ permutes L and changes the sign of v,

$$2\sum_j |u(i) + v(j)| = \sum_j \bigl(|u(i) + v(j)| + |u(i) - v(j)|\bigr) = 2\sum_j \max(|u(i)|, |v(j)|),$$

so $T = \sum_{i,j} \max(|u(i)|, |v(j)|) = 2\sum |f| \leq 24$. All $u(i)$ and $v(j)$ have the same parity, because $u(i) + v(j) = 2f(xij)$. Write $K = \{1, \varepsilon, a, a\varepsilon\}$, $V = \sum_j |v(j)|$ and $\Sigma(s) = \sum_j \max(s, |v(j)|)$; then $T = 2\Sigma(|u(1)|) + 2\Sigma(|u(a)|)$, $\Sigma(s) \geq \max(6s, V)$, and $V \geq 2|v(1)|$ because $|v(\varepsilon')| = |v(1)|$.

If the parity is odd, all 24 terms of T are at least 1, so all equal 1, and we are in case (c). If it is even and $u(1) \geq 1$, then $u(1) \geq 2$ and $24 \geq T \geq 24 + 2V$, so $u(1) = 2$, $v = 0$, and $f(xj) = (u(1) + v(j))/2 = 1$: case (b). If it is even and $u(1) \leq 0$, then $v(1) = 2f(x) - u(1) \geq 2$. If $u(1) \neq 0$, then $T \geq 2 \cdot 12 + 2V > 24$; so $u(1) = 0$, and in the same way $u(a) = 0$. Then $u = 0$ on K, $T = 4V \leq 24$, so $2v(1) \leq V \leq 6$ and $v(1) = 2$, and $f(xi) = (u(i) + v(1))/2 = 1$: case (a). □

3.5. **The core dichotomy.**

Proposition 3.11. *Let $e > 1$ be prime to 6, $U = (\mathbb{Z}/e)^\times$, and let $F\colon U \to \mathbb{Z}$ satisfy $F(-z) = -F(z)$, $\|F\| \leq 12$ and $F(x) \geq 1$ for some x, and be split along $(K_p)_{p|e}$ on every coset of their product H. Then*

(1) *$F \geq 1$ on a line wK_p for $p = 5$ or $p = 7$; or*
(2) *the primes dividing e are 5 and 7, $|K_5| = 4$, $|K_7| = 6$, and there are $y \in U$, $\varepsilon \in K_5$ and $\varepsilon' \in K_7$ with $\varepsilon\varepsilon' = -1$, and functions $u\colon K_5 \to \{\pm 1\}$ and $v\colon K_7 \to \{\pm 1\}$ with $u(i\varepsilon) = -u(i)$, $v(j\varepsilon') = -v(j)$ and $2F(yij) = u(i) + v(j)$ for $i \in K_5$, $j \in K_7$; the mass of F on yK_5K_7 is 12.*

Proof. By Lemma 3.5(a) every $|K_p|$ is at least 4, only K_5 can have order 4, and $|K_p| \le 9$ forces $p \in \{5, 7\}$.

Case (a): $-1 \notin H$. The cosets xH and $-xH$ are distinct and carry the same mass, so $\sum_{xH} |F| \le 6$. By Lemma 3.8, $F = \sigma$ on a line along some K_p with $|K_p| \le 6$, so $p \in \{5, 7\}$, and $\sigma = 1$ because the line passes through x.

Case (b): $-1 \in H$ and $|K_p| \ge 10$ for some p. Write $-1 = \varepsilon\varepsilon'$ with $\varepsilon \in K_p$ and ε' in the product of the other K_q; $\varepsilon \ne 1$ by Lemma 3.5(b). Lemma 3.9 gives (1).

Case (c): $-1 \in H$ and $|K_p| \le 9$ for all p. Then every prime factor of e is 5 or 7. If only one prime p divides e, then $-1 \in K_p$ and F is K_p-invariant on xK_p by (S4), so $F(x) = F(-x) = -F(x)$, which is impossible. So $H = K_5K_7$, and $-1 = \varepsilon\varepsilon'$ with $\varepsilon \in K_5$, $\varepsilon' \in K_7$, both different from 1 by Lemma 3.5(b). From $\varepsilon^2\varepsilon'^2 = 1$ and $K_5 \cap K_7 = 1$ we get $\varepsilon^2 = \varepsilon'^2 = 1$, so $|K_5|$ and $|K_7|$ are even: $|K_5| = 4$ and $|K_7| = 6$. Lemma 3.10 on xK_5K_7 gives (1) or (2). □

3.6. **The sum of the elements.**

Lemma 3.12. *Let $p \mid e$ with $|K_p| \in \{4, 6\}$, and let $F\colon U \to \mathbb{Z}$ be odd with $F \ge 1$ on a line wK_p and $\|F\| \le 2|K_p|$. Then $\|F\| = 2|K_p|$ and $\sum_z F(z)z \ne 0$ in $\mathbb{Z}/e$.*

Proof. If $e = p$ then $K_p = U \ni -1$, and $F(w) \ge 1$, $F(-w) \ge 1$ contradict oddness. So $e/p \ge 5$, $-1 \notin K_p$, and the lines wK_p and $-wK_p$ are disjoint. As $F \ge 1$ on the first and $F \le -1$ on the second, $\|F\| = 2|K_p|$, $F = 1$ on wK_p, $F = -1$ on $-wK_p$, and $F = 0$ elsewhere. Then $\sum_z F(z)z = 2w \sum_{k \in K_p} k \equiv 2|K_p|\, w \pmod{e/p}$. If it vanished, e/p would divide $2|K_p| \in \{8, 12\}$, which is impossible for $e/p > 1$ prime to 6. □

Lemma 3.13. *In case (2) of Proposition 3.11, if $\|F\| = 12$, then $\sum_z F(z)z \ne 0$ in $\mathbb{Z}/e$.*

Proof. The coset yK_5K_7 carries the whole mass, so $F = 0$ off it. Let $\psi\colon \mathbb{Z}/e \to \mathbb{Z}/5$ be reduction. Since $5 \mid e/7$, every $j \in K_7$ has $\psi(j) = 1$. Since $|K_5| = 4$, $5 \parallel e$, and ψ is injective on K_5: an element of K_5 congruent to 1 modulo 5 is congruent to 1 modulo $\operatorname{lcm}(5, e/5) = e$. From $\varepsilon\varepsilon' = -1$, $\psi(\varepsilon) = -1$. Write $K_5 = \{1, \varepsilon, a, a\varepsilon\}$; then $\psi(a) \ne \pm 1$. As $v(j\varepsilon') = -v(j)$, $\sum_j v(j) = 0$. Hence

$$\begin{aligned}\psi\Big(2\sum_z F(z)z\Big) &= \sum_{i \in K_5}\sum_{j \in K_7} \big(u(i) + v(j)\big)\,\psi(y)\psi(i) \\ &= 6\,\psi(y)\sum_i u(i)\psi(i) = 12\,\psi(y)\big(u(1) + u(a)\psi(a)\big),\end{aligned}$$

using $u(\varepsilon) = -u(1)$ and $u(a\varepsilon) = -u(a)$. Here 12 and $\psi(y)$ are units of $\mathbb{Z}/5$. If $\sum_z F(z)z$ vanished, then $u(1) + u(a)\psi(a) = 0$, and $\psi(a) = \pm 1$, a contradiction. □

3.7. **Proof of Theorem 3.1.** By Lemma 3.3, Lemma 3.7 and the remarks on F, the function F satisfies the hypotheses of Proposition 3.11 with $\|F\| \le 2n \le 12$. In case (2) the mass is 12, so $n = 6$ and $\|F\| = 12 = 2n$, and Lemma 3.3 contradicts Lemma 3.13. So $F \ge 1$ on a line wK_p with $p \in \{5, 7\}$. Since F is odd, $-1 \notin K_p$, so wK_p and $-wK_p$ are disjoint and $\|F\| \ge 2|K_p|$.

If $p = 7$, then $|K_7| = 7$ is impossible as $14 > 12$, and $|K_7| = 6$ forces $n = 6$ and $\|F\| = 12 = 2n$, which Lemma 3.3 and Lemma 3.12 exclude. So $p = 5$. If $n = 4$, then $|K_5| = 5$ is impossible as $10 > 8$, and $|K_5| = 4$ forces $\|F\| = 8 = 2n$, excluded in the same way. This proves (i).

Let $n = 6$ and $p = 5$, and put $x = \kappa(w)$. For $k \in K_5$ we have $5k = 5$ in $\mathbb{Z}/e$, so $5\kappa(wk) = \kappa(5w) = 5x$; and $F(wk) \ge 1$ means $\kappa(wk) \in A$. So A contains the $|K_5|$ distinct

elements of $L = \kappa(wK_5) \subseteq \Phi(x)$. By the computation in the proof of Lemma 2.6(e),

$$\sum_{y\in\Phi(x)} \overline{ty} = 2M + \overline{5tx} \qquad \text{for every unit } t.$$

If $|K_5| = 5$, then $L = \Phi(x)$, and the sixth element b of A satisfies $5x + b = 0$; so $A = \sigma_{5,x}$, and $5x = -b \neq 0$.

If $|K_5| = 4$, then $L = \Phi(x) \setminus \{y_0\}$ for one y_0, and $A = L + \{b, c\}$. Here $e \neq 5$, since otherwise $K_5 = U$ would contain -1; so $5x = \kappa(5w) \neq 0$. Hence $y_0 \neq 0$, as $5y_0 = 5x$, and $y_0 \neq 5x$, since otherwise $20x = 0$, so $5x = 0$, as M is odd. The sum of A gives $5x - y_0 + b + c = 0$, and for every unit t,

$$3M = \sum_{a\in A} \overline{ta} = 2M + \overline{5tx} - \overline{ty_0} + \overline{tb} + \overline{tc}.$$

So the multiset $Q = \{5x, b, c, -y_0\}$ of nonzero elements has sum 0, and $\sum_{q\in Q} \overline{tq} = \overline{5tx} + \overline{tb} + \overline{tc} + M - \overline{ty_0} = 2M$ for every t: it is balanced. By (i) it contains a pair, formed by two of its four members. The pair $\{5x, -y_0\}$ is excluded by $y_0 \neq 5x$, and $\{b, c\}$ is excluded because A contains no pair. Each of the remaining four, $b = -5x$, $c = -5x$, $b = y_0$ or $c = y_0$, together with $5x - y_0 + b + c = 0$, gives $\{b, c\} = \{y_0, -5x\}$, so $A = \sigma_{5,x}$. This proves (ii). □

4. Odd degrees divisible by three

In this section M is odd and divisible by 3. Here Theorem 3.1 fails: the set $\{1, 4, 9, 15, 16, 18\}$ of residues modulo 21 is balanced, contains no pair and is not a multiset $\sigma_{5,x}$. So we use the induction of Proposition 2.9: a sextuple either has a move, by means of one multiset $\sigma_{\ell,x}$, that is direct or lowering, or lies in one of three explicit orbits (Theorem 4.3), whose classes we write down directly (Lemma 4.2).

4.1. **Moves by the multisets $\sigma_{\ell,x}$.** Let A be a sextuple at level M, ℓ an odd prime dividing M, and $x \in \mathbb{Z}/M$ with $\ell x \neq 0$. Let $S \subseteq A$ consist of one copy of each element of $\Phi_\ell(x)$ that lies in A, and let $k = |S|$. The move $(\sigma_{\ell,x}, S)$ of Definition 2.7 gives

$$A'' = (A - S) + (-(\Phi_\ell(x) - S)) + \{\ell x\},$$

a multiset of $7 + \ell - 2k$ elements. We say that (ℓ, x) is a *move for A* if this move is direct or lowering, that is, if $|A''| \leq 4$, or $|A''| = 6$ and $o(A'') < o(A)$.

Lemma 4.1. *Let A be a sextuple, $x \in \mathbb{Z}/M$ an element of order e, and $\ell \neq e$ an odd prime dividing e. If A contains at least $(\ell + 1)/2$ distinct elements of $\Phi_\ell(x)$ of order e, then (ℓ, x) is a move for A.*

Proof. Here $\ell x \neq 0$, as $e \nmid \ell$. Since $\ell \mid e$, the element $M/\ell = (M/e)(e/\ell)$ lies in the subgroup of order e, and hence so does $\Phi_\ell(x)$; and ℓx has order e/ℓ. If $k \geq (\ell + 3)/2$, then $|A''| \leq 4$. Otherwise $k = (\ell + 1)/2$, $|A''| = 6$, and S consists of elements of order e. Passing from A to A'' removes these k elements and adds $\ell + 1 - k = k$ elements of order at most e, one of which, ℓx, has order e/ℓ. So A and A'' have the same elements of order larger than e, and A'' has fewer elements of order e; hence $o(A'') < o(A)$. □

4.2. **Three exceptional orbits.** Let

$$O_{21} = \{1, 4, 9, 15, 16, 18\}, \qquad O_{33} = \{1, 4, 16, 22, 25, 31\}, \qquad O_{39} = \{1, 7, 16, 22, 34, 37\},$$

subsets of $\mathbb{Z}/21$, $\mathbb{Z}/33$ and $\mathbb{Z}/39$. In terms of residues, O_{21} consists of the three elements congruent to 1 modulo 3 whose residue modulo 7 is a nonzero square, and the three elements $3v$ with v a nonsquare modulo 7; O_{33} consists of the five elements congruent to 1 modulo 3 whose residue modulo 11 is a nonzero square, and 22; and O_{39} consists of the six elements

congruent to 1 modulo 3 whose residue modulo 13 lies in $\{1,3,9\} \cup \{7,8,11\}$, the union of two cosets of the subgroup generated by 3 in $(\mathbb{Z}/13)^\times$.

Lemma 4.2. *The sets O_{21}, O_{33} and O_{39} are balanced and algebraic.*

Proof. Balance. Let $O = O_M$. By the description by residues, $tO = O$ for the units $t \equiv 1 \pmod 3$ whose residue modulo 7, 11 or 13 is a nonzero square, a square, or lies in $\{1,3,9\}$ respectively. As $\sum_{a\in O} \overline{-ta} = 6M - \sum_{a\in O} \overline{ta}$, it suffices to check $\sum_{a\in O} \overline{ta} = 3M$ for one t in each coset of the group generated by these units and -1. These cosets are represented by $t = 1, 2$ for $M = 21$, by $t = 1, 5$ for $M = 33$, and by $t = 1, 2, 4, 8$ for $M = 39$, and the representatives of tO are

$$2O_{21} = \{2,8,18,9,11,15\}, \qquad 5O_{33} = \{5,20,14,11,26,23\},$$
$$2O_{39} = \{2,14,32,5,29,35\}, \quad 4O_{39} = \{4,28,25,10,19,31\}, \quad 8O_{39} = \{8,17,11,20,38,23\},$$

with sums 63, 99 and 117, as are the sums of O_{21}, O_{33} and O_{39} themselves.

O_{21}. With $\sigma_{3,2} = \{2,9,16\} + \{15\}$ and $Q = \{1,4,18,19\}$ we have

$$O_{21} + \{2,19\} = \sigma_{3,2} + Q.$$

As O_{21}, the pair and $\sigma_{3,2}$ are balanced, so is Q. By Lemma 2.6(a), $\sigma_{3,2}$ and Q are algebraic, so $O_{21} + \{2,19\}$ is algebraic by Lemma 2.6(b), and so is O_{21}.

O_{33}. By Lemma 2.6(d) it suffices to show that $a = 2O_{33} = \{2,8,32,44,50,62\} \subseteq \mathbb{Z}/66$ is algebraic; it is balanced because O_{33} is. In $\mathbb{Z}/66$,

$$a + \{1,65\} + \{25,41\} + \{33,33\} = Q + T(32) + T(8),$$

with $Q = \{1,25,44,62\}$, $T(32) = \{2,32,33,65\}$ and $T(8) = \{8,33,41,50\}$. As before Q is balanced, the three quadruples on the right are algebraic by Lemma 2.6(a), so is their sum, and a is algebraic by Lemma 2.6(b).

O_{39}. In the same way, $a = 2O_{39} = \{2,14,32,44,68,74\} \subseteq \mathbb{Z}/78$ satisfies

$$a + \{5,73\} + \{7,71\} + \{39,39\} = Q + T(32) + T(5),$$

with $Q = \{2,7,73,74\}$, $T(32) = \{14,32,39,71\}$ and $T(5) = \{5,39,44,68\}$, and we conclude as for O_{33}. □

For $M = 21$, Lemma 4.2 is contained in [dSJ21], and for $M = 33$ and $M = 39$ in [Jum26b]: the identity used for O_{33} is [Jum26b, Theorem 1.4], while [Jum26b] treats O_{39} through the correspondence of Shioda and Katsura [Jum26b, Theorem 1.2].

4.3. **The reduction.**

Theorem 4.3. *Let M be odd and divisible by 3, and let A be a sextuple at level M that contains no pair and generates $\mathbb{Z}/M$. Then A has a move, or $M \in \{21,33,39\}$ and $tA = O_M$ for some $t \in (\mathbb{Z}/M)^\times$.*

Corollary 4.4. *The Hodge conjecture holds for X_M^4 for every odd M.*

Proof. By induction on M. For M prime to 3 this is Corollary 3.2. Let $3 \mid M$; the proper divisors of M are odd, and the Hodge conjecture holds for them by induction. By Theorem 4.3, every sextuple at level M that contains no pair and generates $\mathbb{Z}/M$ has a move, which is a chain of one move that succeeds, or satisfies $tA = O_M$, and is then algebraic by Lemma 4.2 and Lemma 2.6(c). Proposition 2.9 gives the claim. □

The rest of this section proves Theorem 4.3. We fix A as there and assume that A has no move.

4.4. **Maximal orders and the character relation.** An order e of an element of A is *maximal* if no element of A has an order that is a proper multiple of e. As A generates $\mathbb{Z}/M$, M is the least common multiple of the maximal orders. For a maximal order e we use the objects of Section 3: the embedding $\kappa\colon \mathbb{Z}/e \to \mathbb{Z}/M$, $U = (\mathbb{Z}/e)^\times$, and the odd function F on U; we let N_e be the number of elements of A of order e. As A contains no pair, $|F(z)| = \mathrm{mult}_A(\kappa z) + \mathrm{mult}_A(-\kappa z)$, so $\|F\| = 2N_e \le 12$, and Lemma 3.3 holds with the same proof. For a prime $p \mid e$ the subgroup $K_p \subseteq U$ and the product H are defined as there; Lemma 3.5(b),(c) hold with the same proofs, and by Lemma 3.5(a), $|K_p| \ge 4$ for $p \ge 5$, with equality only for $p = 5 \parallel e$, while $|K_3| = 3$ if $9 \mid e$ and $|K_3| = 2$ if $3 \parallel e$.

Lemma 4.5. *Let B be a balanced multiset of nonzero elements of $\mathbb{Z}/M$. For $a \in B$ let f_a be its order, and write $a = (M/f_a)z_a$ with $z_a \in (\mathbb{Z}/f_a)^\times$. Let χ be an odd primitive Dirichlet character of conductor $c \mid M$. Then*

$$\sum_{a\in B,\ c\mid f_a} \frac{\bar\chi(z_a)}{\varphi(f_a)} \prod_{\ell\mid f_a,\ \ell\nmid c} (1-\chi(\ell)) = 0, \tag{1}$$

where ℓ runs over primes and $\bar\chi(z_a)$ is the value at the residue of z_a modulo c.

Proof. First, for a multiple f of c,

$$\sum_{\substack{1\le s\le f\\(s,f)=1}} \chi(s)\, s = f\, B_{1,\chi} \prod_{\ell\mid f,\ \ell\nmid c} (1-\chi(\ell)). \tag{2}$$

Indeed, for N divisible by c, $\sum_{s=1}^{N} \chi(s)s = \sum_{r=1}^{c} \chi(r) \sum_{j=0}^{N/c-1} (r+jc) = NB_{1,\chi}$, since $\sum_r \chi(r) = 0$. Let f_0 be the product of the primes dividing f but not c. As $\chi(s) = 0$ unless $(s,c) = 1$, and then $(s,f) = 1$ if and only if $(s,f_0) = 1$, Möbius inversion gives

$$\sum_{(s,f)=1} \chi(s)s = \sum_{d\mid f_0} \mu(d)\chi(d)\, d \sum_{s'=1}^{f/d} \chi(s')s' = fB_{1,\chi} \sum_{d\mid f_0} \mu(d)\chi(d),$$

which is (2). Now regard χ as a character of $(\mathbb{Z}/M)^\times$ through reduction modulo c. Since B is balanced, $\sum_{a\in B} \sum_t \chi(t)\, \overline{ta} = \frac{|B|M}{2} \sum_t \chi(t) = 0$. If $c \nmid f_a$, the inner sum vanishes, as in the proof of Lemma 3.4: χ is not trivial on the units congruent to 1 modulo f_a, since otherwise it would be induced from a character modulo the proper divisor $\gcd(f_a, c)$ of c. If $c \mid f_a$, then $\overline{ta} = (M/f_a)\, \overline{tz_a}$, with $\overline{tz_a} \in \{0, \dots, f_a - 1\}$, and $\chi(t)$ depends only on t modulo f_a; so, substituting $s \mapsto sz_a^{-1}$ and using (2),

$$\sum_t \chi(t)\, \overline{ta} = \frac{M\,\varphi(M)}{f_a\varphi(f_a)}\, \bar\chi(z_a) \sum_{s\in(\mathbb{Z}/f_a)^\times} \chi(s)\, \bar s = \frac{M\,\varphi(M)B_{1,\chi}}{\varphi(f_a)}\, \bar\chi(z_a) \prod_{\ell\mid f_a,\ \ell\nmid c} (1-\chi(\ell)).$$

Adding over a and dividing by $M\varphi(M)B_{1,\chi} \ne 0$ gives (1). □

For a maximal order e and $c = e$, the only elements in (1) are those of order e, and (1) becomes the relation $\sum_z \mathrm{mult}_A(\kappa z)\chi(z)^{-1} = 0$ of the proof of Lemma 3.4. So Lemma 3.4 holds for every maximal order e, and with it Lemma 3.7: F is split along $(K_p)_{p\mid e}$ on every coset of H.

Lemma 4.6. *Let e be a maximal order, p a prime dividing e and $w \in U$. Then $\kappa(wK_p) \subseteq \Phi_p(\kappa w)$, with equality if $p^2 \mid e$. If $e \ne p$, $|K_p| \ge (p+1)/2$ and $F \ge 1$ on wK_p, then A has a move.*

Proof. An element $k \in K_p$ is congruent to 1 modulo e/p, so $\kappa(wk) - \kappa(w) = (M/e)\, w(k-1)$ lies in $(M/e)(e/p)\mathbb{Z} = (M/p)\mathbb{Z}$. If $p^2 \mid e$, both sides have p elements. If $F \ge 1$ on wK_p, then the $|K_p|$ elements $\kappa(wk)$ of order e lie in A, and Lemma 4.1 applies. □

4.5. **Every maximal order is divisible by 3.** Let e be a maximal order prime to 3; then e is prime to 6. The function F is odd, $\|F\| \le 12$, $F(z_0) \ge 1$ when $\kappa(z_0) \in A$ has order e, and F is split, so Proposition 3.11 applies. In its case (2), $\|F\| = 12$, so all six elements of A have order e, and Lemma 3.3 gives $\sum_z F(z)z = 0$, against Lemma 3.13. In case (1), $F \ge 1$ on a line wK_p with $p \in \{5, 7\}$. As F is odd, $-1 \notin K_p$, so $e \ne p$, and $|K_p| \ge 4 \ge (p+1)/2$; by Lemma 4.6, A has a move, which we excluded.

4.6. **No maximal order is divisible by 9.** Let e be a maximal order with $9 \mid e$. Each K_p consists of units congruent to 1 modulo e/p, and $3 \mid e/p$; so H consists of units congruent to 1 modulo 3, and $-1 \notin H$. Let $F(x) \ge 1$ and $C = xH$. The cosets C and $-C$ are distinct and carry the same mass, so $\sum_C |F| \le 6$. Write $K_3 = \{1, k, k^2\}$ and $H' = \prod_{p \ne 3} K_p$, and let $S_c = xcH'$, for $c \in K_3$, be the slices of C, with $\mu(c) = \sum_{S_c} |F|$.

(i) *No line zK_3 contains two points $z_1 \ne z_2$ with $F(z_1)F(z_2) > 0$.* Otherwise, replacing z_1, z_2 by $-z_1, -z_2$ if necessary, $F(z_1), F(z_2) \ge 1$, and $\kappa(z_1), \kappa(z_2)$ are two elements of A of order e in $\Phi_3(\kappa z_1)$ (Lemma 4.6); as $e \ne 3$, Lemma 4.1 gives a move.

(ii) *$\mu(c) \ge 1$ for every c.* Otherwise, by (S2), F is split along $(K_p)_{p \ne 3}$ on $S_1 \ni x$, where its mass is at most 6. These subgroups have orders at least 4, with equality for at most one of them, so by Lemma 3.8, $F = 1$ on a line xK_p with $p \in \{5, 7\}$ and $|K_p| \le 6$, and Lemma 4.6 gives a move.

(iii) For $c \ne c'$ in K_3 the function $G(z) = F(z) - F(zc^{-1}c')$ on S_c is split along $(K_p)_{p \ne 3}$ by (S1), and $\sum_{S_c} |G| \le \mu(c) + \mu(c') \le 5$ by (ii). It is not zero: by (ii) some $z \in S_c$ has $F(z) \ne 0$, and $F(zc^{-1}c') = F(z)$ would contradict (i), as z and $zc^{-1}c'$ lie on the line zK_3. By Lemma 3.8, $|G| = 1$ on a line along some K_p with $|K_p| \le 5$ and $G = 0$ elsewhere, so $p = 5$ and $\mu(c) + \mu(c') \ge |K_5| \ge 4$. Adding over the three pairs $\{c, c'\}$ gives $2\sum_c \mu(c) \ge 12$. As $\sum_c \mu(c) \le 6$, we get $|K_5| = 4$ and $\mu(c) = 2$ for every c.

Now take $c = 1$ and $c' = k$: $G = \sigma$ on a line $L \subseteq S_1$ along K_5 and $G = 0$ elsewhere, for a sign σ. By (i), $F(z)F(zk) \le 0$ for all z. Off L, $F(z) = F(zk)$, so both vanish; on L, $F(z) - F(zk) = \sigma$, so $(F(z), F(zk))$ is $(\sigma, 0)$ or $(0, -\sigma)$. Since $\mu(1) = 2$, $F = \sigma$ at two points of L, and $F = 0$ at the other two points z of L, where $F(zk) = -\sigma$. The same argument with $c' = k^2$ gives a line $L' \subseteq S_1$ and a sign σ' such that F takes values in $\{0, \sigma'\}$ on S_1, with support in L'; so $L' = L$ and $\sigma' = \sigma$, and $F(zk^2) = -\sigma$ at the two points $z \in L$ with $F(z) = 0$. For such a z we get $F(zk) = F(zk^2) = -\sigma$, against (i).

4.7. **The case $3 \parallel e$.** By the last two subsections, every maximal order e satisfies $3 \parallel e$; write $e = 3e^*$. Then $e^* > 1$: for $e = 3$, $U = K_3 = \{\pm 1\}$, and F would be K_3-invariant by (S4), so $F(1) = F(-1) = -F(1)$, while $F \ne 0$. Fix a maximal order e. By the Chinese remainder theorem $U = (\mathbb{Z}/3)^\times \times U^*$ with $U^* = (\mathbb{Z}/e^*)^\times$, and we write $z = (s, u)$ and $\kappa(s, u)$. Then $K_3 = (\mathbb{Z}/3)^\times \times \{1\}$, and for $p \mid e^*$, $K_p = \{1\} \times K_p^*$, where $K_p^* \subseteq U^*$ consists of the units congruent to 1 modulo e^*/p; so $H = K_3 \times H^*$ with $H^* = \prod_{p \mid e^*} K_p^*$.

For $u \in U^*$ the points $(1, u)$ and $(-1, u)$ form a line along K_3. If $F(1, u)F(-1, u) > 0$, then A has a move, as in (i) above. So $F(1, u)F(-1, u) \le 0$ for all u, and we define $D\colon U^* \to \mathbb{Z}$ by $D(u) = F(1, u) - F(-1, u)$. The elements of A of order e are its *tops* at e.

Lemma 4.7. (a) *D is even, $\|D\| = 2N_e$, and D is split along $(K_p^*)_{p \mid e^*}$ on every coset of H^*.*
(b) *For $u \in U^*$, the tops of the form $\kappa(s, \pm u)$ are $|D(u)|$ in number, and each of them has $s = \operatorname{sign} D(u)$.*

Proof. (a) Since F is odd, $D(-u) = F(1, -u) - F(-1, -u) = -F(-1, u) + F(1, u) = D(u)$. As $F(1, u)F(-1, u) \le 0$, $|D(u)| = |F(1, u)| + |F(-1, u)|$, and these add up to $\|F\| = 2N_e$. Applying (S1) with the element $(-1, 1)$ of K_3 to the cosets $(1, u)(\{1\} \times H^*)$ shows that D is split. (b)

If $D(u) > 0$, then $F(1,u) \geq 0 \geq F(-1,u)$; as A contains no pair, it contains $\kappa(1,u)$ with multiplicity $F(1,u)$, $\kappa(1,-u) = -\kappa(-1,u)$ with multiplicity $-F(-1,u)$, and neither $\kappa(-1,u)$ nor $\kappa(-1,-u)$. The case $D(u) < 0$ is symmetric, and if $D(u) = 0$ there are no such tops. □

Lemma 4.8. *Let $e^* > 1$ be prime to 6, and let $D\colon U^* \to \mathbb{Z}$ be even, not zero, with $\|D\| \leq 12$, and split along $(K_p^*)_{p|e^*}$ on every coset of H^*. Then one of the following holds.*

(E1) *$D = \sigma(\mathbf{1}_L + \mathbf{1}_{-L})$ for a sign σ and a line L along K_5^* or K_7^* with $L \cap (-L) = \emptyset$.*
(E2) *e^* is a prime $q \leq 13$ and D is constant.*
(E3) *$e^* = 35$, and $D(u) = g(u \bmod 5) + h(u \bmod 7)$ with even functions g on $(\mathbb{Z}/5)^\times$ and h on $(\mathbb{Z}/7)^\times$.*

Proof. By Lemma 3.5(a), $|K_p^*| \geq 4$, with equality only for $p = 5 \parallel e^*$, and $|K_p^*| \leq 6$ only for $p \in \{5, 7\}$.

e^* is not squarefree, say $p^2 \mid e^*$. Every K_q^* consists of units congruent to 1 modulo e^*/q, hence modulo p; so $-1 \notin H^*$. For a coset C of H^*, the cosets C and $-C$ are distinct and carry the same mass, so $\sum_C |D| \leq 6$. By Lemma 3.8, on each coset C where $D \neq 0$, $D = \sigma_C$ on a line L_C along K_5^* or K_7^* and $D = 0$ elsewhere on C; as D is even, D is then σ_C on $-L_C \subseteq -C$. Each such pair of cosets carries mass at least 8, so there is only one, and (E1) holds.

$e^* = q$ is prime. Then $U^* = K_q^*$, so D is constant by (S4), and $\|D\| = |D(1)|(q-1) \leq 12$ gives $q \leq 13$: (E2).

e^* is squarefree with at least two prime factors, one of them $p \geq 11$. Now $H^* = U^*$, by the Chinese remainder theorem. Put $H'^* = \prod_{q \neq p} K_q^*$, and consider the slices $S_c = cH'^*$, $c \in K_p^*$, with masses $\mu(c)$. Let $\varepsilon_p \in K_p^*$ be congruent to -1 modulo p; then $-1 = \varepsilon_p \varepsilon'$ with $1 \neq \varepsilon' \in H'^*$, so $-1 \notin K_p^*$, and $-S_c = S_{c\varepsilon_p}$, so $\mu(c\varepsilon_p) = \mu(c)$, with $c\varepsilon_p \neq c$. Suppose that every slice carries mass. As $|K_p^*| \geq 10$ and $\sum \mu \leq 12$, at most one slice has $\mu \geq 3$, hence none, by the pairing; and some c_0 has $\mu(c_0) = 1$. For every c, the function $D(z) - D(zc_0^{-1}c)$ on S_{c_0} is split along $(K_q^*)_{q \neq p}$ by (S1) and has mass at most 3, so it vanishes by Lemma 3.8. Hence D is constant on $z_0 K_p^*$, where z_0 is the point of S_{c_0} with $D(z_0) \neq 0$, and vanishes elsewhere. As $D(-z_0) = D(z_0) \neq 0$, this gives $-1 \in K_p^*$, a contradiction. So some slice has $\mu = 0$, and by (S2) D is split along $(K_q^*)_{q \neq p}$ on every slice. As $\mu(c) = \mu(c\varepsilon_p)$, every slice has mass at most 6, and we conclude by Lemma 3.8 as in the first case, with slices in place of cosets: (E1).

Otherwise e^* is squarefree and its prime factors are 5 and 7, so $e^* = 35$ and $U^* = K_5^* K_7^*$. By (S3), $D(ij) = D(i) + D(j) - D(1)$ for $i \in K_5^*$, $j \in K_7^*$; this gives $D(u) = g(u \bmod 5) + h(u \bmod 7)$, with $g(i) = D(i) - D(1)$ and $h(j) = D(j)$, as i and j are determined by u mod 5 and u mod 7. Since D is even, $g(-i) - g(i) = h(j) - h(-j)$ for all i, j; both sides are a constant γ, and replacing i by $-i$ shows $\gamma = -\gamma$. So g and h are even: (E3). □

Lemma 4.9. *Let $M' > 1$ be prime to 6. There are primitive Dirichlet characters modulo M' of both parities. Among those of a given parity, the values at 3 are not all equal, unless $M' = 5$ and the parity is even.*

Proof. A character modulo M' is a product of characters ψ_p modulo the prime powers $p^a \parallel M'$, and it is primitive if and only if every ψ_p is. The characters modulo p^a are the powers θ^j of a generator θ, which is faithful; so $\theta(-1) = -1$, and $\theta(9) \neq 1$ because $9 \not\equiv 1 \pmod p$. The character θ^j has parity $(-1)^j$, and it is primitive if and only if $(p-1) \nmid j$ when $a = 1$, and $p \nmid j$ when $a \geq 2$. Both conditions hold for $j = 1, 2$, so both parities occur. Replacing $\psi_p = \theta^j$ by $\theta^{j \pm 2}$ keeps the parity and multiplies the value at 3 by $\theta(9)^{\pm 1} \neq 1$. One of θ^{j+2} and θ^{j-2} is primitive, since p cannot divide both $j+2$ and $j-2$, and $p-1$ divides both only if $p = 5$ and $j \equiv 2 \pmod 4$. So the values at 3 in a parity class are constant only if every ψ_p is the quadratic character modulo $p^a = 5$, that is, only if $M' = 5$ and the parity is even. □

Lemma 4.10. *Let $e = 3e^*$ be a maximal order and $b \in A$ an element whose order $f \neq e$ is a multiple of e^*. Then $f = e^*$.*

Proof. The order f divides some maximal order e', and $3 \mid e'$; so $e = 3e^*$ divides e', and $e' = e$ by the maximality of e. Hence $e^* \mid f \mid e$ and $f \neq e$, so $f = e^*$. □

4.8. **The cases (E1) and (E3) do not occur.** Let $e = 3e^*$ be a maximal order. By Lemma 4.7(a), Lemma 4.8 applies to D.

Case (E1). Let $L = wK_p^*$ with $p \in \{5, 7\}$. By Lemma 4.7(b), $N_e = |L|$, and the tops are the elements $\kappa(\sigma, \varepsilon_u u)$, $u \in L$, one for each u, with signs ε_u. For a sign ε, the tops with $\varepsilon_u = \varepsilon$ lie in $\kappa\big((\sigma, \varepsilon w)K_p\big) \subseteq \Phi_p(\kappa(\sigma, \varepsilon w))$ by Lemma 4.6, and $e \neq p$; so by Lemma 4.1 fewer than $(p+1)/2$ of them have $\varepsilon_u = \varepsilon$. If $|L| = 5$, this is impossible. Since $2|L| \leq 12$, the remaining cases are $|L| = 4$, $p = 5$, with two tops of each sign, and $|L| = 6$, $p = 7$, with three of each sign.

Let $p = 5$ and $|L| = 4$, so $5 \parallel e^*$. Here $e^* \neq 5$, since $K_5^* = U^*$ would contain -1; so $e^* = 5e'$ with $e' > 1$ prime to 30, and reduction modulo 5 identifies K_5^* with $(\mathbb{Z}/5)^\times$. Put $\varepsilon'(k) = \varepsilon_{wk}$, a function on $(\mathbb{Z}/5)^\times$ with two values 1 and two values -1. Let $\chi = \chi_5\chi'$ be an odd primitive character modulo e^*, with χ_5 modulo 5 and χ' modulo e'. By Lemma 4.10, the elements $a \in A$ with $e^* \mid f_a$ are the tops and the elements of order e^*, and (1) with $c = e^*$, multiplied by $\varphi(e) = 2\varphi(e^*)$, reads

$$(1 - \chi(3)) \sum_{u \in L} \bar\chi(\varepsilon_u u) + 2 \sum_{f_b = e^*} \bar\chi(z_b) = 0.$$

As χ is odd and $\chi(k) = \chi_5(k)$ for $k \in K_5^*$, the first sum is $\bar\chi(w)E(\chi_5)$ with $E(\chi_5) = \sum_k \varepsilon'(k)\bar\chi_5(k)$. Since $\sum_k \varepsilon'(k) = 0$, E vanishes at the trivial character, and, with λ_5 the quadratic character, $E(\lambda_5) = 2(\varepsilon'(1) + \varepsilon'(4))$. So either $\varepsilon' = \pm\lambda_5$, and then E vanishes at the two characters of order 4; or $E(\lambda_5) = 0$, and then E does not vanish at some character of order 4, as $\varepsilon' \neq 0$. If no element of A has order e^*, choose χ_5 with $E(\chi_5) \neq 0$, and by Lemma 4.9 (note $e' \neq 5$) a primitive χ' with $\chi'(-1) = -\chi_5(-1)$ and $\chi'(3) \neq \chi_5(3)^{-1}$; the relation then fails. If exactly one element b has order e^*, choose $\chi_5 \neq 1$ with $E(\chi_5) = 0$ and a primitive χ' with $\chi'(-1) = -\chi_5(-1)$; the relation becomes $2\bar\chi(z_b) = 0$, which is absurd. If two elements have order e^*, then all six elements of A lie in $\kappa(\mathbb{Z}/e)$; writing them as $\kappa(y)$, we get $\sum y = 0$ in $\mathbb{Z}/e$, while the residues of the y modulo 3 are σ for the four tops and 0 for the other two, so $4\sigma \equiv 0 \pmod 3$, which is false.

Let $p = 7$ and $|L| = 6$. Then $N_e = 6$, so A consists of tops, $M = e$, and $\sum_{u \in L} \varepsilon_u u = 0$ in $\mathbb{Z}/e^*$, because $\sum_{a \in A} a = 0$. Here $7 \parallel e^*$ and, as before, $e^* = 7e'$ with $e' > 1$ prime to 42. Put $\varepsilon'(k) = \varepsilon_{wk}$ for $k \in K_7^*$, identified with $(\mathbb{Z}/7)^\times$, and let P be the set where $\varepsilon' = 1$. Reducing modulo 7 gives $w\sum_k \varepsilon'(k)k \equiv 0$, and $\sum_k \varepsilon'(k)k = 2\sum_P k - 21$, so $\sum_P k \equiv 0 \pmod 7$. The only three-element subsets of $\{1, \dots, 6\}$ with this property are $\{1, 2, 4\}$ and $\{3, 5, 6\}$; so $\varepsilon' = \pm\lambda_7$, the Legendre symbol modulo 7, and $\varepsilon_u = \tau\lambda_7(u)$ for a sign τ. If $e' \neq 5$, let ψ be an even primitive character modulo e' with $\psi(3) \neq -1$ (Lemma 4.9) and $\chi = \lambda_7\psi$, which is odd and primitive modulo e^*. Then $\chi(3) = -\psi(3) \neq 1$, and $\bar\chi(\varepsilon_u u) = \varepsilon_u\lambda_7(u)\bar\psi(u) = \tau\bar\psi(w)$ for $u \in L$, so the left side of (1) with $c = e^*$ is a nonzero multiple of $6\tau\bar\psi(w) \neq 0$. If $e' = 5$, then $M = 105$; take $\chi = \lambda_7$, of conductor 7. As 3 and 5 are nonsquares modulo 7, (1), multiplied by $\varphi(105)$, reads $(1 - \lambda_7(3))(1 - \lambda_7(5))\sum_u \lambda_7(\varepsilon_u u) = 4 \cdot 6\tau = 0$, which is false.

Case (E3), not (E1). Here $e^* = 35$. The function D is constant on the six classes $\{(\pm i, \pm j)\}$ of four elements each, $i \in (\mathbb{Z}/5)^\times$, $j \in (\mathbb{Z}/7)^\times$; write its values as a matrix $(D_{\alpha\beta})$ with $\alpha \in \{1, 2\}$ and $\beta \in \{1, 2, 3\}$, so that $D_{\alpha\beta} = g_\alpha + h_\beta$ and $4\sum |D_{\alpha\beta}| = \|D\| \leq 12$. The difference $\delta = D_{1\beta} - D_{2\beta}$ does not depend on β. If $\delta = 0$, the two rows agree, and $\sum |D_{\alpha\beta}| \leq 3$ leaves one column with both entries σ and zeros elsewhere: this is (E1), with a line along K_5^*. If $|\delta| \geq 2$, every column contributes at least 2, which is too much. If $|\delta| = 1$, every column is $(\delta, 0)$ or

$(0,-\delta)$. If all three columns agree we are in (E1), with a line along K_7^*. Otherwise $\|D\| = 12$, so A consists of tops, and D takes one sign on eight elements of U^* and the other on four. By Lemma 4.7(b) four tops have residue ± 1 modulo 3 and two have the opposite residue, so the residues of the z_a modulo 3 add up to ± 2. But $\sum_a z_a = 0$ in $\mathbb{Z}/e$, because $\sum_a a = 0$ and κ is injective.

4.9. **The case (E2), and the end of the proof.** Every maximal order e is therefore in case (E2): $e = 3q$ with a prime $q \in \{5, 7, 11, 13\}$, and $D = c_0$ is constant. By Lemma 4.7(b) all tops at e have the same residue $\sigma = \operatorname{sign} c_0$ modulo 3, there are $|c_0|$ of them over each pair $\pm u$ of residues modulo q, and $N_e = |c_0|(q-1)/2$. For an odd character ψ modulo q, $\psi(3) = 1$ only if ψ is trivial on the subgroup generated by 3.

$q = 13$. Then $|c_0| = 1$ and $N_e = 6$, so A consists of tops and $M = 39$. Let $P \subseteq (\mathbb{Z}/13)^\times$ be the set of residues modulo 13 of the elements of A, one from each pair $\pm u$, and $\pi = \mathbf{1}_P - \mathbf{1}_{-P}$. For odd ψ modulo 13, (1) with $c = 13$ gives $(1-\psi(3))\sum_P \bar\psi = 0$, and $\sum_v \pi(v)\bar\psi(v) = 2\sum_P \bar\psi$; for even ψ that sum vanishes because π is odd. So the Fourier transform of π is supported on the characters trivial on $\langle 3\rangle = \{1, 3, 9\}$, and π is invariant under $\langle 3\rangle$. The cosets of $\langle 3\rangle$ are $C_j = 2^j\langle 3\rangle$, $j \in \mathbb{Z}/4$, with $-C_j = C_{j+2}$; hence $P = C_j \cup C_{j+1}$ for some j. A unit t with $t \equiv \sigma \pmod 3$ and $t \equiv 2^{3-j} \pmod{13}$ gives $tA = O_{39}$, since $C_3 \cup C_0 = \{7, 8, 11\} \cup \{1, 3, 9\}$.

$q = 11$. Then $|c_0| = 1$ and $N_e = 5$. The sixth element $b = -\sum(\text{tops})$ lies in $\kappa(\mathbb{Z}/33)$, so $M = 33$; its residue modulo 3 is $-5\sigma = \sigma \neq 0$, so $3 \mid f_b$, and $f_b = 3$ as $f_b \neq 33$; so $b \equiv 0 \pmod{11}$. With P and π as before, (1) with $c = 11$ involves only the tops, and shows that the Fourier transform of π is supported on the odd characters trivial on $\langle 3\rangle$, the subgroup of squares. The only such character is the Legendre symbol λ_{11}, which is odd; so $\pi = \pm\lambda_{11}$, and P is the set of nonzero squares or of nonsquares. A unit $t \equiv \sigma \pmod 3$, $t \equiv \pm 1 \pmod{11}$ gives $tA = O_{33}$.

So we may assume that every maximal order is 15 or 21.

$q = 7$. If $|c_0| = 2$, then $N_e = 6$, A consists of tops, and $M = 21$. Let $n(v)$ be the number of tops with residue v modulo 7. Since 3 generates $(\mathbb{Z}/7)^\times$, $\psi(3) \neq 1$ for odd ψ, and (1) gives $\sum_v n(v)\bar\psi(v) = 0$; so the odd function $n(v) - n(-v)$ vanishes, and $n(v) = 1$ for every v. Then A contains all six elements $\kappa(\sigma, v)$, which lie in $\Phi_7(\kappa(\sigma, 1))$ (Lemma 4.6), and Lemma 4.1 gives a move. So $|c_0| = 1$ and $N_{21} = 3$.

$q = 5$. If $|c_0| = 3$, then A consists of tops and $M = 15$; as $\psi(3) = \pm i \neq 1$ for the odd characters ψ modulo 5, (1) gives $n(v) = n(-v)$ as before, which is impossible since $n(v) + n(-v) = 3$. So $N_{15} \in \{2, 4\}$.

Both 15 *and* 21 *are maximal.* Then $M = 105$. As $N_{21} = 3$ and $N_{15} + N_{21} \leq 6$, $N_{15} = 2$, and there is one further element b. Let ρ_3 and ρ_5 be the reductions of $\mathbb{Z}/105$ modulo 3 and 5. The three tops at 21 have equal ρ_3, which therefore add up to 0, and $\rho_5 = 0$. The two tops at 15 have ρ_3 equal to the same nonzero value, and their ρ_5 are $2v_1$ and $2v_2$ with $v_1 \in \{\pm 1\}$ and $v_2 \in \{\pm 2\}$ (one top over each pair), whose sum $2(v_1 + v_2)$ is not 0 modulo 5. From $\rho_3(\sum_a a) = 0$ we get $\rho_3(b) \neq 0$, so $3 \mid f_b$; as f_b divides 15 or 21 and is neither, $f_b = 3$ and $\rho_5(b) = 0$. Then $\rho_5(\sum_a a) \neq 0$, a contradiction.

Only 15 *is maximal*, so $M = 15$, and the elements of A other than the $N = N_{15}$ tops have order 3 or 5. For a of order 3 write $a = 5z_a$; as A contains no pair, its r elements of order 3 are equal. The character χ_3 modulo 3 in (1) gives $(1 - \chi_3(5))N\sigma/8 + r\chi_3(z)/2 = 0$, that is $N\sigma = -2r\chi_3(z)$. If $N = 4$, then $r = 2$: the other two elements have order 3, and (1) for the odd characters modulo 5 involves only the tops, so $n(v) = n(-v)$ as before and $n(v) = 1$ for each $v \in (\mathbb{Z}/5)^\times$. The four tops then lie in $\Phi_5(\kappa(\sigma, 1))$, and Lemma 4.1 gives a move. If $N = 2$, then $r = 1$, and the other three elements $a = 3z_a$ have order 5, with no two of the z_a opposite.

Let ψ be the odd character modulo 5 with $\psi(2) = i$, and $z_1 \in \{\pm 1\}$, $z_2 \in \{\pm 2\}$ the residues modulo 5 of the two tops. Then (1), multiplied by 8, reads

$$(1 - \psi(3))\big(\bar\psi(z_1) + \bar\psi(z_2)\big) + 2\sum_{f_a=5} \bar\psi(z_a) = 0.$$

The first term has absolute value $|1 + i| \cdot |\pm 1 \pm i| = 2$. In the sum, the n_1 elements with $z_a \in \{\pm 1\}$ contribute $\pm n_1$ and the n_2 with $z_a \in \{\pm 2\}$ contribute $\pm i n_2$, with $n_1 + n_2 = 3$; so its absolute value is $(n_1^2 + n_2^2)^{1/2} \geq \sqrt{5} > 1$, a contradiction.

Only 21 *is maximal*, so $M = 21$, and the elements of A other than the three tops have order 3 or 7. Let $W \subseteq (\mathbb{Z}/7)^\times$ be the set of residues modulo 7 of the tops, one from each pair $\pm u$. For the character χ_3 the product in (1) vanishes for the tops, as $\chi_3(7) = 1$, so the elements of order 3, which are equal, have $\sum \chi_3(z_a) = 0$: there are none. So the other three elements are $a = 3z_a$ of order 7, and the multiset Z of the z_a contains no two opposite residues. Let $E = \mathbf{1}_W - \mathbf{1}_{-W}$ and $B(v) = \mathrm{mult}_Z(v) - \mathrm{mult}_Z(-v)$; both are odd, $|E| = 1$, and $\sum_v |B(v)| = 6$. For odd ψ modulo 7, (1), multiplied by 24, reads $(1 - \psi(3))\hat E(\psi) + 2\hat B(\psi) = 0$, where $\hat E(\psi) = \sum_v E(v)\bar\psi(v)$; as $\psi(3)\hat E(\psi)$ is the transform of $v \mapsto E(3v)$, and all three functions are odd, $2B(v) = E(3v) - E(v)$ for every v. So $|B| \leq 1$, hence $|B| = 1$ everywhere, and $E(3v) = -E(v)$. As 3 generates $(\mathbb{Z}/7)^\times$, $E = E(1)\lambda_7$ and $B = -E$: W is the set of nonzero squares or of nonsquares, and $Z = -W$. A unit $t \equiv \sigma \pmod 3$, $t \equiv \pm 1 \pmod 7$, with tW the set of squares, gives $tA = O_{21}$.

This proves Theorem 4.3. □

5. Aoki's reduction

The induction of Proposition 2.9 works one level at a time and produces chains of moves. For many levels a sharper tool is available: Aoki showed that, at a given level, all balanced multisets are combinations of pairs, of the standard multisets and of finitely many further classes, one for each divisor of a certain kind [Aok83, Aok00]. This section sets up his reduction in the language of Section 2; Section 6 supplies the classes it leaves.

5.1. **Aoki's groups.** Let $M \geq 3$, and let R_M be the free abelian group on the symbols (y), $y \in \mathbb{Z}/M \setminus \{0\}$. A multiset A of nonzero elements has the class $[A] = \sum_{a\in A}(a)$, so that $[A + B] = [A] + [B]$, and an element of R_M with nonnegative coefficients is the class of exactly one multiset. A unit t acts by $t \cdot (y) = (ty)$, and for $M = gM'$ the homomorphism $g_*\colon R_{M'} \to R_M$, $(y) \mapsto (gy)$, maps $[A']$ to $[gA']$.

Following Aoki [Aok00, §2], let $B_M \subseteq R_M$ be the subgroup generated by the classes of the balanced multisets with an even number of elements, $D_M \subseteq B_M$ the subgroup generated by the classes $(y) + (-y)$ of the pairs, and S_M the subgroup generated by D_M and the classes of the standard multisets $\sigma_{\ell,x}$ (ℓ an odd prime dividing M, $\ell x \neq 0$) and $T(x)$ (M even, $2x \neq 0$). By Lemma 2.6(e) the standard multisets are balanced and algebraic, so $D_M \subseteq S_M \subseteq B_M$.[7] The three groups are stable under the units, and g_* carries each of them at level M' into the same group at level M: for B_M and D_M this is Lemma 2.6(c),(d), and a unit t maps $\Phi_\ell(x)$ onto $\Phi_\ell(tx)$, so that $t\sigma_{\ell,x} = \sigma_{\ell,tx}$, $tT(x) = T(tx)$, and similarly $g\sigma_{\ell,x} = \sigma_{\ell,gx}$ and $gT(x) = T(gx)$.

Lemma 5.1. *Let $\mathcal{C}$ be a set of algebraic balanced multisets with an even number of elements, and let A be a balanced multiset of an even number of nonzero elements of $\mathbb{Z}/M$ with $[A] \in D_M + \sum_{C\in\mathcal{C}} \mathbb{Z}[C]$. Then A is algebraic. If $B_M = S_M + \sum_{C\in\mathcal{C}} \mathbb{Z}[C]$, then every balanced multiset of an even number of nonzero elements of $\mathbb{Z}/M$ is algebraic, and the Hodge conjecture holds for X_M^n for every n.*

[7] In Aoki's notation the standard multisets are $\sigma_{p,i}$ with $0 < i < M/p$; since $\Phi_\ell(x)$ and ℓx depend only on x modulo M/ℓ, and $T(x + M/2) = T(x)$, they are the multisets used here.

Proof. Write $[A] = \sum_l \epsilon_l [C_l] + \sum_y k_y((y) + (-y))$ with $C_l \in \mathcal{C}$, repetitions allowed, $\epsilon_l = \pm 1$ and $k_y \in \mathbb{Z}$. When $\epsilon_l = -1$ we use $-[C_l] = [-C_l] - \sum_{c \in C_l}((c) + (-c))$, where $-C_l$ is algebraic by Lemma 2.6(c). This gives $[A] = \sum_l [C'_l] + \sum_y k'_y((y) + (-y))$ with algebraic C'_l and integers k'_y. Let P_- and P_+ be the sums of $|k'_y|$ copies of the pair $\{y, -y\}$ over the y with $k'_y < 0$ and with $k'_y > 0$. Then $[A + P_-] = [C'_1 + \cdots + C'_L + P_+]$, so $A + P_- = C'_1 + \cdots + C'_L + P_+$ as multisets. The right side is algebraic by Lemma 2.6(a),(b), and so is A, by Lemma 2.6(b).

If $B_M = S_M + \sum_C \mathbb{Z}[C]$, we add the standard multisets to $\mathcal{C}$. For even n, the Hodge classes in $H^n(X^n_M, \mathbb{Q})$ are spanned by a power of the hyperplane class and the primitive Hodge classes, and by Proposition 2.3(ii) the latter span, over $\mathbb{C}$, the lines $V(a)$ whose entries form a balanced multiset; these lines are algebraic. In every other degree $2k$ the group $H^{2k}(X^n_M, \mathbb{Q})$ is spanned by a power of the hyperplane class, by the Lefschetz hyperplane theorem and Poincaré duality. □

5.2. **Two parity invariants.** The standard multisets and the pairs satisfy parity conditions that a balanced multiset need not satisfy, and these detect the classes outside S_M.

Lemma 5.2. *Let ℓ be an odd prime with $\ell \parallel M$, and let $\nu_\ell \colon R_M \to \mathbb{Z}/2$ be the homomorphism with $\nu_\ell((y)) = 1$ if y has order ℓ and $\nu_\ell((y)) = 0$ otherwise. Then ν_ℓ vanishes on S_M.*

Proof. For $k \mid M$ let $\Gamma_k \subseteq \mathbb{Z}/M$ be the subgroup of order k. A pair contains 0 or 2 elements of order ℓ, since y and $-y$ have the same order. The multiset $\sigma_{p,x}$ consists of the coset $x + \Gamma_p = \Phi_p(x)$, which does not contain 0, and of $-px$; the multiset $T(x)$ consists of $x + \Gamma_2$, of $-2x$ and of $M/2$, whose order 2 differs from ℓ. We write $p = 2$ in the second case.

Let $p \neq \ell$. Then $\Gamma_p \cap \Gamma_\ell = 0$, so $x + \Gamma_p$ contains at most one element of Γ_ℓ. It contains one exactly when $px \in \Gamma_\ell$: if $z \in (x + \Gamma_p) \cap \Gamma_\ell$ then $px = pz$, and conversely, as multiplication by p is an automorphism of Γ_ℓ, $px = pz$ for some $z \in \Gamma_\ell$, and then $x - z \in \Gamma_p$. In that case $-px = -pz$ has the same order as z. So the multiset has 0 or 2 elements of order ℓ.

Let $p = \ell$. As $\ell x \neq 0$, $x \notin \Gamma_\ell$, so $x + \Gamma_\ell$ does not meet Γ_ℓ. If $-\ell x$ had order ℓ, then $\ell^2 x = 0$, so $x \in \Gamma_\ell$, since $\Gamma_{\ell^2} = \Gamma_\ell$ when $\ell \parallel M$. So $\sigma_{\ell,x}$ has no element of order ℓ. □

Lemma 5.3. *Let $M = pq$ with $p \neq q$ odd primes, and let $\nu_R \colon R_M \to \mathbb{Z}/2$ be the homomorphism with $\nu_R((y)) = 1$ if $y^2 = 1$ in $\mathbb{Z}/M$ and $\nu_R((y)) = 0$ otherwise. Then ν_R vanishes on S_M.*

Proof. A pair $\{y, -y\}$ has 0 or 2 elements with square 1. As M is odd, the standard multisets are the $\sigma_{p',x}$ with $p' \in \{p, q\}$, say $p' = p$. The element $-px$ is not a unit, so its square is not 1. Under the isomorphism $\mathbb{Z}/M \cong \mathbb{Z}/p \times \mathbb{Z}/q$, the coset $\Phi_p(x)$ is $\mathbb{Z}/p \times \{x \bmod q\}$, and its elements with square 1 are those with first coordinate ± 1 and $x^2 \equiv 1 \pmod q$; there are two of them or none. □

5.3. **Aoki's theorem.** Let P be the set consisting of 4 and of the odd primes, and let $\mathcal{Q}(M)$ be the set of divisors $q > 1$ of M that are products of an even number of distinct elements of P. For instance $\mathcal{Q}(2^a 3^b 5^c 7^d)$, for a, b, c, d large, consists of 12, 15, 20, 21, 28, 35 and $420 = 4 \cdot 3 \cdot 5 \cdot 7$, while 2 never enters.

Theorem 5.4 (Aoki)**.** *For every $q \in \mathcal{Q}(M)$ there is an element $\xi_q \in B_q$ such that*

$$B_M = S_M + \sum_{q \in \mathcal{Q}(M)} \sum_{t \in (\mathbb{Z}/M)^\times} \mathbb{Z}\, t \cdot (M/q)_* \xi_q.$$

Moreover $2x \in S_M$ and $x + t \cdot x \in S_M$ for every $x \in B_M$ and every unit t.

This is [Aok00, Theorem 2.1]; the second assertion is part of the proof of [Aok00, Corollary 2.3]. It rests on [Aok83, Theorem D], where the groups are normalized differently; Peterson explains

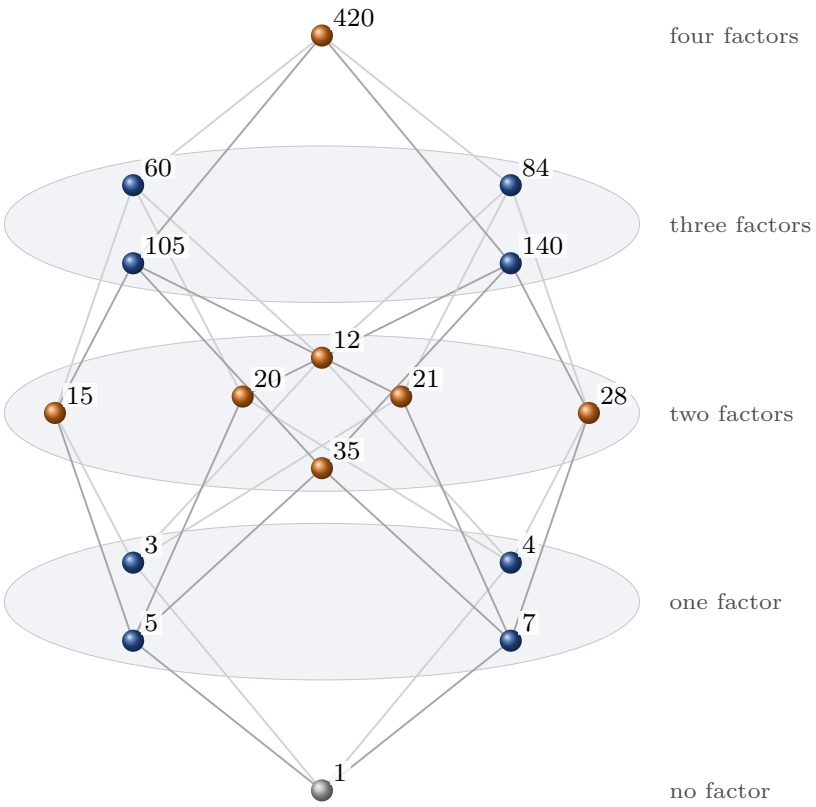


FIGURE 5. The products of distinct elements of P that divide $2^a3^b5^c7^d$ for large exponents, arranged by the number of factors, with an edge for each added factor. The products of an even number of factors, in orange, form $\mathcal{Q}(M)$; each carries one class ξ_q of Theorem 5.4.

the translation [Pet26, §5.2].[8] We use it through the following two consequences. Say that a level M is *complete* if every balanced multiset of an even number of nonzero elements of $\mathbb{Z}/M$ is algebraic. A complete level has the Hodge conjecture for X^n_M in every dimension n, by the end of the proof of Lemma 5.1, and every divisor of a complete level is complete, by Lemma 2.6(d).

Proposition 5.5. *Let $M \geq 3$. If every $q \in \mathcal{Q}(M)$ is a complete level, then M is a complete level.*

Proof. Each $\xi_q \in B_q$ is an integral combination of classes of balanced multisets with an even number of elements, which are algebraic as q is complete. By Lemma 2.6(c),(d), $t \cdot (M/q)_*\xi_q$ is then an integral combination of classes of algebraic balanced multisets at level M. Theorem 5.4 and Lemma 5.1 give the claim. □

Corollary 5.6. *Suppose that $\mathcal{Q}(M)$ has exactly one element, and let $\nu\colon R_M \to \mathbb{Z}/2$ be a homomorphism that vanishes on S_M. If β is a balanced multiset of an even number of nonzero*

[8] Aoki also shows that $B_M/S_M \cong (\mathbb{Z}/2)^{|\mathcal{Q}(M)|}$ [Aok00, Remark 2.4], so that none of the classes ξ_q can be omitted. For $M = p^e$ and $M = 2p^e$, with p an odd prime, $\mathcal{Q}(M)$ is empty, and Theorem 5.4 with Lemma 5.1 gives the Hodge conjecture for X^n_M in every dimension, a theorem of Aoki [Aok87, Aok00].

elements of $\mathbb{Z}/M$ *with* $\nu([\beta]) = 1$*, then* $B_M = S_M + \mathbb{Z}[\beta]$*. This applies to*

$$M = 114,\ \nu = \nu_{19},\ \mathcal{Q} = \{57\}; \qquad M = 102,\ \nu = \nu_{17},\ \mathcal{Q} = \{51\};$$
$$M = 44,\ \nu = \nu_{11},\ \mathcal{Q} = \{44\}; \qquad M = 52,\ \nu = \nu_{13},\ \mathcal{Q} = \{52\};$$
$$M \in \{33, 35, 39\},\ \nu = \nu_R,\ \mathcal{Q} = \{M\}.$$

Proof. Let $\mathcal{Q}(M) = \{q\}$. By Theorem 5.4, B_M/S_M is generated by the images of the elements $t \cdot (M/q)_*\xi_q$, and $t \cdot x \equiv -x \equiv x$ modulo S_M for every $x \in B_M$. So B_M/S_M is generated by one element of order at most 2. As $\nu([\beta]) = 1$, $[\beta] \notin S_M$; hence B_M/S_M has order 2 and is generated by the image of $[\beta]$. The homomorphisms named vanish on S_M by Lemmas 5.2 and 5.3. The elements of P that divide $114 = 2 \cdot 3 \cdot 19$ are 3 and 19, those dividing $102 = 2 \cdot 3 \cdot 17$ are 3 and 17, since $2 \notin \mathsf{P}$, those dividing $44 = 4 \cdot 11$ and $52 = 4 \cdot 13$ are $4, 11$ and $4, 13$, and those dividing 33, 35, 39 are their two prime factors; this gives the values of $\mathcal{Q}$. □

5.4. **Aoki's degrees.** Aoki proves the Hodge conjecture for every abelian variety of Fermat type of degree m when $m = 2^a3^b5^c7^d$ with $c = 0$ or $d = 0$, and when m is p^e or $2p^e$ for a prime p [Aok00, Theorem 0.1]. Among these abelian varieties are the powers of the Jacobian of the Fermat curve X^1_m, and the Fermat varieties of degree m are built from that curve.

Lemma 5.7. *Let* $m \geq 1$*, let* J *be the Jacobian of* $C = X^1_m$*, and suppose that the Hodge conjecture holds for* J^N *for every* N*. Then it holds for* X^n_m *for every* n*.*

Proof. First, the Hodge conjecture holds for C^N. Fix $o \in C$ and let $\iota\colon C \to J$ be the Abel–Jacobi map $x \mapsto [x - o]$; then $\iota^*\colon H^1(J) \to H^1(C)$ is an isomorphism of Hodge structures. By the Künneth formula, $H^*(C^N)$ is the direct sum of the sub-Hodge structures $H_{T,V}$, indexed by disjoint subsets T, V of $\{1, \dots, N\}$, that are the tensor products of $H^1(C)$ at the factors in T, of $H^2(C) = \mathbb{Q}[o]$ at those in V and of $H^0(C)$ at the others. A Hodge class h of C^N is the sum of its components $h_{T,V}$, which are Hodge classes, and $h_{T,V} = \mathrm{pr}_T^* g \cup \prod_{i \in V} \mathrm{pr}_i^*[o]$ for a Hodge class g in the summand $H^1(C)^{\otimes T}$ of $H^{|T|}(C^T)$. This summand is the image, under the isomorphism $(\iota^T)^*$, of the summand $H^1(J)^{\otimes T}$ of $H^{|T|}(J^T)$, so $g = (\iota^T)^* g'$ for a Hodge class g' on J^T, which is algebraic. So g and $h_{T,V}$ are algebraic.

Next, write $X^j = X^j_m$. We show by induction on j that the Hodge conjecture holds for $C^k \times X^j$ for every k; X^0 is a set of m points and $X^1 = C$, so the cases $j \leq 1$ are the first step. Let $j \geq 2$, and let $\psi\colon Z \to X^j$ be the morphism of the inductive structure with $r = 1$ and $s = j - 1$, so that Z is the blow-up of $C \times X^{j-1}$ along $Y \cong X^0 \times X^{j-2}$. Then $W = C^k \times Z$ is the blow-up of $C^{k+1} \times X^{j-1}$ along $C^k \times Y$, a disjoint union of m copies of $C^k \times X^{j-2}$, of codimension two. By the blow-up formula [Voi02, Theorem 7.31], the map

$$H^i(C^{k+1} \times X^{j-1}) \oplus H^{i-2}(C^k \times Y)(-1) \to H^i(W),$$

given by the pull-back and by the pull-back to the exceptional divisor followed by the push-forward, is an isomorphism of Hodge structures. Both components carry algebraic classes to algebraic classes, so by Lemma 2.1 and induction the Hodge conjecture holds for W. The morphism $\phi = \mathrm{id} \times \psi\colon W \to C^k \times X^j$ is surjective between varieties of the same dimension, and $\phi_*\phi^*$ is multiplication by its degree. So a Hodge class h of $C^k \times X^j$ is $(\deg \phi)^{-1}\phi_*\phi^* h$, and $\phi^* h$ is algebraic. The case $k = 0$ is the claim. □

Proposition 5.8. *Every level* $m = 2^a3^b5^c7^d$ *with* $c = 0$ *or* $d = 0$ *is complete. In particular* 12*,* 15*,* 20*,* 21 *and* 28 *are complete levels.*

Proof. The powers of J are abelian varieties of Fermat type of degree m, so [Aok00, Theorem 0.1] and Lemma 5.7 give the Hodge conjecture for X^n_m for every n. If A is a balanced multiset of $n + 2$ nonzero elements of $\mathbb{Z}/m$, the line $V(\alpha)$ of Proposition 2.3(ii) consists of Hodge classes of X^n_m, hence of algebraic classes. □

6. Classes supported on divisors

By Corollary 5.6, at each of the levels 114, 44, 52, 102, 33, 35 and 39 a single balanced multiset β with $\nu([\beta]) = 1$ completes the standard multisets. For 33 and 39 the multiset O_M of Section 4 serves. For the other five we take β to be the join of two characters of the Fermat threefold, or of one character of the threefold and one of the curve, and the main step is to show that the corresponding pieces of H^3 of the threefold are supported on divisors. This follows from a finite set of integers, a block pattern, which makes the quotient of the threefold by the kernel of the character uniruled.

6.1. **Coniveau.** For a smooth projective variety V let $N^1H^k(V,\mathbb{Q})$ be the set of classes in $H^k(V,\mathbb{Q})$ that vanish on $V \setminus D$ for some divisor $D \subseteq V$. It is a subspace, since a union of two divisors is a divisor, and every proper closed subset of V lies in a divisor.

Lemma 6.1. *Let V and W be smooth projective varieties.*

(a) *Let $f\colon V \to W$ be a surjective morphism with $\dim V = \dim W$. Then $f^*N^1H^k(W,\mathbb{Q}) \subseteq N^1H^k(V,\mathbb{Q})$, $f_*N^1H^k(V,\mathbb{Q}) \subseteq N^1H^k(W,\mathbb{Q})$, and f_*f^* is multiplication by $\deg f$.*

(b) *There is a divisor $D \subseteq W$ such that $N^1H^k(W,\mathbb{Q}) = \iota_*H^{k-2}(\tilde{D},\mathbb{Q})$, where $\iota\colon \tilde{D} \to W$ is a resolution of D followed by the inclusion.*

Proof. (a) For an open set $U \subseteq W$ the maps f^* and f_* commute with restriction to U and to $f^{-1}(U)$; for f_* this is the compatibility of proper push-forward in Borel–Moore homology, which on a smooth variety is cohomology by Poincaré duality, with restriction to open sets [Ful98, §19.1]. If y vanishes on $W \setminus D$, then f^*y vanishes on the complement of $f^{-1}(D)$, a proper closed subset as f is surjective. If v vanishes on $V \setminus E$ for a divisor E, then $f(E)$ has dimension less than $\dim W$, so it lies in a divisor D, and f_*v vanishes on $W \setminus D$ because $f^{-1}(W \setminus D) \subseteq V \setminus E$. By the projection formula, $f_*f^*y = f_*(f^*y \cup 1) = y \cup f_*1 = (\deg f)\, y$.

(b) The kernels of the restriction maps $H^k(W,\mathbb{Q}) \to H^k(W \setminus D,\mathbb{Q})$ grow with D, and $H^k(W,\mathbb{Q})$ is finite-dimensional, so $N^1H^k(W,\mathbb{Q})$ is one of these kernels. By a theorem of Deligne [Del74, Corollaire 8.2.8], the sequence $H^{k-2}(\tilde{D},\mathbb{Q}) \xrightarrow{\iota_*} H^k(W,\mathbb{Q}) \to H^k(W \setminus D,\mathbb{Q})$ is exact.[9] □

Lemma 6.2. *If Y is a smooth projective uniruled threefold, then $N^1H^3(Y,\mathbb{Q}) = H^3(Y,\mathbb{Q})$.*

Proof. As Y is uniruled, there are a smooth projective surface S and a dominant rational map $\mathbb{P}^1 \times S \dashrightarrow Y$. Resolving its points of indeterminacy [Hir64], we obtain a smooth projective threefold W, a birational morphism $\sigma\colon W \to \mathbb{P}^1 \times S$ and a surjective morphism $\varphi\colon W \to Y$.

First, $N^1H^3(\mathbb{P}^1 \times S) = H^3(\mathbb{P}^1 \times S)$. By the Künneth formula, $H^3(\mathbb{P}^1 \times S,\mathbb{Q})$ is the sum of $\mathrm{pr}^*H^3(S,\mathbb{Q})$ and $h \cup \mathrm{pr}^*H^1(S,\mathbb{Q})$, where pr is the projection to S and h is the class of $\{p\} \times S$. The second summand vanishes on $(\mathbb{P}^1 \setminus \{p\}) \times S$. Let $C \subseteq S$ be a smooth hyperplane section. By the Lefschetz hyperplane theorem $H_1(C,\mathbb{Q}) \to H_1(S,\mathbb{Q})$ is surjective, so by Poincaré duality every class of $H^3(S,\mathbb{Q})$ is the Gysin image of a class of $H^1(C,\mathbb{Q})$, and it vanishes on $S \setminus C$. So the first summand vanishes on $\mathbb{P}^1 \times (S \setminus C)$.

Next, $N^1H^3(W) = H^3(W)$. The morphism σ is an isomorphism over the complement U of a proper closed subset of $\mathbb{P}^1 \times S$. For $w \in H^3(W,\mathbb{Q})$ the class $\sigma^*\sigma_*w - w$ vanishes on $\sigma^{-1}(U)$, and $\sigma^*\sigma_*w$ lies in N^1 by the first step and Lemma 6.1(a); so $w \in N^1H^3(W,\mathbb{Q})$.

Finally, for $y \in H^3(Y,\mathbb{Q})$ we have $y = (\deg\varphi)^{-1}\varphi_*\varphi^*y$, which lies in $N^1H^3(Y,\mathbb{Q})$ by Lemma 6.1(a). □

[9] The weight filtration on $H^k(W \setminus D)$ starts in weight k, and its lowest piece is the image of $H^k(W)$; the kernel of $H^k(W) \to H^k(W \setminus D)$ is the image of the cohomology with supports $H^k_D(W)$, whose quotient of weight k is the Gysin image of $H^{k-2}(\tilde{D})$, and the purity of $H^k(W)$ forces the kernel into that quotient.

Lemma 6.3. *Let X be a smooth projective threefold, K a finite group acting on X, and $\rho\colon \tilde{Y} \to X/K$ a resolution that is an isomorphism over the smooth locus of X/K. If $N^1H^3(\tilde{Y},\mathbb{Q}) = H^3(\tilde{Y},\mathbb{Q})$, then $H^3(X,\mathbb{Q})^K \subseteq N^1H^3(X,\mathbb{Q})$.*

Proof. We may assume that K acts faithfully. Let $q\colon X \to X/K$ be the quotient map, $X_0 \subseteq X$ the dense open set of points with trivial stabilizer, and $Y_0 = q(X_0)$. Then $X_0 = q^{-1}(Y_0)$, the variety Y_0 is smooth, and $q\colon X_0 \to Y_0$ is a finite étale Galois covering with group K, so that $q^*q_* = \sum_{g\in K} g^*$ on the cohomology of X_0. The rational map $\theta = \rho^{-1}\circ q\colon X \dashrightarrow \tilde{Y}$ is a morphism on X_0. Resolving its indeterminacy [Hir64] gives a smooth projective threefold X'', a birational morphism $\sigma\colon X'' \to X$ that is an isomorphism over X_0, and a surjective morphism $\tau\colon X'' \to \tilde{Y}$ with $\rho\tau = q\sigma$. If $\tau(p) \in \rho^{-1}(Y_0)$, then $q\sigma(p) \in Y_0$ and so $\sigma(p) \in X_0$. Hence $\tau^{-1}(\rho^{-1}(Y_0)) = \sigma^{-1}(X_0)$, and under the isomorphisms $\sigma^{-1}(X_0) \cong X_0$ and $\rho^{-1}(Y_0) \cong Y_0$ the restriction of τ is q.

Let $x \in H^3(X,\mathbb{Q})^K$ and $w = \sigma^*x$. By the compatibility with restriction used in the proof of Lemma 6.1(a), the restriction of $\tau^*\tau_*w$ to $\sigma^{-1}(X_0) \cong X_0$ is $\sum_{g\in K} g^*(x|_{X_0}) = |K|\,x|_{X_0}$. So $\tau^*\tau_*w - |K|\,w$ vanishes on the complement of a proper closed subset of X''. Since $\tau_*w \in N^1H^3(\tilde{Y},\mathbb{Q})$, Lemma 6.1(a) gives $\tau^*\tau_*w \in N^1H^3(X'',\mathbb{Q})$, hence $w \in N^1H^3(X'',\mathbb{Q})$, and $x = \sigma_*\sigma^*x = \sigma_*w$ lies in $N^1H^3(X,\mathbb{Q})$. □

6.2. **Block patterns.** Let $X = X_m^3 \subseteq \mathbb{P}^4$. Its cohomology $H^3(X,\mathbb{Q})$ is primitive, and by Proposition 2.3(i), $H^3(X,\mathbb{C})$ is the direct sum of the lines $V(a)$, $a \in \mathfrak{A}_m^3$, with $V(a) \subseteq H^{3-q,q}$ for $q = \langle a\rangle - 1$. As in the proof of Proposition 2.3(ii), for $a \in \mathfrak{A}_m^3$ the sum $\bigoplus_t V(ta)$ over the units t is defined over $\mathbb{Q}$; let $M_a \subseteq H^3(X,\mathbb{Q})$ be the rational sub-Hodge structure with $M_a \otimes \mathbb{C} = \bigoplus_t V(ta)$.

Fix $\alpha \in \mathfrak{A}_m^3$, with representatives $\bar{\alpha}_k \in \{1,\dots,m-1\}$ of its entries, and write $g\cdot\bar{\alpha} = \sum_k g_k\bar{\alpha}_k$ for $g \in \mathbb{Z}^5$.

Definition 6.4. A *block pattern* for α consists of integers $\delta_1,\dots,\delta_r \geq 1$ and $e_{kl} \geq 0$ ($0 \leq k \leq 4$, $1 \leq l \leq r$) such that

(P1) $D = \sum_l e_{kl}\delta_l$ does not depend on k;
(P2) for every l, some e_{kl} is positive, some e_{kl} vanishes, and the column $e_l = (e_{0l},\dots,e_{4l})$ has $e_l\cdot\bar{\alpha} \equiv 0 \pmod m$;
(P3) $\sum_l \delta_l > D$.

The integers have a geometric meaning: a column e_l with degree δ_l stands for a binary form of degree δ_l, and D is the degree of a rational curve $s \mapsto [u_0(s) : \cdots : u_4(s)]$ with $u_k = \prod_l P_l^{e_{kl}}$ for binary forms P_l of degree δ_l. Condition (P2) makes the monomials $\prod_k u_k^{\bar{\alpha}_k}$ powers of m-th powers, and (P3) says that such curves move in families large enough to sweep out the quotient below; the proof uses only the integers.

Let $T = (\mathbb{C}^\times)^5/\mathbb{C}^\times \subseteq \mathbb{P}^4$ be the open torus. Its character lattice is $M_0 = \{u \in \mathbb{Z}^5 :\ \sum_k u_k = 0\}$, the character u being the function $x^u = \prod_k x_k^{u_k}$. The group $G = \mu_m^5/\mu_m$ is a subgroup of T, and $g \in G$ multiplies x^u by the value at g of the character u mod m of G. Let $K = \ker\alpha \subseteq G$. The characters of G that are trivial on K are the multiples of α, so T/K is a torus with character lattice

$$L_\alpha = \{u \in M_0 :\ u \equiv j\bar{\alpha} \pmod m \text{ for some } j \in \mathbb{Z}\}.$$

The open set $X^\circ = X \cap T$ is stable under G, and X°/K is the hypersurface in T/K defined by the K-invariant Laurent polynomial

$$f = \sum_{k=0}^{4} x^{m(e_k-e_0)} = x_0^{-m}\sum_{k=0}^{4} x_k^m,$$

where $e_0, \dots, e_4$ is the standard basis of $\mathbb{Z}^5$; it is a dense open subset of X/K. The Newton polytope of f is the simplex $P \subseteq L_\alpha \otimes \mathbb{R} = M_0 \otimes \mathbb{R}$ with the five vertices $m(e_k - e_0)$, which lie in L_α. As the coefficients of f at these vertices do not vanish, f is nondegenerate with respect to P [Bat24, Example 5.2]. Following Batyrev [Bat24, Definition 1.2], the *Fine interior* $F(P)$ is the set of points $y \in P$ with $\nu(y) - \min_P \nu \geq 1$ for every nonzero ν in the dual lattice $L_\alpha^\vee = \mathrm{Hom}(L_\alpha, \mathbb{Z})$.[10]

Lemma 6.5. (a) *The nonzero elements of $L_\alpha^\vee$ are the functionals $\nu_g(u) = (g \cdot u)/m$ with $g \in \mathbb{Z}^5$ not constant and $g \cdot \bar{\alpha} \equiv 0 \pmod m$. For $y = \sum_k y_k\, m(e_k - e_0) \in P$, with $y_k \geq 0$ and $\sum_k y_k = 1$, one has $\nu_g(y) - \min_P \nu_g = \sum_k y_k g_k - \min_k g_k$.*

(b) *If α admits a block pattern, then $F(P)$ is empty.*

Proof. (a) A linear form ν on $M_0 \otimes \mathbb{R}$ that is integral on $L_\alpha \supseteq mM_0$ has $m\nu$ integral on M_0, so $m\nu(u) = g \cdot u$ for some $g \in \mathbb{Z}^5$, and $\nu \neq 0$ exactly when g is not constant. The vector $u = \bar{\alpha} - \langle \alpha \rangle m e_0$ lies in L_α, and $g \cdot u \equiv g \cdot \bar{\alpha} \pmod m$; so $g \cdot \bar{\alpha} \equiv 0$. Conversely, if $g \cdot \bar{\alpha} \equiv 0$ and $u \equiv j\bar{\alpha} \pmod m$, then $g \cdot u \equiv jg \cdot \bar{\alpha} \equiv 0$, and ν_g is integral on L_α. Since $\nu_g(m(e_k - e_0)) = g_k - g_0$, we get $\nu_g(y) = \sum_k y_k g_k - g_0$, and the minimum of ν_g on the simplex P is attained at a vertex, so it is $\min_k g_k - g_0$.

(b) Let (δ_l, e_{kl}) be a block pattern and $y \in F(P)$, with barycentric coordinates y_k as in (a). By (P2) and (a), each column e_l defines a nonzero $\nu_{e_l} \in L_\alpha^\vee$ with $\min_k e_{kl} = 0$, so $\sum_k y_k e_{kl} \geq 1$. Multiplying by δ_l, adding over l and using (P1),

$$\sum_l \delta_l \leq \sum_l \delta_l \sum_k y_k e_{kl} = \sum_k y_k \sum_l e_{kl}\delta_l = D \sum_k y_k = D,$$

which contradicts (P3). □

Theorem 6.6. *Let $\alpha \in \mathfrak{A}_m^3$ admit a block pattern, and let $K = \ker \alpha \subseteq G$. Then every smooth projective variety birational to X_m^3/K is uniruled, and $M_\alpha \subseteq N^1 H^3(X_m^3, \mathbb{Q})$.*

Proof. Let $\tilde{Y}$ be a smooth projective variety birational to X/K. By Lemma 6.5(b) and Batyrev's theorem [Bat24, Corollary 5.6], the nondegenerate hypersurface X°/K has negative Kodaira dimension; since X°/K is birational to $\tilde{Y}$, so has $\tilde{Y}$. A smooth projective threefold of negative Kodaira dimension is uniruled: by Mori's theorem [Mor88] it is birational to a projective threefold Y' with terminal singularities that is either minimal, with $K_{Y'}$ nef, or the total space of a Mori fibre space. In the first case Y' would have nonnegative Kodaira dimension by Miyaoka's theorem [Miy88]. In the second, a general fibre F has $-K_F$ ample, so K_F is negative on the curves cut out on F by general ample divisors, and F is uniruled by the criterion of Miyaoka and Mori [MM86]; hence Y' is uniruled, and so is $\tilde{Y}$.

Choose $\tilde{Y}$ to be a resolution of X/K that is an isomorphism over the smooth locus [Hir64]. By Lemma 6.2, $N^1 H^3(\tilde{Y}, \mathbb{Q}) = H^3(\tilde{Y}, \mathbb{Q})$, and by Lemma 6.3, $H^3(X, \mathbb{Q})^K \subseteq N^1 H^3(X, \mathbb{Q})$. For a unit t the character $t\alpha$ is trivial on K, so K acts trivially on $V(t\alpha)$ and on M_α, and $M_\alpha \subseteq H^3(X, \mathbb{Q})^K$. □

6.3. **Joins.** The joins below use the inductive structure of Section 2: for $r, s \geq 1$ the map $f\colon H^{r+s-2}(X_m^{r-1} \times X_m^{s-1}) \to H^{r+s}(X_m^{r+s})$ carries $V(a') \otimes V(b')$ onto $V(a' * b')$ and algebraic classes to algebraic classes.

[10] Every interior lattice point y_0 of a lattice polytope lies in its Fine interior: a nonzero integral functional ν takes integer values at the vertices, so $\min_P \nu \leq \nu(y_0) - 1$. For a reflexive polytope, whose facets lie at lattice distance one from the interior point, the Fine interior is that point alone; these are the polytopes of Batyrev's Calabi–Yau hypersurfaces. The reflexive polytopes of dimension four, which give the Calabi–Yau threefolds of the Kreuzer–Skarke list, are the subject of [Bha26c].

Proposition 6.7. *Let $\alpha, \alpha' \in \mathfrak{A}^3_m$ with M_α and $M_{\alpha'}$ contained in $N^1H^3(X^3_m, \mathbb{Q})$.*

(a) *If $\langle t\alpha\rangle + \langle t\alpha'\rangle = 5$ for every unit t, then the multiset of the ten entries of α and α' is balanced and algebraic.*

(b) *If $b \in \mathfrak{A}^1_m$ and $\langle t\alpha\rangle + \langle tb\rangle = 4$ for every unit t, then the multiset of the eight entries of α and b is balanced and algebraic.*

Proof. (a) Let $X = X^3_m$. The multiset is balanced, because for every unit t the representatives of the entries of $t\alpha$ and $t\alpha'$ add up to $5m$. For every t, one of $V(t\alpha)$, $V(t\alpha')$ lies in $H^{2,1}$ and the other in $H^{1,2}$, as $\langle ta\rangle \in \{1, \dots, 4\}$ and $\langle -ta\rangle = 5 - \langle ta\rangle$ force both values into $\{2, 3\}$. So $L = \bigoplus_t V(t\alpha) \otimes V(t\alpha') \subseteq H^6(X \times X, \mathbb{C})$ has type $(3, 3)$. It is stable under $\mathrm{Gal}(\mathbb{Q}(\mu_m)/\mathbb{Q})$, so $L = L_\mathbb{Q} \otimes \mathbb{C}$ for a space $L_\mathbb{Q} \subseteq M_\alpha \otimes M_{\alpha'}$ of Hodge classes.

By Lemma 6.1(b) there are smooth projective surfaces $\tilde{D}, \tilde{D}'$ with morphisms $\iota\colon \tilde{D} \to X$, $\iota'\colon \tilde{D}' \to X$ such that $M_\alpha \subseteq \iota_* H^1(\tilde{D}, \mathbb{Q})$ and $M_{\alpha'} \subseteq \iota'_* H^1(\tilde{D}', \mathbb{Q})$. The map

$$\gamma = (\iota \times \iota')_*\colon H^1(\tilde{D}, \mathbb{Q}) \otimes H^1(\tilde{D}', \mathbb{Q})(-2) \to H^3(X, \mathbb{Q}) \otimes H^3(X, \mathbb{Q})$$

is a morphism of polarizable Hodge structures of weight 6, given by $\pm\iota_* \otimes \iota'_*$ in the Künneth decomposition, and its image contains $M_\alpha \otimes M_{\alpha'}$. Applying Lemma 2.1 to the restriction of γ to $\gamma^{-1}(M_\alpha \otimes M_{\alpha'})$, every $\upsilon \in L_\mathbb{Q}$ is $\gamma(\upsilon')$ for a Hodge class υ' in $H^1(\tilde{D}) \otimes H^1(\tilde{D}')(-2)$, that is, a Hodge class of degree 2 on $\tilde{D} \times \tilde{D}'$. By Theorem 2.2 it is algebraic, and so is $\upsilon = (\iota \times \iota')_*\upsilon'$. Hence $V(\alpha) \otimes V(\alpha') \subseteq \mathrm{Alg}^3(X \times X) \otimes \mathbb{C}$, and by the inductive structure with $r = s = 4$ the line $V(\alpha * \alpha') = f(V(\alpha) \otimes V(\alpha'))$ is algebraic.

(b) Let $X' = X^1_m$. The multiset is balanced because the entries of $t\alpha$ and tb add up to $4m$. As $\langle tb\rangle \in \{1, 2\}$, we have $\langle t\alpha\rangle \in \{2, 3\}$, and $V(t\alpha) \subseteq H^{2,1}(X)$ exactly when $V(tb) \subseteq H^{0,1}(X')$; so $\bigoplus_t V(t\alpha) \otimes V(tb) \subseteq H^4(X \times X', \mathbb{C})$ has type $(2, 2)$, and as in (a) it is $L_\mathbb{Q} \otimes \mathbb{C}$ for a space $L_\mathbb{Q} \subseteq M_\alpha \otimes H^1(X', \mathbb{Q})$ of Hodge classes. With $\tilde{D}$ and ι as in (a), the map $(\iota \times \mathrm{id})_*\colon H^1(\tilde{D}, \mathbb{Q}) \otimes H^1(X', \mathbb{Q})(-1) \to H^3(X, \mathbb{Q}) \otimes H^1(X', \mathbb{Q})$ has image containing $M_\alpha \otimes H^1(X', \mathbb{Q})$, and Lemma 2.1 lifts every $\upsilon \in L_\mathbb{Q}$ to a Hodge class of degree 2 on the threefold $\tilde{D} \times X'$. It is algebraic by Theorem 2.2, so υ is algebraic, and the inductive structure with $r = 4$ and $s = 2$ shows that $V(\alpha * b)$ is algebraic. □

6.4. **Five pairs of characters.** For a character a of X^3_m or X^1_m we have $\langle -ta\rangle = 5 - \langle ta\rangle$, respectively $3 - \langle ta\rangle$, so it suffices to list the values at the units $t < m/2$. Each value is the sum of the representatives in $\{1, \dots, m-1\}$ of the entries of ta, divided by m.

Lemma 6.8. *In each line of the following list, the characters named satisfy the hypothesis of Proposition 6.7(a) or (b), and the multiset β of their entries is balanced, contains no pair, and has $\nu([\beta]) = 1$ for the homomorphism ν of Corollary 5.6.*

(i) $m = 114$: $\alpha_1 = (1, 25, 43, 57, 102)$, $\alpha_2 = (39, 63, 68, 80, 92)$, $\beta = \{1, 25, 39, 43, 57, 63, 68, 80, 92, 102\}$, $\nu = \nu_{19}$.

(ii) $m = 44$: $\alpha = (1, 5, 8, 33, 41)$, $b = (9, 37, 42)$, $\beta = \{1, 5, 8, 9, 33, 37, 41, 42\}$, $\nu = \nu_{11}$.

(iii) $m = 52$: $\alpha = (1, 5, 16, 37, 45)$, $\alpha' = (29, 18, 42, 50, 17)$, $\beta = \{1, 5, 16, 17, 18, 29, 37, 42, 45, 50\}$, $\nu = \nu_{13}$.

(iv) $m = 102$: $\alpha = (1, 7, 51, 60, 85)$, $\alpha' = (91, 34, 26, 98, 57)$, $\beta = \{1, 7, 26, 34, 51, 57, 60, 85, 91, 98\}$, $\nu = \nu_{17}$.

(v) $m = 35$: $\alpha = (1, 2, 16, 21, 30)$, $b = (17, 22, 31)$, $\beta = \{1, 2, 16, 17, 21, 22, 30, 31\}$, $\nu = \nu_R$.

Proof. The entries are nonzero and the entry sums are multiples of m: 228 and 342 in (i), 88 and 88 in (ii), 104 and 156 in (iii), 204 and 306 in (iv), 70 and 70 in (v). The values at the units $t < m/2$ are:

(i) $m = 114$:

t	1	5	7	11	13	17	23	25	29	31	35	37	41	43	47	49	53	55
$\langle t\alpha_1\rangle$	2	2	2	2	3	2	2	2	3	3	2	2	3	2	2	3	3	2
$\langle t\alpha_2\rangle$	3	3	3	3	2	3	3	3	2	2	3	3	2	3	3	2	2	3

(ii) $m = 44$:

t	1	3	5	7	9	13	15	17	19	21
$\langle t\alpha\rangle$	2	2	3	2	2	2	3	3	2	3
$\langle tb\rangle$	2	2	1	2	2	2	1	1	2	1

(iii) $m = 52$:

t	1	3	5	7	9	11	15	17	19	21	23	25
$\langle t\alpha\rangle$	2	2	2	2	3	2	3	2	3	2	2	3
$\langle t\alpha'\rangle$	3	3	3	3	2	3	2	3	2	3	3	2

(iv) $m = 102$:

t	1	5	7	11	13	19	23	25	29	31	35	37	41	43	47	49
$\langle t\alpha\rangle$	2	2	2	2	3	2	2	3	2	2	2	3	2	3	2	3
$\langle t\alpha'\rangle$	3	3	3	3	2	3	3	2	3	3	3	2	3	2	3	2

(v) $m = 35$:

t	1	2	3	4	6	8	9	11	12	13	16	17
$\langle t\alpha\rangle$	2	2	2	2	2	3	2	2	2	3	3	3
$\langle tb\rangle$	2	2	2	2	2	1	2	2	2	1	1	1

For instance, in (v), $8\alpha = (8, 16, 23, 28, 30)$ and $8b = (31, 1, 3)$ have entry sums 105 and 35.

In each line the representatives of the entries of $t\beta$ add up to $m(\langle t\alpha\rangle + \langle t\alpha'\rangle)$ or $m(\langle t\alpha\rangle + \langle tb\rangle)$, a constant, so β is balanced; and no two entries of β add up to m. The elements of order 19 in $\mathbb{Z}/114$ are the nonzero multiples of 6, and 102 is the only entry of β of this kind in (i); the elements of order 11 in $\mathbb{Z}/44$ and of order 13 in $\mathbb{Z}/52$ are the nonzero multiples of 4, and 8, respectively 16, is the only such entry in (ii), (iii); the elements of order 17 in $\mathbb{Z}/102$ are the nonzero multiples of 6, and 60 is the only such entry in (iv). In (v) the elements of $\mathbb{Z}/35$ with square 1 are 1, 6, 29 and 34, and only 1 is an entry of β. □

Corollary 6.9. *Each of the characters in Table 1 admits the block pattern listed there. In each row the degrees add up to $D + 1$. Hence $M_\alpha \subseteq N^1H^3(X_m^3, \mathbb{Q})$ for each of these characters.*

Proof. Condition (P1) says that each of the five coordinates of $\sum_l \delta_l e_l$ equals D; for instance, in the second row these coordinates are $21 + 3$, $7 + 2 + 15$, $14 + 10$, 24 and $2 + 3 + 19$, and in the last row they are $25 + 10 + 15$, $25 + 20 + 5$, 50, 50 and $10 + 5 + 35$. Condition (P2) is the divisibility of the weights $e_l \cdot \bar{\alpha}$ by m. Row by row, the weights are

114, 228, 114, 228; 114, 342, 114, 228, 1938;
44, 88, 88, 44, 132; 156, 156, 52, 208, 52;
104, 52, 104, 52, 104; 306, 408, 102, 102;
102, 1632, 102, 306, 102; 105, 35, 35, 35, 1050;

for instance $(0, 0, 1, 2, 0) \cdot (51, 21, 86, 14, 56) = 86 + 28$. The degrees add up to 4, 25, 12, 15, 6, 7, 19 and 51, which is (P3). Theorem 6.6 gives the last assertion. □

TABLE 1. Block patterns, a column e_l with degree δ_l being written $e_l{:}\delta_l$

m	α	D	block pattern
114	$13\alpha_2 = (51, 21, 86, 14, 56)$	3	$(0,0,1,2,0){:}1$, $(0,0,2,0,1){:}1$, $(1,3,0,0,0){:}1$, $(2,0,0,1,2){:}1$
114	$\alpha_1 = (1, 25, 43, 57, 102)$	24	$(3,1,2,0,0){:}7$, $(0,1,5,0,1){:}2$, $(0,0,0,2,0){:}12$, $(1,5,0,0,1){:}3$, $(0,0,0,0,19){:}1$
44	$(1, 5, 8, 33, 41)$	11	$(3,0,0,0,1){:}1$, $(0,0,11,0,0){:}1$, $(1,1,0,0,2){:}5$, $(1,2,0,1,0){:}3$, $(0,0,0,4,0){:}2$
52	$(1, 5, 16, 37, 45)$	14	$(0,0,0,3,1){:}4$, $(0,1,1,0,3){:}1$, $(2,1,0,0,1){:}7$, $(0,0,13,0,0){:}1$, $(0,3,0,1,0){:}2$
52	$9\alpha' = (1, 6, 14, 34, 49)$	5	$(0,0,5,1,0){:}1$, $(3,0,0,0,1){:}1$, $(2,0,0,3,0){:}1$, $(0,3,0,1,0){:}1$, $(0,1,0,0,2){:}2$
102	$37\alpha' = (1, 34, 44, 56, 69)$	6	$(0,0,0,3,2){:}1$, $(0,0,3,0,4){:}1$, $(0,3,0,0,0){:}2$, $(2,0,1,1,0){:}3$
102	$5\alpha = (5, 35, 51, 96, 17)$	18	$(0,0,0,0,6){:}2$, $(0,0,0,17,0){:}1$, $(0,0,2,0,0){:}9$, $(0,6,0,1,0){:}1$, $(3,2,0,0,1){:}6$
35	$(1, 2, 16, 21, 30)$	50	$(0,0,0,5,0){:}10$, $(1,1,2,0,0){:}25$, $(1,2,0,0,1){:}10$, $(3,1,0,0,1){:}5$, $(0,0,0,0,35){:}1$

Theorem 6.10. *The levels* 33*,* 35*,* 39*,* 44*,* 52*,* 102 *and* 114 *are complete, and so are* 51 *and* 57*.*

Proof. For $M \in \{33, 39\}$, O_M is balanced and algebraic by Lemma 4.2. The elements of $\mathbb{Z}/33$ with square 1 are 1, 10, 23, 32, and those of $\mathbb{Z}/39$ are 1, 14, 25, 38; of these, $O_{33} = \{1, 4, 16, 22, 25, 31\}$ and $O_{39} = \{1, 7, 16, 22, 34, 37\}$ contain only 1. So $\nu_R([O_M]) = 1$, and by Corollary 5.6, $B_M = S_M + \mathbb{Z}[O_M]$.

For the other five levels let β be the multiset of Lemma 6.8. Since $M_{ta} = M_a$ for every unit t, Corollary 6.9 shows that M_α, and $M_{\alpha'}$ in (i), (iii), (iv), lie in $N^1H^3(X^3_m, \mathbb{Q})$: in (i), $M_{\alpha_2} = M_{13\alpha_2}$; in (iii), $M_{\alpha'} = M_{9\alpha'}$; in (iv), $M_\alpha = M_{5\alpha}$ and $M_{\alpha'} = M_{37\alpha'}$. By Proposition 6.7, β is algebraic, and by Corollary 5.6, $B_m = S_m + \mathbb{Z}[\beta]$.

In all seven cases Lemma 5.1 shows that the level is complete. The levels 57 and 51 divide 114 and 102. □

Corollary 6.11. *The following levels M are complete, and the Hodge conjecture holds for X^n_M for every n:*

(i) $M = 2^a3^b5^c7^d$ *with* $420 \nmid M$*;*
(ii) $M = 2^a3^b11^c$ *and* $M = 2^a3^b13^c$*;*
(iii) $M = 2^a3^b17^c$ *and* $M = 2^a3^b19^c$ *with* $a \leq 1$*.*

Proof. For $M \leq 2$, X^n_M is a linear space or a quadric, and every balanced multiset at level 2 is a sum of pairs $\{1, 1\}$; let $M \geq 3$. In (i), the elements of P dividing M are among $4, 3, 5, 7$, so $\mathcal{Q}(M)$ is contained in $\{12, 15, 20, 21, 28, 35, 420\}$, and $420 \nmid M$ removes the last; in (ii) it is contained in $\{12, 33, 44\}$ or $\{12, 39, 52\}$; in (iii), where $4 \nmid M$, in $\{51\}$ or $\{57\}$. These levels are complete by Proposition 5.8 and Theorem 6.10, and Proposition 5.5 applies. □

Part (i) for the degrees divisible by 35 is due to Peterson [Pet26], who proves the case $M = 35$ with the character α of Lemma 6.8(v) by a family of curves on X^3_{35} and a residue computation; the block pattern replaces that computation. Parts (ii) and (iii) are [Bha26b, Theorem A], where the coniveau step is also proved by families of curves.

7. Even degrees prime to three

In this section and the next, M is even and prime to 3, and A is a sextuple at level M that contains no pair, is not split and generates $\mathbb{Z}/M$. We look for a chain of moves that succeeds, as required by Proposition 2.9. The definitions of Section 4 (maximal orders, the embedding κ, the function F, moves (ℓ, x) for A) and Lemmas 4.1, 4.5 and 4.6 do not use that M is odd, and we use them as they stand.

The *odd part* of a positive integer f is $f/2^{v_2(f)}$. Let e^* be an odd part of an order of an element of A that is maximal for divisibility among these odd parts, and let $2^J e^*$ be the largest order of an element of A with odd part e^*. Then $2^J e^*$ is a maximal order: an order that is a proper multiple of it would have an odd part divisible by e^*, hence equal to e^*, and a larger power of 2. This section treats $J = 1$, and Section 8 treats $J \geq 2$.

7.1. **Every maximal order is even.**

Lemma 7.1. *If A has no move, then every maximal order of A is even. In particular $J \geq 1$.*

Proof. Let e be an odd maximal order. Since $3 \nmid M$, e is prime to 6, and $e > 1$. As in Section 4, F is odd, $\|F\| = 2N_e \leq 12$, $F(z_0) \geq 1$ when $\kappa(z_0) \in A$ has order e, and F is split along $(K_p)_{p|e}$ on every coset of H, so Proposition 3.11 applies. In its case (1), $F \geq 1$ on a line wK_p with $p \in \{5, 7\}$; as F is odd, $-1 \notin K_p$, so $e \neq p$, and $|K_p| \geq 4 \geq (p+1)/2$, so A has a move by Lemma 4.6. In case (2), $\|F\| = 12$, so the six elements of A have order 35 and lie in the subgroup of order 35; they do not generate $\mathbb{Z}/M$, as M is even. □

7.2. **The top order.**

From now on in this section $J = 1$, and we put $e = 2e^*$. If $e^* = 1$, every odd part is 1, the largest order is 2, and A consists of six copies of $M/2$, which contains a pair; so $e^* > 1$, and e^* is prime to 6. Let $U^* = (\mathbb{Z}/e^*)^\times$. For $v \in U^*$ let

$$x_v = \tfrac{M}{e}\, u \quad (u \text{ odd},\ u \equiv v \bmod e^*), \qquad y_v = x_v + \tfrac{M}{2} = \tfrac{M}{e^*} \cdot \tfrac{v}{2},$$

where $v/2$ is taken modulo e^*. The x_v are the elements of order e and the y_v those of order e^*, and

$$-x_v = x_{-v}, \quad -y_v = y_{-v}, \quad 2x_v = y_{2v}, \quad -2x_v = y_{-2v}, \quad T(x_v) = \{x_v,\, y_v,\, y_{-2v},\, M/2\}.$$

Reduction modulo e^* identifies $U = (\mathbb{Z}/e)^\times$ with U^*, and $\kappa(u) = x_{u \bmod e^*}$. So the function F of Section 4 becomes $F(v) = \mathrm{mult}_A(x_v) - \mathrm{mult}_A(x_{-v})$, and for $p \mid e^*$ the subgroup K_p becomes the group of units congruent to 1 modulo e^*/p; we write H^* for the product of these subgroups, which is direct (Lemma 3.5). Put

$$E(v) = \mathrm{mult}_A(y_v) - \mathrm{mult}_A(y_{-v}).$$

Both F and E are odd, and since A contains no pair, $|F(v)| = \mathrm{mult}_A(x_v) + \mathrm{mult}_A(x_{-v})$ and likewise for E; so $\|F\| = 2N_e$ and $\|E\| = 2N_{e^*}$, where N_f is the number of elements of A of order f. The elements of A of order e are the *tops*; we say that v is a top if $x_v \in A$, that is, if $F(v) \geq 1$. The elements of order other than e and e^* are *low*, and there are $6 - N_e - N_{e^*}$ of them.

Lemma 7.2. *Let $q \mid e^*$ be prime and $v \in U^*$. The elements x_w, $w \in vK_q$, lie in $\Phi_q(x_v)$, and so do the elements y_w, $w \in vK_q$, if $e^* \neq q$. If A contains $(q+1)/2$ distinct elements of one of these two kinds, then A has a move.*

Proof. For $w \in vK_q$ we have $w \equiv v$ modulo e^*/q. The representatives u, u' of v, w in the definition of x can be chosen odd and congruent modulo $2e^*/q$, so $x_w - x_v \in (M/e)(e/q)\mathbb{Z} = (M/q)\mathbb{Z}$; likewise $y_w - y_v = (M/e^*)(w - v)/2 \in (M/q)\mathbb{Z}$. The last assertion is Lemma 4.1, applied with $\ell = q$, which differs from the orders e and e^* of these elements. □

Lemma 7.3. *Let $E_1(w) = E(w) - F(w) + F(w/2)$. Then E_1 is odd and split along $(K_p)_{p|e^*}$ on every coset of H^* in U^*. It is the function $w \mapsto \mathrm{mult}_{A_1}(y_w) - \mathrm{mult}_{A_1}(y_{-w})$ of the multiset*

$$A_1 = A - \sum_{v \ top} \{x_v\} + \sum_{v \ top} \{y_{-v},\, y_{2v},\, M/2\},$$

the sums running over the tops with their multiplicities, and A_1 is the result of the chain of moves $(T(x_v), \{x_v\})$ over the tops.

Proof. Let χ be an odd primitive character modulo e^*. The elements a of A whose order is divisible by e^* have odd part e^*, by maximality, so they are the x_v and y_v in A. For y_v we have $z_a = v/2$, and for x_v we have $z_a \equiv v$ modulo e^* and the factor $1-\chi(2)$; and $\varphi(e) = \varphi(e^*)$. Since χ is odd, $\sum_v \mathrm{mult}_A(y_a)\bar\chi(v) = \frac{1}{2}\hat E(\chi)$ with $\hat f(\chi) = \sum_v f(v)\bar\chi(v)$, and likewise for F. So (1) reads $\chi(2)\hat E(\chi) + (1-\chi(2))\hat F(\chi) = 0$, that is, $\hat E(\chi) = (1-\bar\chi(2))\hat F(\chi)$. As $\widehat{F(\cdot/2)}(\chi) = \bar\chi(2)\hat F(\chi)$, we get $\hat E_1(\chi) = 0$. For even χ, $\hat E_1(\chi) = 0$ because E_1 is odd. So $\hat E_1$ vanishes at every primitive character, and the proof of Lemma 3.7, with Lemma 3.5(c) for e^*, shows that E_1 is split on every coset of H^*.

The move $(T(x_v), \{x_v\})$ removes x_v and adds $-y_v = y_{-v}$, $-y_{-2v} = y_{2v}$ and $-M/2 = M/2$. The function of A_1 at w receives $E(w)$ from A, $+1$ for each top $v = -w$ and -1 for each top $v = w$, which is $-F(w)$, and $+1$ for each top $v = w/2$ and -1 for each top $v = -w/2$, which is $F(w/2)$. □

7.3. **Elementary moves.**

Lemma 7.4. *Let v be a top and $x = x_v$. Each of the following is a move that lowers the profile, or leaves at most four elements after the removal of pairs.*

(a) *$y_v = x + M/2 \in A$: with $S = \{x, x+M/2\} \subseteq T(x)$, $A'' = A - x - (x+M/2) + \{2x,\, M/2\}$.*
(b) *$y_{-2v} = -2x \in A$: with $S = \{x, -2x\} \subseteq T(x)$, $A'' = A - x - (-2x) + \{y_{-v},\, M/2\}$.*
(c) *$2v$ is also a top: with $\sigma = T(x_{2v}) + T(x_v)$ and $S = \{x_{2v}, x_v\}$, $A'' = A - x_{2v} - x_v + \{y_{4v},\, y_{-v}\}$.*
(d) *$M/2 \in A$: with $S = \{x, M/2\} \subseteq T(x)$, $A'' = A - x - M/2 + \{y_{-v},\, y_{2v}\}$.*

Proof. These are instances of Definition 2.7. In (c), σ is algebraic by Lemma 2.6(b),(e), $\sigma - S = \{y_{2v}, y_{-4v}, M/2, y_v, y_{-2v}, M/2\}$, and $-(\sigma - S) = \{y_{-2v}, y_{4v}, M/2, y_{-v}, y_{2v}, M/2\}$; after removing the pairs $\{y_{2v}, y_{-2v}\}$ and $\{M/2, M/2\}$ the stated multiset remains. In each case one element of order e is removed and only elements of order e^* or 2 are added, while the elements of order larger than e stay. So if six elements remain they form a sextuple of smaller profile. □

From now on we assume that A has none of the moves of Lemmas 7.2 and 7.4.

Lemma 7.5. *For every $w \in U^*$ the numbers $E(w)$, $-F(w)$ and $F(w/2)$ do not have opposite signs, so*

$$|E_1(w)| = |E(w)| + |F(w)| + |F(w/2)|.$$

Consequently:

(i) *$\|E_1\| = 2N_{e^*} + 4N_e \le 24$;*
(ii) *at every top v, $E_1(v) \le -1$ and $E_1(2v) \ge 1$, so $E_1 \ne 0$;*
(iii) *the elements of order e^* of A_1 contain no pair, and $\mathrm{mult}_{A_1}(y_w)$ is the positive part of $E_1(w)$;*
(iv) *the negative part of E_1 is the counting function of the multiset $\{w :\ y_{-w} \in A\} + \{tops\} + \{-2v :\ v \ top\}$.*

Proof. If $E(w) \ge 1$ and $F(w) \ge 1$, then $y_w = x_w + M/2$ and x_w lie in A, which is Lemma 7.4(a); the same with $-w$ if both are negative. If $F(w) \le -1$ and $F(w/2) \le -1$, then $-w$ and $-w/2$

are tops, which is (c); the same with w if $F(w) \geq 1$ and $F(w/2) \geq 1$. If $E(w) \geq 1$ and $F(w/2) \leq -1$, then y_w and $x_{-w/2}$ lie in A, and $y_w = -2x_{-w/2}$, which is (b); the same with $-w$ in the other case. So the three numbers never have opposite signs, and the displayed identity follows. Summing it gives (i), with $N_e + N_{e^*} \leq 6$. At a top v, $-F(v) \leq -1$ and $F(2v/2) \geq 1$, which gives (ii), and $N_e \geq 1$ since e is an order of A. The multiplicity of y_w in A_1 is the sum of the positive parts of the three numbers, and that of y_{-w} the sum of their negative parts; so one of them vanishes, which is (iii), and (iv) is the same computation for the negative parts. □

7.4. **The compound move.** A *non-cancelling line sum* is a function $f = \sum_i s_i \mathbf{1}_{L_i}$, for a finite family of lines L_i (repetitions allowed) and signs $s_i \in \{\pm 1\}$, such that $|f| = \sum_i \mathbf{1}_{L_i}$.

Lemma 7.6. *If E_1 is a non-cancelling line sum of lines along the subgroups K_p, $p \mid e^*$, and e^* is not prime, then A has a chain of moves that is direct or lowering.*

Proof. Pass from A to A_1 (Lemma 7.3). By Lemma 7.5(iii), the elements of order e^* of A_1 are the y_w for w in the lines L_i with $s_i = 1$, each counted once per line. Let $L = wK_p$ be such a line and $y = y_w$. By Lemma 7.2 the $y_{w'}$, $w' \in L$, are $|K_p|$ distinct elements of order e^* of the fibre $\Phi_p(y)$. If $p^2 \mid e^*$, they are the whole fibre; if $p \parallel e^*$, the fibre consists of them and one element y_0 of order e^*/p. Since e^* is not prime, $py \neq 0$, and the move $(\sigma_{p,y}, S)$ with $S = \{y_{w'} : w' \in L\}$ removes these elements and adds py, and $-y_0$ if $p \parallel e^*$: at most two elements of order dividing e^*/p. We do this for each positive line in turn.

Each line has $|K_p| \geq 4$ points, and the positive lines carry $\|E_1\|/2 = N_{e^*} + 2N_e$ points in all, so there are at most $(N_{e^*} + 2N_e)/4$ of them. The multiset A_1 contains N_e copies of $M/2$, which is not in A by Lemma 7.4(d); after removing pairs, $\epsilon \in \{0, 1\}$ copies remain, with $\epsilon \equiv N_e$ modulo 2. The final multiset therefore has at most

$$(6 - N_e - N_{e^*}) + \epsilon + 2 \cdot \frac{N_{e^*} + 2N_e}{4} = 6 + \epsilon - \frac{N_{e^*}}{2} \leq 7$$

elements. It is balanced (Lemma 2.8), so its size is even and at most 6. It contains the low elements of A, at most one copy of $M/2$ and elements of order smaller than e^*; so it has no element of order e, the same elements of order larger than e as A, and the chain is direct or lowering. □

7.5. **Split functions of mass at most eight.**

Lemma 7.7. *Let $K_1, \dots, K_r$ be subgroups of a finite abelian group whose product H is direct, with pairwise distinct orders $|K_i| \geq 4$, such that the orders 4 and 5 do not both occur. Let C be a coset of H and $f \colon C \to \mathbb{Z}$ split along $(K_1, \dots, K_r)$, not identically zero, with $\sum_C |f| \leq 8$. Then f is one of*

(i) *$\sigma \mathbf{1}_L$, with $\sigma = \pm 1$ and L a line along some K_i;*
(ii) *$2\sigma \mathbf{1}_L$, with L a line along the K_i of order 4;*
(iii) *$\sigma \mathbf{1}_L + \sigma' \mathbf{1}_{L'}$, with $L \neq L'$ lines along the K_i of order 4;*
(iv) *a cross $\sigma(\mathbf{1}_L - \mathbf{1}_{L'})$, with L a line along the K_i of order 6, L' a line along the K_j of order 4, and $|L \cap L'| = 1$.*

In particular $\sum_C |f| \geq \min_i |K_i|$.

Proof. By induction on r; for $r = 0$ there is nothing to prove. For $r = 1$, f is constant on the line C by (S4), with a value $c \neq 0$ and $|c|\,|K_1| \leq 8$; so $|c| = 1$, case (i), or $|c| = 2$ and $|K_1| = 4$, case (ii). Let $r \geq 2$, and number the subgroups so that $K = K_1$ has the largest order; then $|K| \geq 5$ and $|K| > m' = \min_{i \geq 2} |K_i| \geq 4$. We use the slices S_c, $c \in K$, of Section 3.

Case A: f is K-invariant. Then f is constant on the lines along K, a nonzero value c gives $|c|\,|K| \le 8$, so $|c| = 1$, and a second line on which $f \neq 0$ would add $|K| \ge 5$ to the mass. This is (i).

Case B1: $\mu(c_0) = 0$ for some c_0. By (S2), f is split along $(K_2, \dots, K_r)$ on every slice. By induction a nonzero slice carries mass at least $m' \ge 4$, so at most two slices are nonzero. If one is, f is one of (i)–(iv) inside it. If two are, each carries mass 4 and is of the form $\sigma \mathbf{1}_L$ with $|L| = 4$; this is (iii).

Case B2: f is not K-invariant and $\mu \ge 1$ on every slice. As $\sum_c \mu(c) \le 8 < 2|K|$, some c_s has $\mu(c_s) = 1$. For $c \in K$ the function $g_c(z) = f(z) - f(zc^{-1}c_s)$ on S_c is split along $(K_2, \dots, K_r)$ by (S1), with mass at most $\mu(c) + 1 \le 8$. If every g_c vanished, f would be K-invariant. Let D be the set of c with $g_c \neq 0$, and $d = |D| \ge 1$. For $c \in D$ the induction gives $\mu(c) + 1 \ge m'$. Hence

$$8 \ge \sum_c \mu(c) \ge d(m' - 1) + (|K| - d), \qquad \text{so} \qquad |K| + d(m' - 2) \le 8.$$

With $|K| \ge m' + 1$ and $d \ge 1$ this forces $m' = 4$, $d = 1$ and $|K| \le 6$; the value $|K| = 5$ is excluded, as the orders 4 and 5 do not both occur. So $|K| = 6$, all the inequalities are equalities, $\mu(c_1) = 3$ for the element c_1 of D, and $\mu = 1$ on the other five slices. Off S_{c_1}, f is the translate of its restriction to S_{c_s}, so it equals σ at one point of each slice, and these points lie on one line L along K. On S_{c_1}, g_{c_1} has mass 4, so by induction it is $\tau \mathbf{1}_{L'}$ for a line L' along the subgroup of order 4; then $f = \sigma\delta_{z_1} + \tau \mathbf{1}_{L'}$ on S_{c_1}, where z_1 is the point of L in S_{c_1}, and the mass 3 forces $z_1 \in L'$ and $\tau = -\sigma$. This is (iv). The last assertion holds in each case. □

For e^* prime to 6 the subgroups K_p, $p \mid e^*$, satisfy the hypotheses of Lemma 7.7: by Lemma 3.5(a), $|K_p|$ is $p - 1$ or p, these pairs of integers are disjoint for different primes $p \ge 5$, and the order 4 occurs only for $5 \parallel e^*$, the order 5 only for $25 \mid e^*$. A cross needs $5 \parallel e^*$ and $7 \parallel e^*$. The functions (i)–(iii) are non-cancelling line sums, and a cross is not.

7.6. **The cases.** We now show that E_1 is a non-cancelling line sum, except when e^* is 35 or 55, which we treat afterwards.

Lemma 7.8. *e^* is not prime.*

Proof. If $e^* = p$, then $K_p = U^* = H^*$, and E_1 is constant by (S4) and odd, so $E_1 = 0$, against Lemma 7.5(ii). □

Proposition 7.9. *If e^* is not squarefree, then E_1 is a non-cancelling line sum.*

Proof. Let $R = \prod_{p^2 \mid e^*} p^{v_p(e^*)-1}$; then $R \ge 5$, since e^* is prime to 6. Every K_p consists of units congruent to 1 modulo R, as $R \mid e^*/p$, and the group of such units has order $\prod_{p \parallel e^*}(p-1) \prod_{p^2 \mid e^*} p = |H^*|$; so H^* is the group of units congruent to 1 modulo R. Hence $-1, 2, -2, 4 \notin H^*$, since R divides none of $2, 1, 3, 3$, and $-4 \in H^*$ only if $R = 5$.

Let v be a top. The cosets $X_1 = vH^*$, $X_2 = -vH^*$, $X_3 = 2vH^*$, $X_4 = -2vH^*$ are distinct, and $E_1 \neq 0$ on each of them by Lemma 7.5(ii), so each carries mass at least 4 (Lemma 7.7). By oddness $\mu(X_1) = \mu(X_2) = a$ and $\mu(X_3) = \mu(X_4) = b$, and $2a + 2b \le 24$; so $a, b \le 8$, and every other coset, which has the same mass as its negative, carries at most 4. So on every coset E_1 is zero or one of the functions of Lemma 7.7. If no coset carries a cross, E_1 is a non-cancelling line sum.

Suppose a coset carries a cross. Then $5 \parallel e^*$ and $7 \parallel e^*$, so the primes of R are at least 11, $R \neq 5$ and $-4 \notin H^*$. The coset has mass 8, so it is one of the X_i, and $\{a, b\} = \{4, 8\}$. Then $\|E_1\| = 24 = 2N_{e^*} + 4N_e$, so $N_e = 6$, $N_{e^*} = 0$, and E_1 vanishes off the X_i. By Lemma 7.5(iv) the negative part of E_1 counts the multiset $T + (-2T)$, T the six tops. A top $t \in X_3 \cup X_4$ would have $E_1(2t) \neq 0$ with $2t \in \pm 4vH^*$, and this coset is none of the X_i, as $\pm 4, \pm 2 \notin H^*$,

$\mp 4 \notin H^*$. So $T \subseteq X_1 \cup X_2$ and $-2T \subseteq X_3 \cup X_4$, and the negative mass of E_1 on $X_1 \cup X_2$ is $|T| = 6$. By oddness it is the total mass a of E_1 on X_1, which is 4 or 8; a contradiction. □

Proposition 7.10. *If e^* is squarefree with at least three prime factors, then E_1 is a non-cancelling line sum.*

Proof. Here $H^* = U^*$. Let p_1 be the largest prime factor, so $p_1 \geq 11$, and $K = K_{p_1}$, of order $p_1 - 1 \geq 10$. In the coordinates $U^* = (\mathbb{Z}/p_1)^\times \times (\mathbb{Z}/(e^*/p_1))^\times$ the group K is the first factor, the product H' of the other K_p is the second, and the slices are $S_c = \{c\} \times (\mathbb{Z}/(e^*/p_1))^\times$. Write $-1 = \varepsilon\varepsilon'$ with $\varepsilon = (-1, 1) \in K$ and $\varepsilon' = (1, -1) \in H'$. Then $\mu(-c) = \mu(c)$, and for $z \in S_c$, $E_1(z\varepsilon) = E_1(-z\varepsilon') = -E_1(z\varepsilon')$; so by (S1) the function

$$(*) \qquad g(z) = E_1(z) + E_1(z\varepsilon') = E_1(z) - E_1(z\varepsilon)$$

is split along $(K_p)_{p \neq p_1}$ on S_c, with mass at most $2\mu(c)$, and $g(z\varepsilon') = g(z)$. The element ε' lies in no K_p with $p \neq p_1$, since it is congruent to -1 modulo a third prime factor. A top in S_c makes the slices $S_c, S_{-c}, S_{2c}, S_{-2c}$ nonzero (Lemma 7.5(ii)), and they are distinct, as $p_1 \geq 11$.

Case Z: some slice vanishes. By (S2), E_1 is split along $(K_p)_{p \neq p_1}$ on every slice, so nonzero slices carry mass at least 4 (Lemma 7.7). For a top in S_c, the four slices above carry $2\mu(c)+2\mu(2c) \leq 24$, so $\mu(c), \mu(2c) \leq 8$ and every other slice carries at most 4. So Lemma 7.7 applies on every slice, and without a cross E_1 is a non-cancelling line sum. A cross has mass 8, so for each top it lies on one of the four slices of that top; then $\|E_1\| = 24$, $N_e = 6$, $N_{e^*} = 0$, and E_1 vanishes off these four slices, with $\{\mu(c), \mu(2c)\} = \{4, 8\}$. A top t in $S_{\pm 2c}$ would have $2t$ in $S_{\pm 4c}$, which is none of the four slices, as $4 \not\equiv \pm 1, \pm 2$ modulo p_1; but $E_1(2t) \neq 0$. So the six tops lie in $S_c \cup S_{-c}$, where the negative mass of E_1 is then 6 by Lemma 7.5(iv), while by oddness it is $\mu(c) \in \{4, 8\}$.

Case NZ1: some $\mu(c) = 1$. In $(*)$, g has mass at most $2 < 4$, so $g = 0$; then $E_1(z_0\varepsilon') = -E_1(z_0) \neq 0$ at the support point z_0 of S_c, and $z_0\varepsilon' \neq z_0$ is a second point of S_c where $E_1 \neq 0$. A contradiction.

Case NZ2: all $\mu(c) \geq 2$. Then $2(p_1 - 1) \leq 24$, so p_1 is 11 or 13. On a slice of mass 2, g has mass at most 4; if it is not zero, it is $\pm\mathbf{1}_L$ for a line L along K_5 (Lemma 7.7), and $g(z\varepsilon') = g(z)$ gives $L\varepsilon' = L$, so $\varepsilon' \in K_5$, which we excluded. So on every slice of mass 2, $E_1 = \sigma(\delta_{z_0} - \delta_{z_0\varepsilon'})$. Two slices $S_c, S_{c'}$ of mass 2 are translates of each other: the function $h(z) = E_1(z) - E_1(zc^{-1}c')$ on S_c is split along $(K_p)_{p \neq p_1}$ by (S1), has mass at most 4 and satisfies $h(z\varepsilon') = -h(z)$, while a nonzero split function of mass at most 4 is $\pm\mathbf{1}_L$, which is not of this kind.

If $p_1 = 13$, the twelve slices all have mass 2, so E_1 is K-invariant, and $E_1 = \sigma(\mathbf{1}_L - \mathbf{1}_{L\varepsilon'})$ for a line L along K, with $L \neq L\varepsilon'$: a non-cancelling line sum.

If $p_1 = 11$, then $e^* = 385$ and $H' = K_5 \times K_7$, of orders 4 and 6. The ten slices carry mass at most 24, at least 2 each, and masses at least 3 on the others; so at least six slices have mass 2, and the others come in pairs $\pm c$ with $\mu(c) \leq 4$. For such a c, the difference of E_1 on S_c with the translate of a slice of mass 2 is split, nonzero and of mass at most $\mu(c) + 2 \leq 6$, so by Lemma 3.8 it is $\pm\mathbf{1}_{L'}$ for a line L' along K_5 or K_7. The line L' contains at most one of the two support points $z_0, z_0\varepsilon'$ of the translate, as ε' lies in neither K_5 nor K_7; so $\mu(c) \geq |L'| \geq 4$. Hence $\mu(c) = 4$, L' is a line along K_5 through one of the two points, exactly two slices $S_{\pm c_1}$ have mass 4, the total is 24, and $N_e = 6$. In coordinates $(a, b) \in (\mathbb{Z}/11)^\times \times (\mathbb{Z}/35)^\times$ the support of E_1 lies in $\{b = \pm b_0\} \cup \{a = \pm c_1,\ b \in \pm b_0 K_5\}$, where K_5 is the group of units congruent to 1 modulo 7. For $t = (a, b)$ in this set, $2t$ lies outside it: $2 \not\equiv \pm 1$ modulo 35 and modulo 11, and $\pm 2, \pm\frac{1}{2} \notin K_5$. By Lemma 7.5(ii) there is then no top, against $N_e = 6$. □

Proposition 7.11. *Let $e^* = pq$ with primes $5 \leq p < q$. Write $E_1(y, x)$ for $(y, x) \in (\mathbb{Z}/p)^\times \times (\mathbb{Z}/q)^\times = U^*$. Then $E_1(y, x) = \alpha(x) + \beta(y)$ with odd functions α, β, either both with values in*

$\mathbb{Z}$ or both with values in $\frac{1}{2}+\mathbb{Z}$, and

$$\|E_1\| = \sum_{y,x} \max(|\alpha(x)|, |\beta(y)|).$$

If $e^* \notin \{35, 55\}$, or if α or β is zero, then E_1 is a non-cancelling line sum.

Proof. Here $H^* = U^*$, $K_p = (\mathbb{Z}/p)^\times \times \{1\}$ and $K_q = \{1\} \times (\mathbb{Z}/q)^\times$, so a split function is a sum of a function of x and a function of y. Fixing x_1, put $\alpha_1(x) = E_1(y,x) - E_1(y,x_1)$, which does not depend on y, and $\beta_1(y) = E_1(y,x_1)$; both are integral, and $E_1 = \alpha_1(x) + \beta_1(y)$. Oddness gives $\alpha_1(x) + \alpha_1(-x) = -\beta_1(y) - \beta_1(-y)$, a constant $k \in \mathbb{Z}$; so $\alpha = \alpha_1 - k/2$ and $\beta = \beta_1 + k/2$ are odd, with values in $\mathbb{Z}$ if k is even and in $\frac{1}{2}+\mathbb{Z}$ if k is odd. Pairing (y,x) with $(-y,x)$, $|\alpha(x)+\beta(y)| + |\alpha(x)-\beta(y)| = 2\max(|\alpha(x)|, |\beta(y)|)$, which gives the formula for the mass.

Half-integral values. Then $\|E_1\| \geq \varphi(e^*)/2$, so $\varphi(e^*) \leq 48$ and $e^* \in \{35, 55, 65\}$. Let $e^* = 65$. Equality holds, so $|\alpha| = |\beta| = \frac{1}{2}$ everywhere, $N_e = 6$, $N_{e^*} = 0$, and $E_1 = -1$ exactly on $N = N_y \times N_x$, where $N_y \subseteq (\mathbb{Z}/5)^\times$ and $N_x \subseteq (\mathbb{Z}/13)^\times$ are the sets where β and α are negative, of sizes 2 and 6. By Lemma 7.5(ii),(iv), N is the disjoint union of the set T of the six tops and of $-2T$, and $2T \subseteq -N$. So $T \subseteq (N_y \cap 2N_y) \times (N_x \cap (-\frac{1}{2})N_x)$, using $-\frac{1}{2} \equiv 2$ modulo 5. The set N_y contains one element of each pair $\pm y$, and in $(\mathbb{Z}/5)^\times = \langle 2 \rangle$ each of the four such sets meets its double in one point. So $|T| = 6$ forces $N_x = -2N_x$; but -2 generates $(\mathbb{Z}/13)^\times$, so N_x would be empty or everything.

Integral values, $\alpha \neq 0 \neq \beta$. As α and β are odd, each is nonzero at two points at least, and $\max(|\alpha|, |\beta|) \geq 1$ on at least $2(p-1) + 2(q-1) - 4$ points. So $p + q \leq 16$, and $e^* \in \{35, 55\}$.

Integral values, $\alpha = 0$ or $\beta = 0$. Then E_1 is invariant under K_p or under K_q, and it is a non-cancelling line sum of lines along that subgroup, each line repeated $|E_1|$ times. □

7.7. The tops 35 and 55.

Lemma 7.12 (Carry)**.** *Let $70 \mid M$, let B be a multiset in the subgroup of order 70 of $\mathbb{Z}/M$, and let $A = B + \{y_1, y_2\}$ be balanced, with $y_1, y_2 \neq 0$ and $s = y_1 + y_2$ of order 35. If 35 does not divide $f = \mathrm{ord}(y_1)$, then $f \mid 70$.*

Proof. For a unit t, $\langle ty_1 \rangle + \langle ty_2 \rangle = \langle ts \rangle + \delta(t)$ with $\delta(t) = 1$ if $\langle ty_1 \rangle > \langle ts \rangle$ and $\delta(t) = 0$ otherwise. This sum equals $|A|/2 - \sum_{b \in B} \langle tb \rangle$, which depends only on t modulo 70, and so does $\langle ts \rangle$; hence so does $\delta(t)$. Suppose $f \nmid 70$, and let $g_0 = \gcd(f, 70)$ and $g_1 = \gcd(f, 35)$, a divisor of 35 other than 35. Write $y_1 = (M/f)z_1$. As t runs over the units congruent to t' modulo 70, $\langle ty_1 \rangle$ runs over the set P of the numbers z/f with z a unit modulo f and $z \equiv t'z_1$ modulo g_0, by the Chinese remainder theorem; P depends only on t' modulo g_0. By the above, P lies on one side of $\langle t's \rangle$, and this holds for every t' in the class modulo g_0. Writing $t's = (M/35)w$, the numbers $w/35$ run over the units w modulo 35 in a fixed class modulo g_1, which cut $[0,1)$ into intervals of length at most $14/35$ (if $g_1 = 7$; at most $10/35$ if $g_1 = 5$, and $3/35$ if $g_1 = 1$). So P lies in an interval of length less than $1/2$. But P contains two points at distance at least $1/2$:

- if f has a prime factor $l \notin \{2, 5, 7\}$, then $l \geq 11$; with $l^a \parallel f$ and $z'/f \in P$, P contains the points $z'/f + i/l^a$ for all i outside one class modulo l, and these do not lie in an interval of length less than $1 - 2/l \geq 9/11$;
- otherwise 4, 25 or 49 divides f, f and g_0 have the same prime factors, so every $z \equiv t'z_1$ modulo g_0 is a unit modulo f, and P consists of $h = f/g_0 \geq 2$ points spaced by $1/h$, spanning $(h-1)/h \geq 1/2$.

This is a contradiction. □

Proposition 7.13. *Suppose $e^* \in \{35, 55\}$ and E_1 is not a non-cancelling line sum. Then $M = 2e^*$.*

Proof. Let $n = N_e + N_{e^*}$, so $\|E_1\| = 4N_e + 2N_{e^*} \le 4n$. By Proposition 7.11, α and β are both nonzero or both half-integral.

Let $e^* = 55$. A half-integral E_1 has mass at least $\varphi(55)/2 = 20$, and an integral one with $\alpha, \beta \neq 0$ at least $2 \cdot 4 + 2 \cdot 10 - 4 = 24$; so $n \ge 5$, and A has at most one low element, which is minus the sum of the others. All elements of A then lie in the subgroup of order 110, and as A generates $\mathbb{Z}/M$, $M = 110$.

Let $e^* = 35$, with $y \in (\mathbb{Z}/5)^\times$ and $x \in (\mathbb{Z}/7)^\times$. If $n \ge 5$, the same argument gives $M = 70$. A half-integral E_1 has mass at least 12 and an integral one with $\alpha, \beta \neq 0$ at least 16; so $n \ge 3$.

$n = 3$. Then $\|E_1\| = 12$, $(N_e, N_{e^*}) = (3, 0)$, $|\alpha| = |\beta| = \frac{1}{2}$ everywhere, and $E_1 = -1$ exactly on $N = N_y \times N_x$, with $|N_y| = 2$ and $|N_x| = 3$. As in the proof of Proposition 7.11, N is the disjoint union of T and $-2T$, $|T| = 3$, and $T \subseteq (N_y \cap 2N_y) \times (N_x \cap 3N_x)$, using $-\frac{1}{2} \equiv 3$ modulo 7. Since $|N_y \cap 2N_y| = 1$, this forces $N_x = 3N_x$, impossible as 3 generates $(\mathbb{Z}/7)^\times$.

$n = 4$, *α, β integral and nonzero.* The mass is 16, so $(N_e, N_{e^*}) = (4, 0)$, and equality in the count above gives $\alpha = s(\delta_{x_0} - \delta_{-x_0})$, $\beta = s'(\delta_{y_0} - \delta_{-y_0})$ with signs s, s'. Let x_- and y_- be the points where α and β equal -1, and x_+, y_+ those where they equal 1. A top $t = (y, x)$ has $E_1(t) \le -1$ and $E_1(2t) \ge 1$. Since $2 \not\equiv \pm 1$ modulo 5 and modulo 7, at most one of $x, 2x$ lies in $\{\pm x_0\}$, and at most one of $y, 2y$ in $\{\pm y_0\}$. If $\alpha(x) = -1$, then $\alpha(2x) = 0$, so $\beta(2y) = 1$, $\beta(y) = 0$ and $t = (y_+/2, x_-)$. If $\alpha(x) = 0$, then $\beta(y) = -1$, $\beta(2y) = 0$, $\alpha(2x) = 1$ and $t = (y_-, x_+/2)$. The value $\alpha(x) = 1$ is impossible. So there are at most two tops, each with $|E_1(t)| = 1$ and hence $F(t) = 1$ (Lemma 7.5), against $N_e = 4$.

$n = 4$, *half-integral.* A value $|\alpha| \ge \frac{3}{2}$ or $|\beta| \ge \frac{3}{2}$ adds at least 8 to the mass, which would exceed 16; so $|\alpha| = |\beta| = \frac{1}{2}$, the mass is 12, $(N_e, N_{e^*}) = (2, 2)$, and A has two low elements y_1, y_2. The negative set $N = N_y \times N_x$ is the disjoint union of $-W$, T and $-2T$, where W is the set of the w with $y_w \in A$ (Lemma 7.5(iv)). As in the case $n = 3$, $T \subseteq (N_y \cap 2N_y) \times (N_x \cap 3N_x)$ with $|N_y \cap 2N_y| = 1$, so $N_y = \{c, -2c\}$ for some c, and $|N_x \cap 3N_x| \ge 2$. In $(\mathbb{Z}/7)^\times = \langle g \rangle$, $g = 3$, an odd set of three elements meets its product with g in two points exactly when it is $\{g^k, g^{k+1}, g^{k+2}\}$, and otherwise in none. Hence

$$T = \{(c, g^{k+1}), (c, g^{k+2})\}, \quad -2T = \{(-2c, g^k), (-2c, g^{k+1})\}, \quad -W = \{(c, g^k), (-2c, g^{k+2})\},$$

using $-2 \equiv g^5$ modulo 7. The four elements x_t, $t \in T$, and y_w, $w \in W$, lie in the subgroup of order 70. In $\mathbb{Z}/70 = \mathbb{Z}/2 \times \mathbb{Z}/5 \times \mathbb{Z}/7$, x_t has coordinates $(1, t)$ and y_w has $(0, w)$, so their sum has 2-coordinate 0, 5-coordinate $c + c - c + 2c = 3c \neq 0$ and 7-coordinate $g^{k+1} + g^{k+2} - g^k - g^{k+2} = 2g^k \neq 0$. So $s = y_1 + y_2$, which is minus this sum, has order 35. The elements y_i are low, so $35 \nmid \operatorname{ord}(y_i)$: an order divisible by 35 has odd part 35 and is 35 or 70. By Lemma 7.12, applied with B the four elements above, $\operatorname{ord}(y_i) \mid 70$. So A lies in the subgroup of order 70, and $M = 70$. □

Theorem 7.14. *Let M be even and prime to 3, and let A be a sextuple at level M that contains no pair, is not split and generates $\mathbb{Z}/M$. Suppose that A has a maximal odd part e^* whose largest order is $2e^*$. Then A has a chain of moves that succeeds, unless $M \in \{70, 110\}$.*

Proof. If A has a move of Lemma 7.4 or Lemma 7.2, we are done. Otherwise $e^* > 1$ is not prime (Lemma 7.8), and E_1 is a non-cancelling line sum by Propositions 7.9 to 7.11, except in the case of Proposition 7.13, where $M \in \{70, 110\}$. In the first case Lemma 7.6 gives a direct or lowering chain. □

8. Top orders divisible by four

We keep the setting of Section 7: M is even and prime to 3, A is a sextuple at level M that contains no pair, is not split and generates $\mathbb{Z}/M$, e^* is a maximal odd part of A and $2^J e^*$ is

its largest order. Here $J \geq 2$, so $4 \mid M$. Replacing A by $-A$ does not change the problem (Lemma 2.6(c) with $t = -1$).

8.1. **The case $J = 2$: two relations.** For $v \in U^* = (\mathbb{Z}/e^*)^\times$ let

$$x_v^+,\ x_v^- = \tfrac{M}{4e^*}\,u \quad (u \equiv 1,\ \text{resp. } u \equiv 3 \bmod 4,\ u \equiv v \bmod e^*),$$
$$y_v = \tfrac{M}{2e^*}\,u \quad (u \text{ odd},\ u \equiv v \bmod e^*),$$

and $z_v = (M/e^*)v$, of orders $4e^*$, $2e^*$ and e^*. Then

$$-x_v^+ = x_{-v}^-, \quad x_v^- = x_v^+ + \tfrac{M}{2}, \quad \pm 2x_v^\pm = y_{\pm v}, \quad y_v + \tfrac{M}{2} = z_{v/2}, \quad z_v + \tfrac{M}{2} = y_{2v}, \quad -2y_v = z_{-v},$$

where $\pm 2x_v^\pm = y_{\pm v}$ means $2x_v^+ = 2x_v^- = y_v$ and $-2x_v^\pm = y_{-v}$. By the maximality of e^* and of J, these are the elements of odd part e^*; the other elements are *low*. Let N_x, N_y, N_z be the numbers of elements of A of orders $4e^*$, $2e^*$, e^*, and put

$$P(v) = \operatorname{mult}_A(x_v^+) - \operatorname{mult}_A(x_{-v}^-),$$
$$Y(v) = \operatorname{mult}_A(y_v) - \operatorname{mult}_A(y_{-v}), \qquad Z(v) = \operatorname{mult}_A(z_v) - \operatorname{mult}_A(z_{-v}).$$

Y and Z are odd, and as A contains no pair, $\sum_v |P(v)| = N_x$, $\|Y\| = 2N_y$ and $\|Z\| = 2N_z$. With $\hat f(\chi) = \sum_v f(v)\bar\chi(v)$ we have $\widehat{f(2\,\cdot)} = \chi(2)\hat f$.

Lemma 8.1. (R1) $\hat Z(\chi) + (1-\chi(2))\big(\hat Y(\chi) + \hat P(\chi)\big) = 0$ *for every odd primitive character χ modulo e^*;*
(R2) $\hat P(\psi) = 0$ *for every even primitive character ψ modulo e^*.*

Consequently $D(v) = P(v) + P(-v)$ and $G = Z + Y - Y(2\,\cdot) + P - P(2\,\cdot)$ are split along $(K_p)_{p|e^*}$ on every coset of H^*, and $e^* > 1$.

Proof. (R1) In (1) with $c = e^*$, the elements whose order is divisible by e^* are the $x_v^\pm$, y_v, z_v. The element z_v contributes $\bar\chi(v)/\varphi(e^*)$, the element y_v contributes $\bar\chi(v)(1-\chi(2))/\varphi(e^*)$, and $x_v^\pm$ contributes $\bar\chi(v)(1-\chi(2))/(2\varphi(e^*))$. Since χ is odd, $\sum_v \operatorname{mult}_A(z_v)\bar\chi(v) = \hat Z(\chi)/2$, likewise for Y, and $\sum_v(\operatorname{mult}_A(x_v^+) + \operatorname{mult}_A(x_v^-))\bar\chi(v) = \hat P(\chi)$.

(R2) A primitive character of conductor $4e^*$ is $\chi_4\psi$ with ψ primitive modulo e^*, and it is odd exactly when ψ is even. Only the elements of order $4e^*$ enter (1), and $\overline{\chi_4\psi}(u) = \pm\bar\psi(v)$ for $x_v^\pm$; so $\sum_v(\operatorname{mult}_A(x_v^+) - \operatorname{mult}_A(x_v^-))\bar\psi(v) = 0$, which is $\hat P(\psi) = 0$ as ψ is even.

For a primitive χ: if χ is odd, $\hat D(\chi) = 0$ as D is even, and $\hat G(\chi) = \hat Z + (1-\chi(2))(\hat Y + \hat P) = 0$ by (R1); if χ is even, $\hat D(\chi) = 2\hat P(\chi) = 0$, and $\hat Z = \hat Y = 0$, so $\hat G(\chi) = (1-\chi(2))\hat P(\chi) = 0$ by (R2). The proof of Lemma 3.7 gives the splitting. If $e^* = 1$, all odd parts are 1, $M = 4$, and (R2) at the conductor 4 reads $\operatorname{mult}_A(M/4) = \operatorname{mult}_A(3M/4)$, so $N_x = 0$ as A contains no pair; but $J = 2$. □

So $e^* \geq 5$ is prime to 6, and the subgroups K_p satisfy the hypotheses of Lemma 7.7 and Lemma 3.8.

Lemma 8.2. *Each of the following is a move that leaves the elements of order larger than $4e^*$ in place and either lowers the number of elements of order $4e^*$, or keeps it and lowers the number of elements of order $2e^*$; so it lowers the profile, unless at most four elements remain.*

(a) *$x_v^+, x_v^- \in A$: with $x = x_v^+$ and $S = \{x, x + M/2\} \subseteq T(x)$, $A'' = A - x_v^+ - x_v^- + \{y_v, M/2\}$. And $y_{2v}, z_v \in A$: with $S = \{z_v, z_v + M/2\} \subseteq T(z_v)$, $A'' = A - y_{2v} - z_v + \{z_{2v}, M/2\}$.*
(b) *$x = x_v^\pm$ and $y_{-v} = -2x$ in A: with $S = \{x, -2x\}$, $A'' = A - x - y_{-v} + \{-x + M/2,\ M/2\}$. And $y_v, z_{-v} \in A$: with $S = \{y_v, -2y_v\} \subseteq T(y_v)$, $A'' = A - y_v - z_{-v} + \{z_{-v/2}, M/2\}$.*
(c) *$y_v, y_{2v} \in A$: with $\sigma = T(y_{2v}) + T(y_v)$ and $S = \{y_{2v}, y_v\}$, $A'' = A - y_v - y_{2v} + \{z_{2v}, z_{-v/2}\}$.*

(d) *For a prime $q \mid e^*$ and a line vK_q: if A contains $(q+1)/2$ distinct elements of one of the four kinds x_w^+, x_w^-, y_w, or z_w with $e^* \neq q$, for $w \in vK_q$, then A has a move.*

Proof. (a)–(c) are instances of Definition 2.7; in (c), $-(\sigma - S) = \{z_{-v}, z_{2v}, M/2, z_{-v/2}, z_v, M/2\}$, from which the pairs $\{z_v, z_{-v}\}$ and $\{M/2, M/2\}$ are removed. In (b) the added element $-x + M/2$ has order $4e^*$, and the element y_{-v} of order $2e^*$ is removed. For (d), as in Lemma 7.2, the elements of one kind with $w \in vK_q$ differ by elements of $(M/q)\mathbb{Z}$, so they lie in one fibre of Φ_q, and Lemma 4.1 applies, since q differs from their order. □

From now on A has none of these moves.

Lemma 8.3. (i) *$P(v)$ and $P(-v)$ never have opposite signs. So $|D(v)| = |P(v)| + |P(-v)|$, $\|D\| = 2N_x \leq 12$, $D \neq 0$, and where $P(v) \neq 0$ it has the sign of $D(v)$.*
(ii) *$Z(w)$, $Y(w)$ and $-Y(2w)$ never have opposite signs. So $|Z + Y - Y(2\cdot)| = |Z| + |Y| + |Y(2\cdot)|$.*

Proof. (i) $P(v) \geq 1$ and $P(-v) \leq -1$ mean $x_v^+, x_v^- \in A$, and $P(v) \leq -1$ and $P(-v) \geq 1$ mean $x_{-v}^-, x_{-v}^+ \in A$; both are Lemma 8.2(a). And $N_x \geq 1$ as $J = 2$. (ii) Opposite signs of $Z(w)$ and $Y(w)$ give $z_{\mp w}$ and $y_{\pm w}$ in A, which is (b); of $Z(w)$ and $-Y(2w)$ give $z_{\pm w}$ and $y_{\pm 2w}$, which is (a); of $Y(w)$ and $-Y(2w)$ give $y_{\pm w}$ and $y_{\pm 2w}$, which is (c). □

Lemma 8.4. *Up to replacing A by $-A$, D has one of three shapes:*
(I) *$D = \mathbf{1}_L + \mathbf{1}_{-L}$ with $L \neq -L$ a line along K_q, $|K_q| \leq 6$;*
(II) *$e^* = p$ is prime and D is a positive constant;*
(III) *$e^* = 35$ and $N_x = 6$.*

Proof. Nonzero split functions of mass at most 8 have mass at least 4 (Lemma 7.7). Let C be a coset of H^* on which $D \neq 0$.

Case (a): $-1 \notin H^*$. The cosets C and $-C$ are distinct with the same mass, so D has mass at most 6 on C, and by Lemma 3.8 $D = \sigma \mathbf{1}_L$ on C, with L a line along some K_q with $|K_q| \leq 6$, and $D = \sigma \mathbf{1}_{-L}$ on $-C$. These use mass $2|L| \geq 8$, and the other cosets, which come in pairs, carry nothing: shape I.

Case (b1): $-1 \in H^*$ and $e^* = p$ is prime. Then $H^* = K_p = U^*$, and D is constant by (S4); its value is nonzero, and we make it positive: shape II. (If $e^* = p^k$ with $k \geq 2$, then $H^* = K_p$ consists of units congruent to 1 modulo p^{k-1}, and $-1 \notin H^*$.)

Case (b2): $-1 \in H^*$ and e^* has at least two prime factors. Let $K = K_p$ have the largest order, H' the product of the other K_q, and write $-1 = \varepsilon\varepsilon'$ with $\varepsilon \in K$, $\varepsilon' \in H'$; by Lemma 3.5(b), $\varepsilon \neq 1 \neq \varepsilon'$. For the slices S_c of C, $c \in K$, multiplication by $\varepsilon\varepsilon' = -1$ maps S_c onto $S_{c\varepsilon}$, and as D is even, $\mu(c\varepsilon) = \mu(c)$. A coset $C' \neq C$ carrying a nonzero part of D satisfies $C' = -C'$, and an even nonzero split function of mass 4 on it would be $\pm\mathbf{1}_{L''}$ with $L'' = -L''$, that is, $-1 \in K_q$, which Lemma 3.5(b) excludes; so other cosets carry mass at least 6, or nothing.

(i) If some slice vanishes, D is split along $(K_q)_{q \neq p}$ on every slice by (S2). A nonzero slice has $\mu(c) = \mu(c\varepsilon) \leq 6$, and by Lemma 3.8 $D = \sigma$ on a line L along some K_q, $q \neq p$, with $|K_q| \leq 6$, inside S_c and 0 elsewhere on S_c; on $S_{c\varepsilon}$, $D = \sigma$ on $-L$. This uses mass $2|L| \geq 8$, so nothing else carries mass: shape I.

(ii) If all slices are nonzero and $\mu(c_s) = 1$, then $\mu(c_s\varepsilon) = 1$, and $z \mapsto D(z) - D(z\varepsilon)$ on S_{c_s} is split along $(K_q)_{q \neq p}$ by (S1), of mass at most 2, hence zero. For the support point z_0 of S_{c_s}, the points $z_0\varepsilon$ and $z_0\varepsilon\varepsilon' = -z_0$ of $S_{c_s\varepsilon}$ both carry $D(z_0) \neq 0$; so they coincide and $\varepsilon' = 1$, a contradiction.

(iii) If all $\mu(c) \geq 2$, then $2|K| \leq 12$, so every $|K_q| \leq 6$ and the primes of e^* are 5 and 7, with $7 \parallel e^*$. If $25 \mid e^*$, then $|K_5| = 5$, $K = K_7$, and $\varepsilon' \in K_5$ satisfies $\varepsilon'^2 = 1$ (the product

being direct), so $\varepsilon' = 1$, a contradiction. So $e^* = 35$, $|K| = 6$, every $\mu(c) = 2$, and the mass is $12 = 2N_x$: shape III. □

8.2. **Shape I.**

Lemma 8.5. *In shape* I*, A has a chain of moves that is direct or lowering.*

Proof. By Lemma 8.3(i), $P = \mathbf{1}_S$ with $S \subseteq L \cup (-L)$ containing exactly one of $v, -v$ for each $v \in L$; so the elements of order $4e^*$ are the x_w^+, $w \in S$, each once, and $N_x = |L|$. By Lemma 8.2(d), $S \cap L$ and $S \cap (-L)$ have at most $(q-1)/2$ points each. If $25 \mid e^*$ and $q = 5$, then $|L| = 5 > 4$; so $q \parallel e^*$, $|L| = q - 1$, and both sets have exactly $(q-1)/2$ points.

Let $x = x_{v_0}^+$ with $v_0 \in S \cap L$. The fibre $\Phi_q(x)$ consists of the x_w^+, $w \in L$, and of one element x_0 of order $4e^*/q$. The move $(\sigma_{q,x}, S')$ with $S' = \{x_w^+ : w \in S \cap L\}$ gives

$$A'' = A - S' + \{-x_w^+ : \ w \in L \setminus S\} + \{-x_0,\, qx\}.$$

For $w \in L \setminus S$ we have $-w \in S$ and $-x_w^+ = x_{-w}^- = x_{-w}^+ + M/2$. So A'' contains the $(q-1)/2$ pairs $\{x_u^+, x_u^-\}$, $u \in S \cap (-L)$, and the moves of Lemma 8.2(a) replace each of them by $\{y_u, M/2\}$. The result,

$$A - \{x_w^+ : \ w \in S\} + \{y_u : \ u \in S \cap (-L)\} + \tfrac{q-1}{2}\{M/2\} + \{-x_0,\, qx\},$$

has eight elements, among them at least two copies of $M/2$; after removing a pair at most six remain, with no element of order $4e^*$ and none of larger order that A did not have. □

8.3. **Shape II.**

Lemma 8.6. *In shape* II*, either A has a chain of moves that is direct or lowering, or, up to multiplication by a unit congruent to* 1 *modulo* 4*, the elements of A of odd part p are one of*

(a) $\{x_1^+, x_2^+, z_3\}$, $p = 5$*;*
(b) $\{x_1^+, x_2^+, x_4^+\}$, $p = 7$*;*
(c) $\{x_1^+, x_2^+, x_3^+, z_4, z_5\}$, $p = 7$*;*
(d) $\{x_1^+, x_2^+, x_4^+, x_8^+, x_5^+, z_6\}$, $p = 11$.

Proof. Let $D = c > 0$. By Lemma 8.3(i), $0 \le P(v) \le c$ and $P(v) + P(-v) = c$. The w with $P(w) \ge 1$ index distinct elements x_w^+ of the line $U^* = K_p$, so there are at most $(p-1)/2$ of them by Lemma 8.2(d), and at least one in each pair $\pm v$. Hence $P = c\,\mathbf{1}_S$ for a set S of representatives of $U^*/\{\pm 1\}$, and $N_x = c(p-1)/2 \le 6$.

$c \ge 2$. Let $x = x_{v_0}^+$, $v_0 \in S$. Now $\Phi_p(x)$ consists of all the x_w^+ and one element x_0 of order 4, and as in Lemma 8.5 the move $\sigma_{p,x}$ followed by the moves Lemma 8.2(a) on the pairs $\{x_u^+, x_u^-\}$, $u \in S$, gives

$$A - 2\{x_u^+ : u \in S\} + \{y_u : u \in S\} + \tfrac{p-1}{2}\{M/2\} + \{-x_0,\, px\},$$

with eight elements, at least two copies of $M/2$ and fewer elements of order $4p$ than A.

$c = 1$. G is constant on U^* by (S4), and $\sum_v G(v) = 0$, as Z, Y are odd and $\sum P = \sum P(2\,\cdot)$; so $G = 0$. With $\psi = \mathbf{1}_S - \mathbf{1}_{-S}$ we have $P = (1 + \psi)/2$, so

$$Z + Y - Y(2\,\cdot) = P(2\,\cdot) - P = \tfrac{1}{2}(\psi(2\,\cdot) - \psi),$$

which is 1 on $\Phi = \{w \notin S : \ 2w \in S\}$, -1 on $-\Phi$ and 0 elsewhere. If $y_v \in A$, then by Lemma 8.3(ii) the left side is at least 1 at v, so $v \notin S$, $-v \in S$, and $y_v = -2x_{-v}^+$ with $x_{-v}^+ \in A$, which is Lemma 8.2(b). So $N_y = 0$, $Z = \mathbf{1}_\Phi - \mathbf{1}_{-\Phi}$, and the elements of order p are the z_w, $w \in \Phi$, each once. Along the cycles of $w \mapsto 2w$ on $(\mathbb{Z}/p)^\times$, ψ changes sign $2|\Phi|$ times, and $(p-1)/2 + |\Phi| \le 6$, so $p \le 13$. The units of $\mathbb{Z}/M$ congruent to 1 modulo 4 act on U^* through all of $(\mathbb{Z}/p)^\times$ and preserve the kinds x^+, z.

- $p = 5$: one cycle of length 4 with $-1 = 2^2$, so ψ takes the values $a, b, -a, -b$ along it and changes sign twice; $|\Phi| = 1$, and up to units $S = \{1, 2\}$, $\Phi = \{3\}$. This is (a).
- $p = 7$: two cycles $\{1, 2, 4\}$ and $-\{1, 2, 4\}$, on which ψ takes opposite values; $|\Phi|$ is the number of sign changes on one cycle, 0 or 2. Up to units, $S = \{1, 2, 4\}$, $\Phi = \emptyset$, which is (b), or $S = \{1, 2, 3\}$, $\Phi = \{4, 5\}$, which is (c).
- $p = 11$ and $p = 13$: one cycle of length $p-1$ with $-1 = 2^{(p-1)/2}$, so $\psi(2^{(p-1)/2}w) = -\psi(w)$ and the number of sign changes along a half of the cycle is odd; $|\Phi|$ is odd. For $p = 13$, $6 + |\Phi| > 6$. For $p = 11$, $|\Phi| = 1$, $\psi = 1$ on five consecutive powers of 2, and up to units $S = \{1, 2, 4, 8, 5\}$, $\Phi = \{6\}$. This is (d). □

8.4. **Shape III.**

Lemma 8.7. *Shape* III *does not occur.*

Proof. Here $N_x = 6$, so $Y = Z = 0$, $G = P - P(2\,\cdot)$, and $\hat{G}(\chi) = (1 - \chi(2))\hat{P}(\chi)$ vanishes at every primitive character χ modulo 35. Such a character is $\chi_5\chi_7$ with both factors nontrivial; $\chi_5(2)$ has order 2 or 4, as 2 generates $(\mathbb{Z}/5)^\times$, and $\chi_7(2)$ has order 1 or 3, as 2 has order 3 modulo 7; so $\chi(2) \neq 1$. Hence $\hat{P}$ vanishes at every primitive character, P is split on $U^* = H^*$, nonzero, of mass 6, and by Lemma 3.8 $P = \sigma\mathbf{1}_L$ for a line L along K_7. Then A contains six distinct elements x_w^+, $w \in L$, or six distinct elements x_{-w}^-, $w \in L$, on one line along K_7, which Lemma 8.2(d) excludes. □

8.5. **The case $J \geq 3$.**

Lemma 8.8. *If $J \geq 3$, then A has a chain of moves that is direct or lowering.*

Proof. The elements of order $e = 2^J e^*$ are $x_{t,v} = (M/e)u$ with $u \equiv t$ modulo 2^J and $u \equiv v$ modulo e^*, for $(t, v) \in (\mathbb{Z}/2^J)^\times \times U^*$. Put $\tilde{F}(t, v) = \mathrm{mult}_A(x_{t,v}) - \mathrm{mult}_A(x_{-t,-v})$. Since e is a maximal order, (1) with $c = e$ involves only these elements, as in Section 4, so $\hat{\tilde{F}}$ vanishes at every primitive character modulo e (at the even ones because $\tilde{F}$ is odd). A character modulo e that is not primitive is trivial on the units congruent to 1 modulo e/l for some prime $l \mid e$; for $l = 2$ these form $\tilde{K}_2 = \{1, \eta\} \times \{1\}$ with $\eta = 1 + 2^{J-1}$, and for $l = p \mid e^*$ they form $\{1\} \times K_p$. So $\tilde{F}$ is split along $\tilde{K}_2$ and the K_p. As $x_{t,v} + M/2 = x_{t\eta,v}$, the pairs $\{x_{t,v}, x_{t\eta,v}\}$ are the pairs $\{x, x + M/2\}$.

Suppose A contains no such pair, as otherwise the move $(T(x), \{x, x + M/2\})$ lowers the profile. By (S1), $g_t(v) = \tilde{F}(t, v) - \tilde{F}(t\eta, v)$ is split along $(K_p)_{p|e^*}$ on every coset of H^*. As $J \geq 3$, the orbit $O = \{t, t\eta, -t, -t\eta\}$ has four elements. Of the four elements $x_{t,v}$, $x_{t\eta,v}$, $x_{-t,-v}$, $x_{-t\eta,-v}$, two opposite ones form a pair and two adjacent ones a pair $\{x, x + M/2\}$, so A contains copies of at most one of them or of the diagonal $\{x_{t,v}, x_{-t\eta,-v}\}$ or $\{x_{t\eta,v}, x_{-t,-v}\}$, and $|g_t(v)|$ is the number of elements of A among the four. Summing over v, the mass of g_t is the number of elements of A of order e whose first coordinate lies in O. A nonzero split function has mass at least 4, and the number of elements of order e is at most 6; so exactly one orbit O is occupied, and for $t_0 \in O$, $g_{t_0} = \sigma\mathbf{1}_L$ on one coset with L a line along some K_q, $|L| \leq 6$ (Lemma 3.8). If $e^* = 1$, the family (K_p) is empty, $g_{t_0} = 0$, and A has no element of order e, a contradiction.

Replacing t_0 by $t_0\eta$ we may take $\sigma = 1$. Then for $v \in L$ exactly one of $x_{t_0,v}$ and $x_{-t_0\eta,-v}$ lies in A, once, and A has no other element of order e. The $x_{t_0,v}$, $v \in L$, lie in one fibre of Φ_q, and so do the $x_{-t_0\eta,-v}$; so by Lemma 4.1 each kind occurs at most $(q-1)/2$ times, $q \parallel e^*$, $|L| = q - 1$, and both occur $(q-1)/2$ times. The move $\sigma_{q,x}$ with $x = x_{t_0,v_0} \in A$ adds $-x_{t_0,w} = x_{-t_0,-w}$ for the $w \in L$ with $x_{t_0,w} \notin A$, and these form pairs $\{x_{-t_0,-w}, x_{-t_0\eta,-w}\}$ with elements of A. As in Lemma 8.5, the moves T on these pairs leave at most six elements after the removal of a pair of copies of $M/2$, none of order e, and no element of larger order that A did not have. □

8.6. **The global step.**

Proposition 8.9. *Let M be even and prime to 3, with $M \notin \{70, 110\}$. Every sextuple A at level M that contains no pair, is not split and generates $\mathbb{Z}/M$ has a chain of moves that succeeds.*

Proof. Suppose not. By Lemma 7.1, every maximal odd part e^* of A has $J \geq 1$; $J = 1$ is excluded by Theorem 7.14 and $J \geq 3$ by Lemma 8.8. So every maximal odd part has $J = 2$, and $4 \mid M$. By Lemmas 8.4 to 8.7, applied to A or to $-A$, the elements of odd part e^* form, up to units and sign, one of the configurations (a)–(d) of Lemma 8.6, of sizes 3, 3, 5, 6.

If A has two maximal odd parts, their configurations have three elements each, with different primes: (a) for 5 and (b) for 7. Then A lies in the subgroup of order 140, so $M = 140$.

If A has one maximal odd part $e^* = p \in \{5, 7, 11\}$, the low elements have odd part 1. Let 2^K be the largest order among them, and suppose $K \geq 3$. In (1) at the conductor 2^K only the low elements of order 2^K enter, as the elements of odd part p have 2-part at most 4; so the odd counting function of these elements on $(\mathbb{Z}/2^K)^\times$ has a vanishing transform at every primitive character, and the other characters are trivial on $\{1, 1 + 2^{K-1}\}$. Hence it is invariant under multiplication by $1 + 2^{K-1}$, and an element $a \in A$ of order 2^K comes with $(1 + 2^{K-1})a = a + M/2$; the move $(T(a), \{a, a + M/2\})$ lowers the profile. So the low elements lie in $\{M/4, M/2, 3M/4\}$, A lies in the subgroup of order $4p$, and $M \in \{20, 28, 44\}$.

It remains to examine these levels. At $t = 1$, balance says that the representatives in $[1, M-1]$ of the elements of A add up to $3M$, and we may normalise A by a unit and a sign.

- $M = 20$, (a) $= \{1, 17, 12\}$: the three low elements lie in $\{5, 10, 15\}$ and add up to 30, so they are $\{5, 10, 15\}$ or $\{10, 10, 10\}$, and A contains a pair.
- $M = 28$, (b) $= \{1, 9, 25\}$: the low elements lie in $\{7, 14, 21\}$ and add up to 49, so they are $\{7, 21, 21\}$ or $\{14, 14, 21\}$, with a pair.
- $M = 28$, (c) $= \{1, 9, 17, 16, 20\}$: the low element is 21. The moves $(T(1), \{1\})$ and $(T(6), \{20, 16, 14\})$ turn $A = \{1, 9, 16, 17, 20, 21\}$ into $\{2, 9, 13, 14, 16, 17, 20, 21\}$ and then into $\{2, 9, 17\} + \{13, 21, 22\}$, a split sextuple; this chain succeeds.
- $M = 44$, (d) $= \{1, 13, 37, 41, 5, 24\}$: the sum is $121 \neq 132$.
- $M = 140$, (a) for 5 and (b) for 7: up to units and sign, A is $\{7, 119, 84\}$ together with $\{5, 45, 125\}$ or with $\{55, 75, 115\}$, of sums 385 and 455, not 420.

So in every case A contains a pair, is not balanced, or has a chain that succeeds. □

Theorem 8.10. *The Hodge conjecture holds for X^4_M when $3 \nmid M$ and $110 \nmid M$.*

Proof. By induction on M. For M odd this is Corollary 4.4, and for $M = 70$ it is Corollary 6.11. Let M be even, prime to 3, not divisible by 110 and different from 70. The proper divisors of M satisfy the same conditions, so the Hodge conjecture holds for them, and Proposition 8.9 with Proposition 2.9 gives it for M. □

9. Proof of the main theorem

Proof of Theorem A. Let $M \geq 1$ with $6 \nmid M$ and $110 \nmid M$. If $3 \nmid M$, the Hodge conjecture for X^4_M is Theorem 8.10. If $3 \mid M$, then M is odd, and it is Corollary 4.4. □

For the degrees of Corollary 6.11 the conjecture holds for X^n_M in every dimension. For the other degrees the proof gives only the fourfold: it shows that every Hodge sextuple is algebraic, and says nothing about the classes that Aoki's reduction attaches to the divisors of M in higher dimension.

Remark 9.1. The elements of P that divide 110 are 5 and 11, so $\mathcal{Q}(110) = \{55\}$. Let

$$A_{110} = \{19, 29, 39, 55, 79, 109\},$$

a sextuple at level 110; without the element 55 it is the representative of Aoki's class ξ_{55} listed in [MMRV26, Table 4], and that it is balanced is a direct check over the units. Let $\nu\colon R_{110} \to \mathbb{Z}/2$ be the homomorphism with $\nu((y)) = 1$ exactly when $\bar{y} \equiv 4$, 5 or 6 modulo 10. Then ν vanishes on S_{110}. It depends only on y modulo 10, and $\nu((y)) = \nu((-y))$, which settles the pairs. The coset $\Phi_{11}(x)$ consists of eleven elements congruent to x modulo 10, and $-11x \equiv -x$, so $\sigma_{11,x}$ has $12\nu((x))$ elements counted by ν. The coset $\Phi_5(x) = \{x + 22j\}$ runs through the five residues modulo 10 of the parity of x; for x even it contains two elements congruent to 4 or 6 and $-5x \equiv 0$, and for x odd it contains one element congruent to 5 and $-5x \equiv 5$. Finally, if $x \equiv r$ modulo 10, then $T(x) = \{x, x + 55, -2x, 55\}$ has residues r, $r + 5$, $-2r$, 5, and among r, $r + 5$, $-2r$ exactly one lies in $\{4, 5, 6\}$ for every r. In each case the count is even.

Since $\nu([A_{110}]) = 1$, the argument of Corollary 5.6 gives $B_{110} = S_{110} + \mathbb{Z}[A_{110}]$. Hence the Hodge conjecture for X^4_{110} is equivalent to the algebraicity of the single line of A_{110}, and if that line is algebraic, then 110 is a complete level and the conjecture holds for X^n_{110} in every dimension n.[11]

10. Degrees divisible by six

In this section $6 \mid M$, $110 \nmid M$, and A is a sextuple at level M that contains no pair, is not split and generates $\mathbb{Z}/M$. In Sections 4, 7 and 8 the elements of largest order carry the argument. Here an order has a 2-part and a 3-part, both of which can be large, and we separate them from the rest of the order. This reduces the Hodge conjecture for X^4_M to a statement about configurations of at most six residues, Hypothesis 10.4 (Theorem 10.8). We prove the part of that statement that concerns large 2- and 3-parts (Corollary 10.20); the rest is open, and Theorem A does not cover these degrees.

10.1. **Cores.** The *core* $f(y)$ of a nonzero $y \in \mathbb{Z}/M$ is the largest divisor of its order that is prime to 6. Let E be the set of the cores of the elements of A that are maximal for divisibility among these cores. For $e' \in E$ the elements of A of core e' form the *e'-part* $A_{e'}$, and the elements whose core lies outside E form the *low part*. Let $D_{e'}$ be the 2, 3-part of the least common multiple of the orders of the elements of $A_{e'}$, so that $A_{e'}$ lies in the subgroup of order $D_{e'}e'$.

Lemma 10.1. *Let $e' \in E$ and let χ be an odd primitive character of conductor $c \mid M$ whose largest divisor prime to 6 is e'. The elements of A that enter (1) at χ lie in $A_{e'}$.*

Proof. An element a enters only if $c \mid f_a$; then $e' \mid f(a)$, and $f(a) = e'$ by maximality. □

The relations at these conductors thus concern $A_{e'}$ alone, and so do the moves that act on $A_{e'}$. We formalise this.

Definition 10.2. Let $e' > 1$ be prime to 6, $D = 2^v 3^w$ and $N = De'$. A *local configuration* at level N is a multiset B of at most six elements of $\mathbb{Z}/N$ of core e' such that

(i) (1) holds for B at every odd primitive character whose conductor divides N and has largest divisor prime to 6 equal to e';
(ii) B contains no pair, and, if N is even, no two elements x and $x + N/2$;

[11] The homomorphism ν is not one of those of Lemmas 5.2 and 5.3: the parities ν_5 and ν_{11} of the numbers of elements of order 5 and 11 vanish on $[A_{110}]$, and the parity of the number of elements with square 1 does not vanish on S_{110}, because $T(1)$ contains exactly one such element.

(iii) for $\ell = 3$ and for every prime $\ell \mid e'$, B does not contain $(\ell+1)/2$ distinct elements of one order $n \neq \ell$, with $\ell \mid n$, that lie in one coset $\Phi_\ell(x)$.

A *local chain* for B is a chain of moves (σ, S) at level N, each with $\sigma = T(x)$ or $\sigma = \sigma_{\ell,x}$, $\ell \in \{3\} \cup \{\ell \mid e'\}$, and S the largest multiset contained both in σ and in the current multiset, with every pair removed after each move, that ends in a multiset $B^\flat$ such that either $|B^\flat| \leq |B| - 2$, or $|B^\flat| = |B|$ and, at the largest integer at which B and $B^\flat$ have different numbers of elements of that order, B has more.

Every move changes the number of elements by an even number, so $|B^\flat|$ and $|B|$ have the same parity. For $g \mid M$ the map $\iota_g \colon \mathbb{Z}/(M/g) \to \mathbb{Z}/M$, $y \mapsto gy$, preserves orders and cores, and $\iota_g \Phi_\ell(x) = \Phi_\ell(\iota_g x)$, $\iota_g \sigma_{\ell,x} = \sigma_{\ell,\iota_g x}$ and $\iota_g T(x) = T(\iota_g x)$.

Lemma 10.3. *Suppose that A has no chain of moves that succeeds, let $e' \in E$ with $e' > 1$, and let D be a multiple of $D_{e'}$ with $De' \mid M$. Then $A_{e'} = \iota_g B$, $g = M/(De')$, for a local configuration B at level De' that has no local chain.*

Proof. As $A_{e'}$ lies in the subgroup of order De', $A_{e'} = \iota_g B$ for a multiset B of elements of core e'. The terms of (1) depend only on the order f_a and on z_a, which ι_g preserves, so (i) follows from Lemma 10.1. A contains no pair. If B contained x and $x + N/2$, the move $(T(\iota_g x), \{\iota_g x, \iota_g x + M/2\})$ would replace them by $2\iota_g x$ and $M/2$; if n is the order of x, the orders removed and added are $\{n, n\}$ and $\{n/2, 2\}$ when $4 \mid n$, $\{n, n/2\}$ and $\{n/2, 2\}$ when $2 \parallel n$, and $\{n, 2n\}$ and $\{n, 2\}$ when n is odd, so this chain of one move is lowering, or direct if a pair cancels. If (iii) failed, Lemma 4.1 would give a move for A, a chain of one move that succeeds. So B is a local configuration. A local chain for B, carried by ι_g to level M, is a chain of moves for A that leaves the other elements of A untouched, apart from pairs that cancel against them. If it ends in at most $|B| - 2$ elements, the chain for A is direct. Otherwise the orders of A and of the final multiset differ exactly as those of B and $B^\flat$, and the chain is lowering, or direct if further pairs cancel. □

Hypothesis 10.4 (Local step)**.** *Let $e' > 1$ be prime to 6, $D = 2^v 3^w$, and let B be a local configuration at level De' that has no local chain. Let D_B be the $2,3$-part of the least common multiple of the orders of the elements of B.*

(a) *The core e', the number of elements of B and D_B are as in Table 2. At $e' = 35$ the configurations with four elements are, up to units, the image of $\{1, 24, 61, 62\} \subseteq \mathbb{Z}/70$, and those with five elements have $D_B = 2$.*

(b) *If B has at most three elements, then B is a unit multiple of $\iota_h B_0$, $h = De'/N$, for a row B_0 of Table 3 at level $N = D_B e'$, and B_0 together with the added elements of its row is a balanced multiset of four elements. When B_0 has two elements, the added elements are τ and $N/2$, and their orders are smaller than the largest order of an element of B_0.*

We first derive the Hodge conjecture for the degrees divisible by 6 from Hypothesis 10.4, and then, in Section 10.6, prove the part of it that bounds the 2- and 3-parts of the orders: in a local configuration without a local chain every order divides $72e'$. What remains of Hypothesis 10.4 are the configurations at the levels De' with $D \mid 72$, for every e' prime to 6.

10.2. **The low part.**

Lemma 10.5. *Let $A' \subseteq A$, let N be the least common multiple of the orders of the elements of $A - A'$, and let W be the group of units w with $w \equiv 1 \pmod N$. Then $s(t) = \sum_{a \in A'} \langle ta \rangle$ satisfies $s(tw) = s(t)$ for every unit t and every $w \in W$. If $A' = \{a\}$, then $M \mid \operatorname{lcm}(N, 2)$.*

TABLE 2. Local configurations without a local chain

e'	number of elements	D_B
5	$2, 3, 4, 5, 6$	$3, 4, 6, 8, 12, 18, 24$
7	$3, 4, 5, 6$	$3, 4, 6, 8, 12, 24$
11	$3, 4, 5, 6$	$4, 6$
13	$3, 4, 6$	6
17, 19	6	6
35	$4, 5$	2
35	6	$1, 2, 3, 6$
55	$5, 6$	2

Proof. Every $w \in W$ fixes the elements of $A - A'$, and $h_A(tw) = h_A(t) = 3$; this is the first assertion. Let $A' = \{a\}$, $n = \mathrm{ord}(a)$ and $g = \gcd(n, N)$. As A generates $\mathbb{Z}/M$, $M = \mathrm{lcm}(N, n)$, and by the Chinese remainder theorem the residues modulo n of the elements of W are all the units u with $u \equiv 1 \pmod g$. If such a u moves a, then $\langle wa\rangle \neq \langle a\rangle$ for a lift $w \in W$, against $s(w) = s(1)$. So the kernel of $(\mathbb{Z}/n)^\times \to (\mathbb{Z}/g)^\times$ is trivial, which happens only for $n = g$ and for $n = 2g$ with g odd. In both cases $n \mid \mathrm{lcm}(N, 2)$, and so does M. □

10.3. **One maximal core.**

Proposition 10.6. *Suppose that Hypothesis 10.4 holds, that A has no chain of moves that succeeds and that $E = \{e'\}$. Then M is one of the levels of Corollary 6.11.*

Proof. Every core divides e', so $M = 2^a 3^b e'$. If $e' \in \{1, 5, 7, 11, 13\}$, Corollary 6.11(i),(ii) applies. Otherwise, by Lemma 10.3 and Hypothesis 10.4, e' is 17, 19, 35 or 55 and $A_{e'}$ is listed in Table 2. For $e' = 17, 19$ it has six elements and $D_{e'} = 6$, so $A = A_{e'}$ lies in the subgroup of order $6e'$ and $M = 6e'$, which is in Corollary 6.11(iii). For $e' = 55$, $D_{e'} = 2$ and $110 \mid M$, which is excluded. Let $e' = 35$. If $420 \nmid M$, Corollary 6.11(i) applies; we show that $420 \mid M$ is impossible. If $|A_{35}| = 6$, then A lies in the subgroup of order $35D_{35}$ with $D_{35} \mid 6$, so $M \mid 210$. If $|A_{35}| = 5$, then $D_{35} = 2$ and Lemma 10.5 with A' the low element gives $M \mid 70$.

Let $|A_{35}| = 4$. Then $A_{35} = u\,\iota_g\{1, 24, 61, 62\}$ for a unit u, $g = M/70$, and the low part is $\{a_1, a_2\}$, of cores in $\{1, 5, 7\}$. Write $M = 35C$ with C the 2, 3-part of M, so $12 \mid C$, and

$$a_i/M \equiv \alpha_i/C + \gamma_i \pmod 1, \qquad \alpha_i \in \mathbb{Z}/C,\ \gamma_i \in \tfrac{1}{35}\mathbb{Z}/\mathbb{Z},$$

where the denominator f_i of γ_i is the core of a_i. Lemma 10.5 with $A' = \{a_1, a_2\}$ and $N = 70$ applies; the group W fixes the γ_i and acts on the α_i through all of $(\mathbb{Z}/C)^\times$, as every unit modulo C is odd.

If both α_i had order at most 2, the orders of all elements of A would have 2, 3-part at most 2, against $12 \mid C$. If exactly one, say α_1, had order larger than 2, a $w \in W$ acting on α_1 by -1 would change $\langle a_1\rangle$ and fix a_2, against $s(w) = s(1)$. So both orders exceed 2. Modulo 1, $s(tw) - s(t) \equiv t(w-1)(\alpha_1 + \alpha_2)/C$, and $w \equiv -1$ modulo C gives $2(\alpha_1 + \alpha_2) = 0$, so $\alpha_2 = -\alpha_1 + \delta C$ with $\delta \in \{0, 1/2\}$.

The 3-part of C is the 3-part of the order of a_1 or of a_2, as the elements of A_{35} have orders dividing 70, and adding δC does not change the 3-part of an order; so both orders are divisible by 3. At the odd character χ of conductor 3 only a_1 and a_2 enter (1), and the factor $\prod_\ell(1 - \chi(\ell))$ of a_i contains $1 - \chi(5) = 2$ if $f_i = 5$ and $1 - \chi(7) = 0$ if $f_i = 7$, the other factors being nonzero. So $\{f_1, f_2\} \neq \{5, 7\}$, and the least common multiple of f_1 and f_2 is 1, 5 or 7.

Since t is odd, $s(t) \equiv t(\alpha_1 + \alpha_2)/C + t(\gamma_1 + \gamma_2) \equiv \delta + t(\gamma_1 + \gamma_2) \pmod 1$, a fraction whose denominator divides 10 or 14. At $t = u^{-1}$, however, the elements of tA_{35} contribute

$(1+24+61+62)/70 = 148/70$ to $h_A(t) = 3$, so $s(t) = 3 - 148/70 = 31/35$, whose denominator is 35. This contradiction excludes $|A_{35}| = 4$. □

10.4. **Two maximal cores.** A balanced multiset of four elements is algebraic, so it can serve as σ in a move. If $B \subseteq A$ and a multiset T with $|B| + |T| = 4$ makes $B + T$ balanced, the move $(B + T, B)$ replaces B by $-T$. This is how the small configurations are removed.

TABLE 3. The local configurations of at most three elements, up to units, and the elements that complete them to balanced multisets of four elements

e'	N	B	added	e'	N	B	added
5	30	$\{1, 19\}$	25, 15	5	60	$\{2, 21, 57\}$	40
5	30	$\{1, 24\}$	20, 15	7	28	$\{1, 9, 25\}$	21
5	60	$\{1, 49\}$	40, 30	7	42	$\{1, 24, 38\}$	21
5	15	$\{1, 6, 13\}$	10	7	42	$\{1, 25, 37\}$	21
5	20	$\{1, 12, 17\}$	10	7	42	$\{1, 30, 32\}$	21
5	30	$\{1, 7, 27\}$	25	11	66	$\{1, 25, 62\}$	44
				13	78	$\{1, 55, 61\}$	39

Proposition 10.7. *Suppose that Hypothesis 10.4 holds. If A has no chain of moves that succeeds, then E has one element.*

Proof. If $1 \in E$, every core is 1 and $E = \{1\}$. Suppose that E has two elements $e_1 \neq e_2$, both larger than 1. By Lemma 10.3, applied with $D = D_{e_i}$, $A_{e_i} = \iota_{g_i} B_i$ for a local configuration B_i without a local chain at level $N_i = D_{e_i} e_i$, and by Hypothesis 10.4(a) each B_i has at least two elements. The parts A_{e_1} and A_{e_2} are disjoint, so one of them, say A_{e_1}, has at most three elements. By Hypothesis 10.4, $e_1 \in \{5, 7, 11, 13\}$ and $B_1 = uB$ for a unit u and a row B of Table 3 at level $N = N_1$; let T be the added elements of that row and $g = g_1$.

The multiset $B + T$ is balanced, and so is $u(B + T)$. Its image $\sigma = \iota_g u(B + T)$ is a balanced multiset of four nonzero elements at level M by Lemma 2.6(c),(d), algebraic by Lemma 2.6(a), and it contains $S = A_{e_1}$. The move (σ, S) gives $A'' = (A - S) + (-\iota_g uT)$. If $|S| = 3$, then $|A''| = 4$ and the move is direct. If $|S| = 2$, then $e_1 = 5$ and $T = \{\tau, N/2\}$, so $-\iota_g uT = \{-gu\tau, M/2\}$. The orders of these two elements are those of τ and $N/2$, and by Hypothesis 10.4(b) they are smaller than the largest order n of an element of S. So A'' has the same elements as A of order larger than n and fewer of order n, and the move is lowering, or direct if A'' contains a pair. In both cases A has a chain of one move that succeeds, a contradiction. □

10.5. **The reduction.**

Theorem 10.8. *If Hypothesis 10.4 holds, then the Hodge conjecture holds for X^4_M whenever $6 \mid M$ and $110 \nmid M$.*

Proof. By induction on M. The proper divisors of M are not divisible by 110, and the Hodge conjecture holds for them by Theorem 8.10, Corollary 4.4 and induction. Let A be a sextuple at level M that contains no pair, is not split and generates $\mathbb{Z}/M$, and suppose that it has no chain of moves that succeeds. Then E has one element by Proposition 10.7, and M is a level of Corollary 6.11 by Proposition 10.6, so A is algebraic. Proposition 2.9 gives the claim. □

10.6. **The 2- and 3-parts of a local configuration.** Throughout this subsection $e' > 1$ is prime to 6, $D = 2^v 3^w$, $N = De'$, $U = (\mathbb{Z}/e')^\times$, and B is a local configuration at level N. We show that B has a local chain as soon as one of its elements has an order divisible by 16 or by 27, so that the orders in a configuration without a local chain divide $72e'$.

Coordinates. An element x of core e' has order $n = 2^j 3^i e'$, and $x = (N/n)u$ for a unit u modulo n. We write $x = [t, \beta, \mu]_{j,i}$, where t, β and μ are the residues of u modulo 2^j, 3^i and e', and call j the 2-*level* and i the 3-*level* of x; the coordinate t is omitted when $j = 0$, and β when $i = 0$. By the Chinese remainder theorem these data determine x. Put $\eta_j = 1 + 2^{j-1}$. The following rules come from $N/2 = (N/n)(n/2)$ and the analogous identities for $N/3$ and N/q.

(C1) $-x = [-t, -\beta, -\mu]_{j,i}$. If $j \geq 2$, then $x + N/2 = [t\eta_j, \beta, \mu]_{j,i}$; if $j = 0$, then $x + N/2 = [1, 2\beta, 2\mu]_{1,i}$.
(C2) If $i \geq 1$, then $3x = [t, \beta \bmod 3^{i-1}, \mu]_{j,i-1}$. If $j \geq 1$, then $2x = [t \bmod 2^{j-1}, \beta, \mu]_{j-1,i}$.
(C3) If $i \geq 2$, the coset $\Phi_3(x)$ consists of the three elements $[t, \beta', \mu]_{j,i}$ with $\beta' \equiv \beta \pmod{3^{i-1}}$. If $i = 1$, it consists of $[t, \pm\beta, \mu]_{j,1}$ and $[t/3, \mu/3]_{j,0}$.
(C4) For a prime $q \mid e'$, the elements of order n in $\Phi_q(x)$ are the $[t, \beta, \mu']_{j,i}$ with $\mu' \in \mu K_q$, where $K_q \subseteq U$ is the subgroup of Lemma 3.5. If $q \parallel e'$, the remaining element of $\Phi_q(x)$ has order n/q and core e'/q.

In these terms, Definition 10.2(ii) says in particular that B does not contain both x and $-x$, nor, when $j \geq 2$, both $[t, \beta, \mu]_{j,i}$ and $[t\eta_j, \beta, \mu]_{j,i}$. By (C3) and (C4), Definition 10.2(iii) says that B does not contain two distinct elements $[t, \beta, \mu]_{j,i}$ and $[t, \beta', \mu]_{j,i}$ with $i \geq 1$ and $\beta' \equiv \beta \pmod{3^{i-1}}$, and that for no prime $q \mid e'$ does B contain $(q+1)/2$ distinct elements $[t, \beta, \mu']_{j,i}$ whose coordinates μ' lie in one coset of K_q.

The relations of B on boxes. Let $G_2 = (\mathbb{Z}/2^v)^\times$ and $G_3 = (\mathbb{Z}/3^w)^\times$, with Haar measures of total mass 1. The *box* of $x = [t, \beta, \mu]_{j,i}$ is

$$Q(x) = \{(s, a) \in G_2 \times G_3 : \ s \equiv t \ (2^j), \ a \equiv \beta \ (3^i)\},$$

of volume $1/\varphi(2^j)\varphi(3^i)$. We work in the group algebra $\mathbb{Q}[U]$, with basis $[\mu]$ for $\mu \in U$, and regard $f = \sum f(\mu)[\mu]$ as a function on U. Put $\|f\| = \sum |f(\mu)|$, $c_*[\mu] = [c\mu]$ for $c \in U$, and $\psi(f) = \sum f(\mu)\psi(\mu)$ for a character ψ of U. Call f *split* if $\psi(f) = 0$ for every primitive character ψ modulo e', and write $f \sim g$ when $f - g$ is split. For a submultiset $B' \subseteq B$ let

$$R_{B'}(s, a) = \sum_{x \in B'} \Big(\mathbf{1}_{Q(x)}(s, a)\,[\mu_x] - \mathbf{1}_{Q(x)}(-s, -a)\,[-\mu_x] \Big) \in \mathbb{Z}[U],$$

where μ_x is the coordinate μ of x, and $R = R_B$. Thus $R(-s, -a) = -(-1)_* R(s, a)$.

Lemma 10.9. *Let B_3 and B_2 consist of the elements of B of 3-level at least 1 and of 2-level at least 1.*

(a) $R(s, a) - R(s', a) - R(s, a') + R(s', a') \sim 0$ *for* $s, s' \in G_2$ *and* $a, a' \in G_3$.
(b) $F_3(s) = \int_{G_3} \big(R(s, a) - (1/3)_* R_{B_3}(3s, a)\big)\, da$ *satisfies* $F_3(s) \sim F_3(s')$ *for* $s, s' \in G_2$.
(c) $F_2(a) = \int_{G_2} \big(R(s, a) - (1/2)_* R_{B_2}(s, 2a)\big)\, ds$ *satisfies* $F_2(a) \sim F_2(a')$ *for* $a, a' \in G_3$.

Proof. Let χ_2 and χ_3 be characters of G_2 and G_3, of conductors c_2 and c_3, let ψ be primitive modulo e', and let $\chi = \chi_2\chi_3\psi$, a primitive character of conductor $c = c_2c_3e'$ dividing N. If χ is odd, so is $\bar\chi$, and (1) holds for B at $\bar\chi$ by Definition 10.2(i). For $x = [t, \beta, \mu]_{j,i}$ in B we have $c \mid f_x$ exactly when $c_2 \mid 2^j$ and $c_3 \mid 3^i$, and then z_x has the residues t, β, μ and

$$\frac{\chi(z_x)}{\varphi(f_x)} = \frac{1}{\varphi(e')} \iint \chi_2(s)\chi_3(a)\, \mathbf{1}_{Q(x)}(s, a)\, \psi([\mu])\, ds\, da;$$

otherwise the integral vanishes, because χ_2 or χ_3 is nontrivial on the units congruent to 1 modulo 2^j or 3^i. The primes in the product of (1) are 2, when $c_2 = 1$ and $j \geq 1$, and 3, when $c_3 = 1$ and $i \geq 1$. Finally, for every function R' built like R from a function $R'^+(s,a) = \sum \mathbf{1}_Q(s,a)[\mu]$, the substitution $(s,a) \mapsto (-s,-a)$ gives

$$\iint \chi_2\chi_3\,\psi(R') = (1-\chi(-1)) \iint \chi_2\chi_3\,\psi(R'^+),$$

which vanishes when χ is even.

(a) Let χ_2 and χ_3 be nontrivial. Then $6 \mid c$, the product in (1) is empty, and (1) at $\bar\chi$ says that $\iint \chi_2\chi_3\,\psi(R^+) = 0$ when χ is odd. Hence $\iint \chi_2\chi_3\,\psi(R) = 0$ for all nontrivial χ_2 and χ_3: the function $\psi(R(s,a))$ on $G_2 \times G_3$ is a sum $g(s)+h(a)$, and its second differences vanish. This holds for every primitive ψ.

(b) Let χ_2 be nontrivial and $\chi_3 = 1$. For $i \geq 1$ the factor of x in (1) at $\bar\chi$ is $1-\bar\chi(3)$, and $\chi(z_x)\bar\chi(3) = \chi_2(t/3)\psi(\mu/3)$, while $\mathbf{1}_{Q(x)}(3s,a)$ is the indicator of the box of x with t replaced by $t/3$. So (1) says that $\iint \chi_2(s)\,\psi(R^+(s,a) - (1/3)_* R^+_{B_3}(3s,a)) = 0$ when χ is odd, and as in (a), $\int \chi_2(s)\,\psi(F_3(s))\,ds = 0$ for every nontrivial χ_2. So $\psi(F_3(s))$ does not depend on s.

(c) The same, with $\chi_2 = 1$, χ_3 nontrivial and the factor $1-\bar\chi(2)$, for $j \geq 1$. □

Lemma 10.10. (a) *A line wK_q along some K_q is split, and c_*f is split when f is.*

(b) *A nonzero split $f \in \mathbb{Z}[U]$ has $\|f\| \geq 4$. If $\|f\| \leq 6$, then $f = \pm\mathbf{1}_L$ for a line L along K_5, or along K_7 when $7 \parallel e'$, and $|L|$ is 4 if $5 \parallel e'$, 5 if $25 \mid e'$, and 6 for K_7.*

(c) *Two distinct lines along subgroups K_q and $K_{q'}$ meet in at most one point.*

Proof. (a) K_q is the kernel of $U \to (\mathbb{Z}/(e'/q))^\times$, so a primitive ψ is nontrivial on K_q and $\psi(\mathbf{1}_{wK_q}) = \psi(w)\sum_{k\in K_q}\psi(k) = 0$. Also $\psi(c_*f) = \psi(c)\psi(f)$.

(b) By Fourier inversion on U, f is a combination of characters that are not primitive, each trivial on some K_p by Lemma 3.5(c). So on every coset of $H = \prod_p K_p$ the function f is split along $(K_p)_{p\mid e'}$ in the sense of Definition 3.6, and Lemma 3.5(a) gives the hypothesis of Lemma 3.8. On a coset where f does not vanish its mass is at least 7, or by Lemma 3.8 equal to some $|K_p| \geq 4$. If $\|f\| \leq 6$, then f lives on one coset, and Lemma 3.8 gives $f = \pm\mathbf{1}_L$ for a line along a K_q with $|K_q| \leq 6$, that is $q = 5$, or $q = 7$ and $7 \parallel e'$, by Lemma 3.5(a).

(c) Lines along one K_q are equal or disjoint, and for $q \neq q'$ the set $wK_q \cap w'K_{q'}$ is empty or a coset of $K_q \cap K_{q'}$, which is trivial by Lemma 3.5(b). □

The lines of size at most 6 are the *short lines*. There are none when $5 \nmid e'$ and 7 does not divide e' exactly once, and then every nonzero split $f \in \mathbb{Z}[U]$ has $\|f\| \geq 7$.

Tops and orbits. From here to the end of the proof of Proposition 10.14, $J \geq 4$ is the largest 2-level of an element of B, so $v \geq J$, and $\eta = \eta_J$. A *top* is an element of B of 2-level J. The function $R(s,a)$ depends on s only modulo 2^J, and a box of 2-level smaller than J contains (s,a) and $(s\eta,a)$ together. The *orbit* of a unit t modulo 2^J is $O_t = \{\pm t, \pm t\eta\}$, a coset of the subgroup $\{\pm1, \pm\eta\}$.

Fix a *base* t of an orbit O. A top x whose coordinate t_x lies in O is a *top of* O. Its *type* $(\varepsilon_x, \sigma_x)$ is $(+,+)$, $(-,+)$, $(-,-)$ or $(+,-)$ according as t_x is t, $t\eta$, $-t$ or $-t\eta$; its *signed box* P_x is the ball of the $a \in G_3$ with $a \equiv \sigma_x\beta_x \pmod{3^i}$, i its 3-level ($P_x = G_3$ when $i = 0$); and its *label* is $\lambda_x = \sigma_x\mu_x$. We call ε_x the *sign* of x. Since the other elements contribute equally at $s = t$ and $s = t\eta$, and a top x contributes $\mathbf{1}_{Q(x)}(s,a)[\mu_x]$ at $s = t_x$ and $-\mathbf{1}_{Q(x)}(-s,-a)[-\mu_x]$ at $s = -t_x$,

$$\Phi_O(a) := R(t,a) - R(t\eta,a) = \sum_{x \text{ top of } O} \varepsilon_x\,\mathbf{1}_{P_x}(a)\,[\lambda_x]. \tag{3}$$

Replacing the base t by $t\eta$ changes every sign, keeps the signed boxes and the labels, and replaces Φ_O by $-\Phi_O$.

The group $(\mathbb{Z}/2^J)^\times = \{\pm 1\} \times \langle 5\rangle$ has $\{\pm 1, \pm\eta\}$ as the product of $\{\pm 1\}$ with the subgroup of order 2 of $\langle 5\rangle$, so the orbits form a cyclic group of order $K = 2^{J-3} \geq 2$, generated by the class of 3 (as $-3 \equiv 5 \pmod 8$), and $3^K \equiv \eta \pmod{2^J}$ because $3^{2^m} - 1$ is exactly divisible by 2^{m+2} for $m \geq 1$. For a base t let $O_k = O_{3^k t}$, with base $3^k t$. Then $O_0, \dots, O_{K-1}$ are distinct, and O_K is O_0 with the base $t\eta$, so $\Phi_{O_K} = -\Phi_{O_0}$.

Lemma 10.11. *Let O be an orbit with base t, and $O' = O_{3t}$ with base $3t$.*

(a) *$\Phi_O(a) \sim \Phi_O(a')$ for $a, a' \in G_3$.*

(b) *$I_O \sim (1/3)_* W_{O'}$, where $I_O = \int_{G_3} \Phi_O(a)\, da$ and $W_{O'} = \sum \varepsilon_x \varphi(3^i)^{-1}[\lambda_x]$, the sum over the tops x of O' of 3-level $i \geq 1$.*

(c) *Two tops of O with the same 3-level, the same signed box and the same label have the same sign.*

Proof. (a) is Lemma 10.9(a) with $s = t$ and $s' = t\eta$. (b) By Lemma 10.9(b), $F_3(t) \sim F_3(t\eta)$, and $\int (R(t,a) - R(t\eta, a))\, da = I_O$. In $R_{B_3}(3t, a) - R_{B_3}(3t\eta, a)$ only the tops of O' of 3-level at least 1 contribute, by (3) for O' and its base $3t$, and a signed box of 3-level i has measure $\varphi(3^i)^{-1}$. So $F_3(t) - F_3(t\eta) = I_O - (1/3)_* W_{O'}$. (c) Let x and x' be such tops with opposite signs. If $\sigma_x = \sigma_{x'}$, they have the same coordinates β and μ, and their coordinates t are t and $t\eta$, or $-t$ and $-t\eta$; then $x' = x + N/2$ by (C1), against Definition 10.2(ii). If $\sigma_x = -\sigma_{x'}$, then $\beta_{x'} = -\beta_x$, $\mu_{x'} = -\mu_x$, and the coordinates t are t and $-t$, or $t\eta$ and $-t\eta$; then $x' = -x$, against Definition 10.2(ii). □

For a top $y = [s, \alpha, \mu]_{J,1}$ put $y^* = [-s\eta, \alpha, -\mu]_{J,1}$, so that $y^{**} = y$. If y has the type (ε, σ), then y^* has the type $(\varepsilon, -\sigma)$, the same label and the opposite signed box; so a pair $\{y, y^*\} \subseteq B$ contributes the constant $\varepsilon[\lambda_y]$ to Φ_O. We say that B contains (C) if B contains y, y^* and $-3y = [-s, -\mu]_{J,0}$ for some top y of 3-level 1.

Lemma 10.12. *Let O be an orbit, $\lambda \in U$ and $m \geq 1$, and suppose that the coefficient of $[\lambda]$ in Φ_O is constant on every ball of 3-level $m-1$ in G_3. Then no top of O with label λ has 3-level at least $\max(m, 2)$. If $m = 1$, the tops of O of 3-level 1 with label λ form pairs $\{y, y^*\}$, counted with multiplicity.*

Proof. Let k be the largest 3-level of a top of O with label λ, and suppose that $k \geq 1$ and $k \geq m$. Let P be a ball of 3-level $k-1$. On P the tops of 3-level smaller than k contribute constants, so the tops of 3-level k and label λ contribute equally to the children of P, the balls of 3-level k in P; there are three of them if $k \geq 2$, and two, Q and $-Q$, if $k = 1$. For such a child Q' let $n_{Q'}$ be the sum of the signs of these tops with signed box Q'; it is $\pm$ their number by Lemma 10.11(c), and the $n_{Q'}$ are equal. Suppose they are nonzero, of sign ε. The tops of sign ε have the two types $(\varepsilon, \pm)$, and two tops of one type with label λ at two different children have the same coordinates t and μ and coordinates β that agree modulo 3^{k-1}, which Definition 10.2(iii) excludes. For $k \geq 2$ this is impossible with three children, so the $n_{Q'}$ vanish; then no top of 3-level k and label λ lies in P, for every P, against the choice of k. This proves the first claim. For $k = m = 1$, one type occurs at Q and only the other at $-Q$; the tops of type (ε, σ) at Q are copies of one element y, those of type $(\varepsilon, -\sigma)$ at $-Q$ are copies of y^*, and there are equally many. □

An orbit O is *flat* if Φ_O is constant. By Lemma 10.12 with $m = 1$, the tops of a flat orbit have 3-level 0 or 1, and those of 3-level 1 form pairs $\{y, y^*\}$. Let Z_O be the sum of $\varepsilon_x[\lambda_x]$ over the tops of O of 3-level 0, Y_O the sum of $\varepsilon_y[\lambda_y]$ over the pairs, and z_O, y_O the numbers of these

tops and pairs. Then $\Phi_O = I_O = Z_O + Y_O$ and $W_O = Y_O$. By Lemma 10.11(c), $\|Z_O\| = z_O$ and $\|Y_O\| = y_O$. If Z_O and Y_O have opposite signs at a label λ, then B contains (C): by (C1) and (C2) the two tops of 3-level 0 with label λ and sign opposite to that of a pair $\{y, y^*\}$ at λ are $-3y$ and $-3y^*$.

Chains.

Lemma 10.13. *In each of the following cases B has a local chain. In* (c), (d) *and* (e), *O is an orbit with base t.*

(a) *For a prime $q \parallel e'$, a line $L = wK_q$ is the disjoint union of two sets L_1 and L_4 of $(q-1)/2$ elements, and B contains the elements $[t, b, \lambda]_{j,i}$ for $\lambda \in L_1$ and $[-t\eta_j, -b, -\lambda]_{j,i}$ for $\lambda \in L_4$, where $j \geq 2$.*

(b) *B contains* (C).

(c) *For some label λ and some $i \geq 2$, B contains a top z_1 of O of 3-level $i-1$, sign $+$, signed box P and label λ, and two tops of O of 3-level i and label λ, of the types $(-,+)$ and $(-,-)$, whose signed boxes are two different balls contained in P.*

(d) *For some label μ^*, B contains a top s of O of 3-level 1 and sign $-$ and a top z_0 of O of 3-level 0 and sign $+$, both with label μ^*, and a top e_6 of $O_{t/3}$, with base $t/3$, of 3-level 0, sign $-$ and label $\mu^*/3$; or the same holds with all signs reversed.*

(e) *$5 \parallel e'$, $L = wK_5$ contains ν, and B consists of three tops of O of 3-level 0 and sign $+$ with the labels in $L - \{\nu\}$, a pair $\{y, y^*\}$ of tops of O of sign $+$ and label ν, and a top of $O_{t/3}$ of 3-level 0.*

Proof. In each case we list the moves and the final multiset $B^\flat$. One checks from Definition 10.2(ii),(iii) that an element added by a move is never the negative of an element that a later move uses, so the pairs removed between the moves, among them the copies of $N/2$, only lower $|B^\flat|$ by an even number.

(a) Let n be the order of these elements and $x = [t, b, \lambda_1]_{j,i}$ with $\lambda_1 \in L_1$. By (C4), $\Phi_q(x)$ consists of the $[t, b, \lambda]_{j,i}$ with $\lambda \in L$ and an element x_0 of core e'/q. The elements $[t, b, \lambda]_{j,i}$ with $\lambda \in L_4$ are not in B by Definition 10.2(iii), and x_0, $-qx$ are not in B as their cores differ from e'. So the move by $\sigma_{q,x}$ removes the elements with $\lambda \in L_1$ and adds $z_\lambda = [-t, -b, -\lambda]_{j,i}$ for $\lambda \in L_4$, together with $-x_0$ and qx. By (C1), $z_\lambda + N/2 = [-t\eta_j, -b, -\lambda]_{j,i}$ lies in B, and the move by $T(z_\lambda)$ removes z_λ and $z_\lambda + N/2$ and adds $2z_\lambda$ and $N/2$ (or removes $-2z_\lambda$ and adds $N/2$, if $-2z_\lambda$ is present). After these $1 + (q-1)/2$ moves, $(q-1)$ elements of order n have been removed, and at most $(q-1)/2$ elements of order $n/2$, two of order n/q and one copy of $N/2$ added. As $q \geq 5$ and $|B^\flat| \equiv |B| \pmod 2$, we get $|B^\flat| \leq |B|$, and if equality holds, B has more elements of order n than $B^\flat$ and as many of every larger order.

(b) Let $y = [s, \alpha, \mu]_{J,1}$ with $y, y^*, -3y \in B$, and $n = \operatorname{ord}(y)$. By (C3), $\Phi_3(y) = \{y, y', y_0\}$ with $y' = [s, -\alpha, \mu]_{J,1}$, which is not in B by Definition 10.2(iii), and $y_0 = [s/3, \mu/3]_{J,0}$. The move by $\sigma_{3,y}$ removes y and $-3y$, and y_0 if it lies in B, and adds $-y'$, and $-y_0$ if $y_0 \notin B$. By (C1), $-y' + N/2 = [-s\eta, \alpha, -\mu]_{J,1} = y^*$, so the move by $T(-y')$ removes $-y'$ and y^* and adds $-2y'$ and $N/2$. The result is $B - \{y, y^*, -3y\} + \{-y_0, -2y', N/2\}$, with the orders $n, n, n/3$ removed and $n/3, n/2, 2$ added, or $B - \{y, y^*, y_0, -3y\} + \{-2y', N/2\}$, with $|B| - 2$ elements.

(c) Let P be the ball of the $a \equiv p \pmod{3^{i-1}}$. The top of type $(-,+)$ is $[t\eta, b_1, \lambda]_{J,i}$, and by (C1), (C2) its image under -3 is $[-t\eta, -p, -\lambda]_{J,i-1}$, of type $(+,-)$, signed box P and label λ. The top of type $(-,-)$ is $[-t, -b_2, -\lambda]_{J,i}$, and its image under -3 is $[t, p, \lambda]_{J,i-1}$, of type $(+,+)$, signed box P and label λ. These are the two tops of 3-level $i-1$ with sign $+$, signed box P and label λ, so $z_1 = -3z_2$ for exactly one of the two tops of 3-level i, say z_2; let z_3 be the other one and $n = \operatorname{ord}(z_2)$. By (C3), $\Phi_3(z_2)$ consists of z_2 and two siblings, which are not in B by

Definition 10.2(iii), and one of them is $-(z_3 + N/2)$: for instance, if $z_2 = [-t, -b_2, -\lambda]_{J,i}$, then $z_3 + N/2 = [t, b_1, \lambda]_{J,i}$ by (C1), and $b_1 \equiv b_2 \pmod{3^{i-1}}$. The move by σ_{3,z_2} removes z_2 and z_1 and adds $z_3 + N/2$ and the negative z' of the other sibling; the move by $T(z_3)$ removes z_3 and $z_3 + N/2$ and adds $2z_3$ and $N/2$. The result is $B - \{z_1, z_2, z_3\} + \{z', 2z_3, N/2\}$, with the orders $n, n, n/3$ removed and $n, n/2, 2$ added.

(d) Let $s = [s_2, \gamma, \nu]_{J,1}$, $n = \mathrm{ord}(s)$, and $s_0 = [s_2/3, \nu/3]_{J,0}$. By (C1) and (C2), the two tops of O of 3-level 0 with sign + and label μ^* are $-3s$ and $3s + N/2$, and the two tops of $O_{t/3}$ of 3-level 0 with sign − and label $\mu^*/3$ are s_0 and $-s_0 + N/2$; we take $z_0 = -3s$ if $-3s \in B$ and $e_6 = s_0$ if $s_0 \in B$. By (C3), $\Phi_3(s) = \{s, s', s_0\}$ with $s' \notin B$. The move by $\sigma_{3,s}$ removes s, and z_0 if $z_0 = -3s$, and e_6 if $e_6 = s_0$; it adds $-s'$, and $3s$ if $z_0 \neq -3s$, and $-s_0$ if $e_6 \neq s_0$. In these two cases follow it by the move by $T(3s)$, which removes $3s$ and $z_0 = 3s + N/2$, and by the move by $T(-s_0)$, which removes $-s_0$ and $e_6 = -s_0 + N/2$; each adds an element of order $n/6$ and $N/2$. The result is $B - \{s, z_0, e_6\} + \{-s'\}$ plus one element of order $n/6$ for each move by T and the copies of $N/2$: it has $|B| - 2$ elements if no move by T is needed, and otherwise $|B|$ elements, with the orders $n/3, n/3$ removed and only smaller orders added, and as many elements of order n as B. Replacing t by $t\eta$ reverses all signs in O and in $O_{t/3}$, and the moves do not depend on the base.

(e) By (C1), the tops of 3-level 0, sign + and label λ are $[t, \lambda]_{J,0}$ and $[-t\eta, -\lambda]_{J,0}$, and by Definition 10.2(iii) and (C4) at most two of the three tops of 3-level 0 are of each of these forms. Suppose two of them are $[t, \lambda]_{J,0}$ for $\lambda \in \{\lambda_1, \lambda_2\}$, the third being $[-t\eta, -\lambda_3]_{J,0}$, so that $L = \{\lambda_1, \lambda_2, \lambda_3, \nu\}$, and let $y = [t, \alpha, \nu]_{J,1}$ be the element of type $(+, +)$ of the pair. The move by $\sigma_{5,x}$ with $x = [t, \lambda_1]_{J,0}$ removes the two tops $[t, \lambda_1]_{J,0}$, $[t, \lambda_2]_{J,0}$ and adds $[-t, -\lambda_3]_{J,0}$, $[-t, -\nu]_{J,0}$ and two elements of core $e'/5$. The move by $T([-t, -\lambda_3]_{J,0})$ removes $[-t, -\lambda_3]_{J,0}$ and $[-t\eta, -\lambda_3]_{J,0}$ and adds an element of order $n_0/2$, $n_0 = \mathrm{ord}(x)$, and $N/2$. Now $[-t, -\nu]_{J,0} = -3y$, so the multiset contains (C), and the two moves of (b) follow. If the top of $O_{t/3}$ is y_0, the result has $4 = |B| - 2$ elements. Otherwise it has six, the two elements of order $3n_0$ of B are removed, and the added elements have orders at most $3n_0/2$. If two of the three tops have the form $[-t\eta, -\lambda]_{J,0}$, the same moves with $x = [-t\eta, -\lambda_1]_{J,0}$ and with y^* in place of y apply, as $-3y^* = [t\eta, \nu]_{J,0}$. □

Proposition 10.14. *If some element of B has 2-level at least 4, then B has a local chain.*

Proof. Suppose that B has no local chain, and let $J \geq 4$ be the largest 2-level. Every orbit has at most six tops.

Step 1: if there are no short lines, every orbit is flat. For an orbit O and $a, a' \in G_3$, the function $\Phi_O(a) - \Phi_O(a')$ is split by Lemma 10.11(a), has integer coefficients and mass at most the number of tops of O, at most 6; so it vanishes by Lemma 10.10(b).

Step 2: a non-flat orbit O, with base t. By Lemma 10.11(a) and Lemma 10.10(b), every nonzero difference $\Phi_O(a) - \Phi_O(a')$ is $\pm\mathbf{1}_L$ for a short line L. So O has at least four tops, and the other orbits are flat. If Φ_O took three values, two nonzero differences $\pm\mathbf{1}_{L_1}$ and $\pm\mathbf{1}_{L_2}$ would add up to a third, $\pm\mathbf{1}_{L_3}$; but for $L_1 = L_2$ the sum is 0 or has coefficients ± 2, and for $L_1 \neq L_2$, by Lemma 10.10(c), it has a coefficient ± 2, or coefficients of both signs, or at least eight nonzero coefficients. So $\Phi_O(a) = \varphi_0 + \mathbf{1}_V(a)\mathbf{1}_L$ for one short line L, a proper nonempty subset V of G_3 and $\varphi_0 \in \mathbb{Z}[U]$.

Every $\lambda \in L$ carries a top of O of 3-level at least 1, as the coefficient of $[\lambda]$ is not constant. Since $|L| \geq 4$ and there are at most six tops, at least $2|L| - 6 \geq 2$ labels of L carry exactly one top x, and there $\varepsilon_x \mathbf{1}_{P_x} = \varphi_0(\lambda) + \mathbf{1}_V$; so V is P_x or its complement. Replacing t by $t\eta$ replaces V by its complement, so we may assume that V is a ball of some 3-level $i \geq 1$. By Lemma 10.12, applied with $m = i + 1$ at each label, no top of O has 3-level larger than i. A

top of 3-level i, sign $+$ and signed box V is an S^+ *element*; for $i = 1$, a top of 3-level 1, sign $-$ and signed box $-V$ is an S^- *element*.

The case $i = 1$. At $\lambda \in L$ let n_V and n_{-V} be the sums of the signs of the tops of 3-level 1 with signed boxes V and $-V$. They have one sign each by Lemma 10.11(c), and $n_V - n_{-V} = 1$. So either $n_V \geq 1$, and λ carries S^+ elements and no S^- element, or $n_{-V} \leq -1$, and the reverse. In the first case the types at V and at $-V$ are different, by Definition 10.2(iii); since $-V$ carries tops when $n_V \geq 2$, the tops at V are then copies of one element y and those at $-V$ copies of y^*. So λ carries one S^+ element and $n_V - 1$ pairs $\{y, y^*\}$ of sign $+$; likewise in the second case, with sign $-$.

The case $i \geq 2$. A label of L with a single top carries an S^+ element, since the complement of a ball of 3-level at least 2 is not a ball. Suppose that $\lambda_4 \in L$ carries no S^+ element. Let P be the parent of V, of 3-level $i-1$, and V', V'' its other children. At λ_4 the tops of 3-level smaller than i contribute equally to V, V' and V'', so the sums of signs at 3-level i satisfy $n_V - n_{V'} = n_V - n_{V''} = 1$, and $n_V \leq 0$. So V' and V'' carry tops of sign $-$, of different types by Definition 10.2(iii), and then V carries no top of sign $-$: $n_V = 0$ and $n_{V'} = n_{V''} = -1$. In every other ball of 3-level $i-1$ the tops of 3-level i at λ_4 vanish as in the proof of Lemma 10.12. Hence the tops of 3-level smaller than i at λ_4 contribute $c + \mathbf{1}_P$ for a constant c, and there is at least one of them: λ_4 carries at least three tops and each of the other $|L| - 1 \geq 3$ labels at least one. So $|L| = 4$, the other three labels carry one S^+ element each, and λ_4 carries exactly one top z_1 of 3-level smaller than i, with $\varepsilon_{z_1}\mathbf{1}_{P_{z_1}} = c + \mathbf{1}_P$. If $\varepsilon_{z_1} = +$, then $P_{z_1} = P$, and this is Lemma 10.13(c). Otherwise the complement of P is the ball P_{z_1}, so $i = 2$, z_1 has 3-level 1, and $c = -1$; then the six tops all lie in O, $\varphi_0 = -[\lambda_4]$, and $I_O = \varphi_0 + \frac{1}{6}\mathbf{1}_L \sim -[\lambda_4]$ by Lemma 10.10(a). The orbit $O_{3t} \neq O$ has no top, so Lemma 10.11(b) gives $[\lambda_4] \sim 0$, against Lemma 10.10(b).

Every label of L carries an S^+ element, or, for $i = 1$, every label carries an S^- element. In the second case replace t by $t\eta$, which turns the S^- elements for V into S^+ elements for $-V$. Let b be the centre of V. The S^+ elements of the types $(+,+)$ and $(+,-)$ are $[t, b, \lambda]_{J,i}$ and $[-t\eta, -b, -\lambda]_{J,i}$. If L is a line along K_q, then by (C4) and Definition 10.2(iii) at most $(q-1)/2$ labels of L carry S^+ elements of each type, so $|L| \leq q - 1$; hence $|L| = q - 1$, $q \parallel e'$, and the two types occupy disjoint sets L_1 and L_4 of $(q-1)/2$ labels. This is Lemma 10.13(a) with $j = J$.

The mixed case: $i = 1$, and $L = L_+ \sqcup L_-$ with nonempty sets $L_\pm$ of labels that carry $S^\pm$ elements, of sizes $\ell_\pm$. At $a \in -V$ the S^+ elements contribute 0 and the S^- elements -1, so $\varphi_0 = -\mathbf{1}_{L_-} + Z + Y$, where Z is the sum of $\varepsilon_x[\lambda_x]$ over the tops of O of 3-level 0 and Y the sum of $\varepsilon_y[\lambda_y]$ over the pairs $\{y, y^*\}$ of O. Also $I_O = \varphi_0 + \frac{1}{2}\mathbf{1}_L$ and $W_O = Y - \mathbf{1}_{L_-} + \frac{1}{2}\mathbf{1}_L$. The orbits O_{3t} and $O_{t/3}$ are flat, and Lemma 10.11(b) for O and for $O_{t/3}$ (with base $t/3$), with Lemma 10.10(a), gives

$$G = -\mathbf{1}_{L_-} + Z + Y - (1/3)_* Y_{O_{3t}} \sim 0, \tag{G1}$$

$$E = \Phi_{O_{t/3}} - (1/3)_* Y \sim -(1/3)_* \mathbf{1}_{L_-}. \tag{G2}$$

Both G and E have integer coefficients.

If $|L| = 6$, the six $S^\pm$ elements are all of B, so $E = 0$ and $(1/3)_*\mathbf{1}_{L_-}$ is split with at most five nonzero coefficients. By Lemma 10.10(b) it is the indicator of a line along K_5, which meets the line $(1/3)L$ along K_7 in at most one point by Lemma 10.10(c): impossible. If $|L| = 5$, then $25 \mid e'$, there are no short lines of size four, and B has at most one element w besides the five $S^\pm$ elements. If w is a top of $O_{t/3}$ of 3-level 0, then $Z = Y = Y_{O_{3t}} = 0$ and (G1) says $\mathbf{1}_{L_-} \sim 0$ with $1 \leq \ell_- \leq 4$; otherwise $E = 0$ and (G2) says $(1/3)_*\mathbf{1}_{L_-} \sim 0$. Both contradict Lemma 10.10(b).

Let $|L| = 4$, so $5 \parallel e'$. Besides the four $S^\pm$ elements, B has at most two elements, and they make up E, so $\|E\| \leq 2$. The split function $E + (1/3)_*\mathbf{1}_{L_-}$ has integer coefficients

and mass at most 5, so by Lemma 10.10(b) it is 0 or $\pm\mathbf{1}_{L'}$ for a line L' of size 4. For $L' \neq (1/3)L$, Lemma 10.10(c) gives $\|E\| \geq 3$; for $L' = (1/3)L$ the sign $-$ gives $\|E\| \geq 4$. Hence $E = -(1/3)_*\mathbf{1}_{L_-}$ with $\ell_- \leq 2$, or $E = (1/3)_*\mathbf{1}_{L_+}$ with $\ell_+ \leq 2$, and in particular $E \neq 0$.

- $\ell_+ = \ell_- = 2$. Then $\|E\| = 2$; a pair contributes 1 to $\|E\|$ with two elements, so both further elements are tops of $O_{t/3}$ of 3-level 0. Then $Z = Y = 0$, and O_{3t} has no pair (for $K = 2$ it is $O_{t/3}$), so (G1) says $\mathbf{1}_{L_-} \sim 0$: impossible.
- $\ell_- = 1$, $L_- = \{\mu^*\}$. Then $E = -[\mu^*/3]$. A pair of O contributes $-(1/3)_*(\varepsilon_y[\lambda_y])$ to E, which would need sign $+$ at μ^*, whereas the pairs at μ^* have the sign $-$ of the S^- element there. A pair of $O_{t/3}$ uses both further elements; then $Z = Y = 0$ and G is $-[\mu^*]$, or, for $K = 2$, where O_{3t} is $O_{t/3}$ with the base $(t/3)\eta$, $-[\mu^*] - [\mu^*/9]$: a nonzero split function of mass at most 2. So one further element e_6 is a top of $O_{t/3}$ of 3-level 0, sign $-$ and label $\mu^*/3$, and G consists of $-[\mu^*]$ and the contribution of the other further element; as $G \sim 0$ has mass at most 2, it vanishes, and that element is a top of O of 3-level 0, sign $+$ and label μ^*. With the S^- element at μ^* this is Lemma 10.13(d).
- $\ell_+ = 1$. Replacing t by $t\eta$ exchanges the S^+ and S^- elements and reverses all signs, and the previous case gives Lemma 10.13(d) with all signs reversed.

Step 3: all orbits flat. By Lemma 10.13(b), B does not contain (C). Fix a base t of an orbit with a top, and write Z_k, Y_k, z_k, y_k for O_k, $0 \leq k < K$, and $Y_K = -Y_0$. By Lemma 10.11(b),

$$E_k = Z_k + Y_k - (1/3)_*Y_{k+1} \sim 0,$$

with integer coefficients. Without (C), $\|Z_k + Y_k\| = z_k + y_k$, so $\sum_k \|E_k\| \leq \sum_k (z_k + 2y_k) \leq 6$.

If every E_k vanishes, then $z_k + y_k = y_{k+1}$ for all k, with $y_K = y_0$; summing over k gives $\sum z_k = 0$, so all y_k are equal to some $p \geq 1$, and $2Kp \leq 6$ forces $K = 2$ and $p = 1$. Then $Y_0 = (1/3)_*Y_1$ and $Y_1 = (1/3)_*Y_2 = -(1/3)_*Y_0$, so $Y_0 = -(1/9)_*Y_0$ for $Y_0 = \pm[\nu]$, which is impossible. If there are no short lines, every nonzero E_k has mass at least 7; so this case shows that B contains (C), a contradiction.

So there are short lines, and as each nonzero E_k has mass at least 4, exactly one of them is nonzero; choose t so that it is E_0. For $1 \leq k < K$, $z_k + y_k = y_{k+1}$, so $y_1 \leq y_2 \leq \cdots \leq y_K = y_0$ and $\sum_{1 \leq k < K} z_k = y_0 - y_1$. With $4 \leq \|E_0\| \leq z_0 + y_0 + y_1$ and

$$\sum_k (z_k + 2y_k) = z_0 + 3y_0 + y_1 + 2\sum_{2 \leq k < K} y_k \leq 6$$

this gives $y_0 + (K-2)y_1 \leq 1$.

- $y_0 = 0$. All y_k, and all z_k with $k \geq 1$, vanish, and $E_0 = Z_0$ is $\pm\mathbf{1}_L$ for a short line L. After replacing t by $t\eta$, $Z_0 = \mathbf{1}_L$, so L is covered by the labels of tops $[t, \lambda]_{J,0}$ and $[-t\eta, -\lambda]_{J,0}$, one at each label; as in Step 2, (C4) and Definition 10.2(iii) give Lemma 10.13(a).
- $y_0 = 1$ and $y_1 = 0$. Then $y_k = 0$ for $1 \leq k < K$, $z_{K-1} = 1$ and $z_k = 0$ for $1 \leq k < K-1$. So $E_0 = Z_0 + Y_0$ with $4 \leq \|E_0\| \leq z_0 + 1$ and $z_0 + 3 \leq 6$: $z_0 = 3$, and $E_0 = \pm\mathbf{1}_L$ for a line L of size four, so $5 \parallel e'$. After replacing t by $t\eta$, $E_0 = \mathbf{1}_L$, and B consists of three tops of O_0 of 3-level 0 and sign $+$, a pair of O_0 of sign $+$ at the fourth label of L, and one top of 3-level 0 of $O_{K-1} = O_{t/3}$. This is Lemma 10.13(e).
- $K = 2$ and $y_0 = y_1 = 1$. Then $z_1 = 0$, $z_0 = 2$, and B consists of two tops of O_0 of 3-level 0 and the four elements of a pair of O_0 and a pair of O_1. Let $y = [t_y, \alpha, \mu_y]_{J,1}$ lie in the pair of O_0, fix $b \in G_3$, and put $\Delta(s) = R(s, b) - R(s, -b)$. By Lemma 10.9(a), $\Delta(s) \sim \Delta(s')$. Only the four elements of the two pairs contribute to Δ, and only y among them has its coordinate t equal to $\pm t_y$. So $\Delta(t_y) - \Delta(-t_y) = d([\mu_y] - [-\mu_y])$

with $d = \mathbf{1}_\alpha(b) - \mathbf{1}_\alpha(-b) = \pm 1$, where $\mathbf{1}_\alpha$ is the indicator of $a \equiv \alpha \pmod 3$: a nonzero split function of mass 2, against Lemma 10.10(b).

In every case B has a local chain, which proves the proposition. □

Elements of large 3-level. From here to the end of the proof of Proposition 10.19, every element of B has 2-level at most 3, and $I \geq 3$ is the largest 3-level of an element of B, so $w \geq I$. When N is odd every element has 2-level 0, and what follows about elements of positive 2-level is void. Put $\kappa = 1 + 3^{I-1}$, so that $\kappa^3 \equiv 1 \pmod{3^I}$. A *top* is now an element of B of 3-level I. The function $R(s,a)$ depends on a only modulo 3^I, and a box of 3-level smaller than I contains (s,a), $(s,a\kappa)$ and $(s,a\kappa^2)$ together.

The cosets of $\{\pm 1, \pm\kappa, \pm\kappa^2\}$ in $(\mathbb{Z}/3^I)^\times$ form a cyclic group of order $K = 3^{I-2} \geq 3$, generated by the class of 2, and $2^K \equiv -\kappa^2 \pmod{3^I}$, since $2 = -(1-3)$ and $(1-3)^{3^{I-2}} \equiv 1 - 3^{I-1} \equiv \kappa^2 \pmod{3^I}$. Fix a unit b modulo 3^I, put $b_k = 2^k b$, and call $\Omega_k = \{\pm b_k \kappa^r\}$, with the *base* b_k, the 3-*orbits*; then $\Omega_{k+K} = \Omega_k$ and $b_{k+K} = -\kappa^{-1} b_k$. A top $x = [t, \beta, \mu]_{j,I}$ with $\beta \in \Omega_k$ is a *top of* Ω_k. Relative to the base b_k it has a *position* $r \in \mathbb{Z}/3$ and a *sign* σ with $\beta = \sigma b_k \kappa^r$, the *label* $\lambda_x = \sigma\mu$ and the *signed box* $C_x = \{s \in G_2 : s \equiv \sigma t \ (2^j)\}$, which is G_2 when $j \leq 1$. Relative to the base b_{k+K} the same top has the position $r+1$, the sign $-\sigma$, the label $-\lambda_x$ and the signed box $-C_x$; replacing b by $-b$ keeps the positions and replaces the signs, labels and signed boxes by their negatives. Put $T_k(r,s) = R(s, b_k\kappa^r)$. As in (3),

$$T_k(r,s) = h_k(s) + \sum_x \sigma_x \, \mathbf{1}_{C_x}(s) \, [\lambda_x], \tag{4}$$

the sum over the tops of Ω_k at the position r, where $h_k(s)$ collects the elements of 3-level smaller than I and does not depend on r.

Lemma 10.15. (a) $T_k(r,s) - T_k(r,s') - T_k(r',s) + T_k(r',s') \sim 0$.

(b) $G_k(r) \sim G_k(r')$, *where*

$$G_k(r) = \sum_x \frac{\sigma_x}{\varphi(2^{j_x})}[\lambda_x] - (1/2)_* \sum_y \frac{\sigma_y}{\varphi(2^{j_y})}[\lambda_y],$$

the sums over the tops x of Ω_k at r and over the tops y of Ω_{k+1} of 2-level at least 1 at r, with the data of y relative to b_{k+1}.

(c) *Let x and x' be tops of one 3-orbit, with data relative to one base.*
- (i) *If they have the same sign, 2-level, signed box and label, they have the same position.*
- (ii) *If they have the same position, 2-level, signed box and label, they have the same sign.*
- (iii) *If they have the same position, sign, 2-level $j \geq 2$ and label, their signed boxes are not C and $C\eta_j$.*
- (iv) *If $q \mid e'$ is prime and the tops of Ω_k of one position, sign, 2-level and signed box have labels in one coset of K_q, they have at most $(q-1)/2$ different labels.*

(d) *A top of Ω_k of 2-level 0 at (r, σ, λ) and a top of Ω_{k+1} of 2-level 1 at $(r, \sigma, 2\lambda)$, relative to b_{k+1}, do not both lie in B.*

Proof. (a) is Lemma 10.9(a) at $a = b_k\kappa^r$ and $a' = b_k\kappa^{r'}$. (b) By Lemma 10.9(c), $F_2(b_k\kappa^r) \sim F_2(b_k\kappa^{r'})$. In $\int R(s, b_k\kappa^r)\, ds$ a top x of Ω_k at r contributes $\sigma_x \varphi(2^{j_x})^{-1}[\lambda_x]$ by (4), and the elements of 3-level smaller than I contribute the same for every r. In $R_{B_2}(s, 2b_k\kappa^r) = R_{B_2}(s, b_{k+1}\kappa^r)$ the tops of Ω_{k+1} at r relative to b_{k+1} contribute in the same way. Subtracting gives (b). (c) In (i) the two tops have the same coordinates t and μ, and coordinates β that agree modulo 3^{I-1}; by (C3) and Definition 10.2(iii) they are equal. In (ii), opposite signs would give $x' = -x$. In (iii), the coordinates t would be t and $t\eta_j$, so $x' = x + N/2$ by (C1). (iv)

is (C4) with Definition 10.2(iii). (d) By (C1), a top $z = [\beta, \mu]_{0,I}$ has $z + N/2 = [1, 2\beta, 2\mu]_{1,I}$, whose data relative to b_{k+1} are $(r, \sigma, 2\lambda)$. □

Lemma 10.16. *Let $\lambda \in U$, let Y be a multiset of tops of Ω_k with label λ, let $j \geq 2$ be the largest 2-level in Y, and let $F(r, s)$ be the coefficient of $[\lambda]$ in $\sum_{x \in Y} \sigma_x [r = r_x] \mathbf{1}_{C_x}(s)$. If P is a ball of 2-level $j - 1$ such that $F(r, s) - F(r, s')$ does not depend on r for $s, s' \in P$, then no element of Y of 2-level j has its signed box in P.*

Proof. Let C_1 and $C_2 = C_1\eta_j$ be the two balls of 2-level j in P, and $s_i \in C_i$. The elements of Y of smaller 2-level contribute equally at s_1 and s_2. So, if $n_r(C)$ is the sum of the signs of the elements of Y of 2-level j at the position r with signed box C, then $n_r(C_1) - n_r(C_2) = \delta$ for every r, with δ independent of r. By Lemma 10.15(c)(i), each of the four kinds of such elements, with signed box C_1 or C_2 and sign $+$ or $-$, occupies at most one position. The positive contributions to δ come from two kinds, so from at most two of the three positions, and $\delta > 0$ is impossible; so is $\delta < 0$. Hence $n_r(C_1) = n_r(C_2)$ for every r. By Lemma 10.15(c)(ii) each $n_r(C_i)$ is $\pm$ a number of elements, and two equal nonzero values would put elements of one sign with the signed boxes C_1 and $C_1\eta_j$ at one position, against Lemma 10.15(c)(iii). So $n_r(C_1) = n_r(C_2) = 0$, and by Lemma 10.15(c)(ii) no element of 2-level j has its signed box in P. □

A 3-orbit Ω_k is *flat* if every coefficient of T_k is a sum $g(r) + h(s)$. Tops of 2-level at most 1 contribute functions of r alone to T_k; conversely, by Lemma 10.16, applied at every label to all the tops there and to every ball, a flat 3-orbit has no top of 2-level at least 2.

Lemma 10.17. *Let $\lambda \in U$, $m \in \{2, 3\}$, V a ball of 2-level m, P the ball of 2-level $m - 1$ containing it and $V' = V\eta_m$ the other ball of 2-level m in P. Let Y be a multiset of tops of Ω_k with label λ and 2-level at most m, and suppose that the coefficient of $[\lambda]$ in $\sum_{x \in Y} \sigma_x [r = r_x] \mathbf{1}_{C_x}(s)$ is $A(r, s) + [r = r^*]\mathbf{1}_V(s)$ with A additive. Then the elements of Y of 2-level m are*

- *one top at $(r^*, +)$ with signed box V* (type S^+)*, or*
- *one top at $(r^*, -)$ with signed box V'* (type S^-)*, or*
- *two tops, at $(r_1, -)$ with signed box V and at $(r_2, +)$ with signed box V', where $\{r^*, r_1, r_2\} = \mathbb{Z}/3$* (type D)*;*

and the other elements of Y contribute, up to an additive function, 0, $[r = r^*]\mathbf{1}_P$ or $-[r = r_2]\mathbf{1}_P$ in the three cases.

Proof. On a ball of 2-level $m - 1$ other than P the function $\mathbf{1}_V$ vanishes, and Lemma 10.16 leaves no element of 2-level m there. Inside P, with $s_V \in V$, $s_{V'} \in V'$ and n_r as in the proof of Lemma 10.16,

$$n_r(V) - n_r(V') = \delta + [r = r^*], \qquad \delta = A(r, s_V) - A(r, s_{V'}),$$

and δ does not depend on r. The kinds $(V, +)$ and $(V', -)$ give the positive contributions and $(V, -)$, $(V', +)$ the negative ones, each kind at one position, so $\delta \geq 1$ and $\delta \leq -2$ are impossible. If $\delta = 0$, the counts at $r \neq r^*$ vanish as in the proof of Lemma 10.16, and at r^* we have $n(V) - n(V') = 1$ where the two counts are not nonzero of one sign, by Lemma 10.15(c)(iii); so $n(V) = 1$ and $n(V') = 0$, or $n(V) = 0$ and $n(V') = -1$, and by Lemma 10.15(c)(ii) these counts are numbers of tops: types S^+ and S^-. If $\delta = -1$, both positions other than r^* need a negative contribution, so the kinds $(V, -)$ and $(V', +)$ sit at one of them each. At the position of $(V, -)$, Lemma 10.15(c)(ii) excludes $(V, +)$ and (c)(iii) excludes $(V', -)$, so the count -1 there is one top; likewise at the position of $(V', +)$. The kinds $(V, +)$ and $(V', -)$ can then sit only at r^*, where $n(V) - n(V') = 0$ excludes them: type D. Subtracting the part of 2-level m from $[r = r^*]\mathbf{1}_V$ leaves 0, $[r = r^*](\mathbf{1}_V + \mathbf{1}_{V'})$ and $[r = r^*]\mathbf{1}_V + [r = r_1]\mathbf{1}_V - [r = r_2]\mathbf{1}_{V'} = \mathbf{1}_V(s) - [r = r_2]\mathbf{1}_P(s)$. □

Lemma 10.18. *In each of the following cases B has a local chain.*

(a) *B contains an element x of 2-level $j \geq 1$ and $-2x$.*
(b) *$5 \parallel e'$, x_1 has 2-level 1 and order n, the elements of order n in $\Phi_5(x_1)$ are x_1, x_2, y_1, y_2, and $B = \{x_1, x_2, 2y_1, 2y_2, N/2 - y_1, N/2 - y_2\}$.*

Proof. (a) The move by $T(x) = \{x, x + N/2, -2x, N/2\}$ removes x and $-2x$ and adds $-x + N/2$ and $N/2$, as $x + N/2 \notin B$. If $j \geq 2$, then $-x + N/2$ has the order n of x, and the orders $n, n/2$ are replaced by $n, 2$; if $j = 1$, it has order $n/2$, and $n, n/2$ are replaced by $n/2, 2$. Either way the result has $|B|$ elements and fewer of the order $n/2$ or n.

(b) By (C4), $\Phi_5(x_1)$ consists of x_1, x_2, y_1, y_2 and an element x_0 of core $e'/5$. Of these and $-5x_1$, only x_1 and x_2 lie in B, so the move by σ_{5,x_1} removes x_1, x_2 and adds $-y_1$, $-y_2$, $-x_0$, $5x_1$. The move by $T(-y_1) = \{-y_1, N/2 - y_1, 2y_1, N/2\}$ removes three elements and adds $N/2$. The result $\{-y_2, 2y_2, N/2 - y_2, -x_0, 5x_1, N/2\}$ has $|B|$ elements; the orders $n, n/2, n/2$ of B are replaced by $n/5, n/5, 2$. □

Proposition 10.19. *If every element of B has 2-level at most 3 and some element has 3-level at least 3, then B has a local chain.*

Proof. Suppose that B has no local chain. By Lemma 10.18(a), B contains no element x of positive 2-level together with $-2x$. Note that if x is a top of 2-level 1 at (r, σ, λ), then $-2x$ is the top of 2-level 0 of the same 3-orbit at $(r, -\sigma, \lambda)$, by (C2).

Step 1: every 3-orbit is flat. Suppose that Ω_0 is not, and let $F_\lambda(r, s)$ be the coefficient of $[\lambda]$ in $T_0(r, s)$. By Lemma 10.15(a) and Lemma 10.10(b), each nonzero second difference of T_0 is $\pm\mathbf{1}_L$ for a short line L, as its mass is at most the number of tops of Ω_0. The labels at which F_λ is not additive form a union of such lines, and each of them carries a top of Ω_0 of 2-level at least 2. Two different short lines have at least seven points together by Lemma 10.10(c), so these labels form one short line L, every nonzero second difference is $\pm\mathbf{1}_L$, and $F_\lambda - F_{\lambda'}$ is additive for $\lambda, \lambda' \in L$. At least $2|L| - 6 \geq 2$ labels of L carry exactly one top of Ω_0, of 2-level at least 2, and there $F_\lambda = h(s) + \sigma_x[r = r_x]\mathbf{1}_{C_x}(s)$. For two such labels with different positions, the additive difference vanishes at the third position, so it depends on r alone, and $\mathbf{1}_{C_x}$ would be constant. So these tops share a position r^*, and $\sigma_x\mathbf{1}_{C_x} - \sigma_{x'}\mathbf{1}_{C_{x'}}$ is constant: either $(\sigma_{x'}, C_{x'}) = (\sigma_x, C_x)$, or $\sigma_{x'} = -\sigma_x$ and C_x, $C_{x'}$ are the two balls of 2-level 2. After replacing b by $-b$ if necessary,

$$F_\lambda(r, s) = A_\lambda(r, s) + [r = r^*]\mathbf{1}_V(s) \quad (\lambda \in L),$$

with A_λ additive and V a ball of some 2-level $m \in \{2, 3\}$, and F_λ is additive for $\lambda \notin L$. By Lemma 10.16 the tops of Ω_0 have 2-level at most 1 at the labels outside L, and at most m at the labels of L, since $\mathbf{1}_V$ is constant on every ball of 2-level at least m.

Let P and V' be as in Lemma 10.17, which applies at every $\lambda \in L$ to all the tops of Ω_0 there. Let a, c and d be the numbers of labels of L of the types S^+, S^- and D, so that $a + c + d = |L|$, and let $L = wK_q$. With t a residue modulo 2^m defining V and $b' = b\kappa^{r^*}$, the tops of type S^+ are the $[t, b', \lambda]_{m,I}$ and those of type S^- the $[-t\eta_m, -b', -\lambda]_{m,I}$; by Lemma 10.15(c)(iv), a and c are at most $(q-1)/2$. If $m = 3$, then P is a ball of 2-level 2, and at a label of type S^- or D the tops of 2-level at most 2 contribute $[r = r']\mathbf{1}_W$ up to an additive function, with $W = P$, $r' = r^*$, or $W = G_2 - P$, $r' = r_2$; so Lemma 10.17 with $m = 2$ gives one more top there. A top of 2-level 2 enters G_0 with the weight $1/2$. Let p and p' be the positions other than r^*; the tops of the types $S^\pm$ lie at r^*, and for $m = 2$ a label of type D contributes $\mp\frac{1}{2}$ times its point to $G_0(p)$ and $\pm\frac{1}{2}$ times it to $G_0(p')$.

- $m = 2$, $|L| = 6$, so $7 \parallel e'$: $a + c + 2d \leq 6$ forces $d = 0$ and $a = c = 3$, and Lemma 10.13(a) applies.

- $m = 2$, $|L| = 5$, so $25 \mid e'$: $a, c \leq 2$ force $d = 1$, $a = c = 2$, and these six tops are all of B. Then $G_0(p) - G_0(p')$ is $\pm$ one point, against Lemma 10.15(b) and Lemma 10.10(b).
- $m = 2$, $|L| = 4$, so $5 \parallel e'$ and $a, c \leq 2$. For $d = 0$, Lemma 10.13(a) applies. For $d = 2$ the six tops are all of B, and $G_0(p) - G_0(p')$ is a sum of two points with signs, again impossible. For $d = 1$, B has at most one element w besides the five tops in L, and $2(G_0(p) - G_0(p'))$ is split, with integer coefficients, mass at most 4 and at most two nonzero coefficients, so it vanishes, and $G_0(p) = G_0(p') = \pm\frac{1}{2}[\lambda_D]$ for the label λ_D of type D. Then $2(G_0(r^*) - G_0(p))$ is split, with integer coefficients, and has the four points of L with both signs, against Lemma 10.10(b).
- $m = 3$: a label of type S^- carries at least two tops and one of type D at least three, so $a + 2c + 3d \leq 6$. With $a, c \leq (q-1)/2$ this leaves only $|L| = 4$, $d = 0$ and $a = c = 2$, and Lemma 10.13(a) with $j = 3$ applies.

So every 3-orbit is flat, and every top has 2-level at most 1.

Step 2: cells. For $k \in \mathbb{Z}/K$ and $\lambda \in U$, the *cell* (k, λ) consists of the tops of Ω_k with label λ and of its *upper tops*, the tops of Ω_{k+1} of 2-level 1 with label 2λ relative to b_{k+1}. Let $v(r) = v_0(r) + v_1(r)$ and $u(r)$ be the sums of the signs of the tops of Ω_k of 2-level 0 and 1 in the cell, and of its upper tops, at the position r. By Lemma 10.15 the coefficient of $[\lambda]$ in $G_k(r)$ is $v(r) - u(r)$, and:

- each of the kinds *top of* 2*-level* 0 *of a given sign*, *top of* 2*-level* 1 *of a given sign*, *upper top of a given sign* occupies at most one position;
- at each position the tops of Ω_k of the cell have one sign, by (c)(ii) and the absence of x with $-2x$, and so have the upper tops;
- a top of 2-level 0 and an upper top of the same sign do not share a position, by (d).

The upper tops of (k, λ) are the tops of 2-level 1 of the cell $\pi(k, \lambda) = (k + 1, 2\lambda)$, with labels relative to b_{k+1}; by the change of base, $\pi(K - 1, \lambda) = (0, -2\lambda)$. Let n_1 be the number of tops of 2-level 1 of a cell.

Constant cells. Suppose $v - u = c$ is constant. If $c = 0$, the cell has no top of 2-level 0, since such a top of sign σ at r makes $v(r)$, hence $u(r)$, of sign σ; so $v_1 = u$. If $c \neq 0$, reverse the signs in the cell to get $c > 0$. Every position needs a positive contribution, from the three kinds: tops of 2-level 0 of sign $+$, tops of 2-level 1 of sign $+$, and upper tops of sign $-$; so they sit at three different positions p_0, p_1, q. Upper tops of sign $+$ can only sit where $v > 0$, and not at p_0, so at p_1; tops of Ω_k of sign $-$ can only sit where $u < 0$, at q, and have 2-level 1. So the cell has exactly c tops of 2-level 0, all at p_0, and at p_1 and q the numbers a_1, b_1 of tops of 2-level 1 of signs $\pm$ and a_u, b_u of upper tops satisfy $a_1 - a_u = c = b_u - b_1$. In both cases $n_1(C) = n_1(\pi(C))$; a constant cell without upper tops, or without tops of 2-level 1, is empty; and in a constant cell with one top of 2-level 1 and one upper top, the upper top has the position and the sign of that top and there is no top of 2-level 0 ($c = 0$), or it has the opposite sign and another position and there is exactly one top of 2-level 0 ($c \neq 0$).

Defective cells. The function G_k has integer coefficients, and $\|G_k(r) - G_k(r')\|$ is at most the number of tops of Ω_k and of tops of 2-level 1 of Ω_{k+1}. As in Step 2 of the proof of Proposition 10.14, Lemma 10.15(b) and Lemma 10.10 give

$$G_k(r) = g_k + \varepsilon_k[r = \rho_k]\mathbf{1}_{L_k}, \qquad \varepsilon_k \in \{0, \pm 1\},$$

with a short line L_k when $\varepsilon_k \neq 0$. Let $\mathcal{D}$ be the set of k with $\varepsilon_k \neq 0$ and $\ell_k = |L_k| \geq 4$. The cells (k, λ) with $k \in \mathcal{D}$ and $\lambda \in L_k$ are *defective*, with $v - u = c + \varepsilon_k[r = \rho_k]$; all other cells are constant. The same counting as for constant cells, with ε_k in place of $+1$, shows:

- a defective cell with one top is of type α (a top of Ω_k at (ρ_k, ε_k)) or β (an upper top at $(\rho_k, -\varepsilon_k)$);

- a defective cell without upper tops is of type α, or of type A: a top of 2-level 0 and a top of 2-level 1, both of sign $-\varepsilon_k$, at the two positions other than ρ_k;
- a defective cell without tops of 2-level 1 is of type α with a top of 2-level 0, of type β, or of type B': an upper top of sign ε_k and a top of 2-level 0 of sign $-\varepsilon_k$, at the two positions other than ρ_k;
- a defective cell with neither is of type α with a top of 2-level 0.

Indeed, with $\varepsilon_k = 1$, in the second and third cases each sign comes from two kinds at one position each, which excludes $c \geq 1$ and $c \leq -2$; $c = 0$ leaves type α or β, by the last two rules above at the positions $r \neq \rho_k$ and at ρ_k, and $c = -1$ puts the two negative kinds at the two positions other than ρ_k, one top each, and leaves no room for the positive kinds.

Each defective cell of k contains a top, of Ω_k or of 2-level 1 of Ω_{k+1}, and distinct cells of k have no common top. For $k, k' \in \mathcal{D}$ with $k' \notin \{k-1, k, k+1\}$ this needs eight tops, so $\mathcal{D} \subseteq \{k, k+1\}$ for some k, or $K = 3$ and $\mathcal{D} = \mathbb{Z}/3$. Let $N_1(k)$ be the number of tops of 2-level 1 of Ω_k; as π maps the cells of k bijectively onto those of $k+1$, $N_1(k) = N_1(k+1)$ for $k \notin \mathcal{D}$. Let T_0 and T_1 be the numbers of tops of 2-level 0 and 1, so $T_0 + T_1 \leq 6$. If $L_k = wK_q$, then by Lemma 10.15(c)(iv) at most $(q-1)/2 \leq \ell_k - 2$ defective cells of k contain a top of one given kind at one given position.

Step 3: $\mathcal{D} = \emptyset$. Then n_1 is constant along every π-orbit of cells, and the π-orbit of $(0, \lambda)$ has $K \cdot \operatorname{ord}(\gamma)$ cells, $\gamma = -2^K \in U$. If there is a top of 2-level 1, this gives $K \operatorname{ord}(\gamma) \leq 6$, so $K = 3$, $64 \equiv 1 \pmod{e'}$, $e' = 7$ and $\gamma = -1$ has order 2. Then the six cells of the π-orbit have one top of 2-level 1 each, there is no top of 2-level 0, every cell has $c = 0$, and the top of 2-level 1 of $\pi(C)$ has the position and sign of that of C. Passing around the π-orbit, the top of $(0, \lambda)$ at (r, σ) forces one at $(r-1, -\sigma)$ in $(0, -\lambda)$ and one at $(r-2, \sigma)$ in $(0, \lambda)$, which has only one. If there is no top of 2-level 1, every cell has $c = 0$ and no top at all, which is impossible.

$K = 3$ *and* $\mathcal{D} = \mathbb{Z}/3$. A top of 2-level 0 lies in the cells of one k, a top of 2-level 1 in those of two, so $12 \leq \sum_k \ell_k \leq T_0 + 2T_1 \leq 12$. Hence $T_1 = 6$, $\ell_k = 4$, and every defective cell has one top. By the last remark of Step 2 each k has defective cells of both types α and β; so the tops of Ω_{k+1} are at $(\rho_k, -\varepsilon_k)$ and at $(\rho_{k+1}, \varepsilon_{k+1})$, relative to b_{k+1}, whence $\rho_{k+1} = \rho_k$ and $\varepsilon_{k+1} = -\varepsilon_k$ for $k = 0, 1$. For $k = 2$, the tops of $\Omega_3 = \Omega_0$ are at $(\rho_0, -\varepsilon_2) = (\rho_0, -\varepsilon_0)$ relative to b_3, that is at $(\rho_0 - 1, \varepsilon_0)$ relative to b_0, while they are at (ρ_0, ε_0): impossible.

$\mathcal{D} = \{k\}$. Then N_1 is constant, equal to a_1, and $T_1 = Ka_1$.

- $a_1 = 0$, which holds when $K \geq 9$: the defective cells of k are of type α with tops of 2-level 0, and ℓ_k of them give too many labels at one position and sign.
- $K = 3$, $a_1 = 2$: $T_0 = 0$, the defective cells of k contain the four tops of Ω_k and of 2-level 1 of Ω_{k+1}, one each, and $\ell_k = 4$; so the tops of Ω_k are at (ρ_k, ε_k) and those of Ω_{k+1} at $(\rho_k, -\varepsilon_k)$. The cells of $k+1$ and $k+2$ are constant with $c = 0$, so the tops of Ω_{k+2}, and those of $\Omega_{k+3} = \Omega_k$ relative to b_{k+3}, are at $(\rho_k, -\varepsilon_k)$, that is at $(\rho_k - 1, \varepsilon_k)$ relative to b_k: impossible.
- $K = 3$, $a_1 = 1$: let x, y, z be the tops of 2-level 1 of Ω_k, Ω_{k+1}, Ω_{k+2}, and $(\rho, \varepsilon) = (\rho_k, \varepsilon_k)$. A defective cell of k containing neither x nor y is of type α with a top of 2-level 0, so there are at most $(q-1)/2$ of them and at least $\ell_k - 2$; hence $\ell_k = 4$, there are exactly two, and x and y lie in two further defective cells: the cell of x has no upper top and is of type α (x at (ρ, ε)) or A (x of sign $-\varepsilon$, with a top of 2-level 0), and the cell of y has no top of 2-level 1 and is of type β (y at $(\rho, -\varepsilon)$) or B' (y of sign ε, with a top of 2-level 0). With z these are at least five tops, so A and B' do not both occur. The cells of $k+1$ and $k+2$ are constant; the cell of y has the upper top z, and the cell of z the upper top x, relative to b_{k+3}. In each of these two cells the upper top keeps the position and sign of the top of 2-level 1, or the cell has one more top of 2-level 0

and the sign changes; passing from b_{k+3} to b_k changes the sign again and lowers the position by 1. For α and β (five tops, so at most one further top of 2-level 0), x is at $(\rho-1,\varepsilon)$ or has the sign $-\varepsilon$; for α and B' (six tops), x has the sign $-\varepsilon$; for A and β (six tops), x has the sign ε. Each contradicts the type of the cell of x.

$\mathcal{D}=\{k,k+1\}$. Here $N_1(k')=a_1$ for $k'\neq k+1$ and $N_1(k+1)=b_1$, so $T_1=(K-1)a_1+b_1$. Let S_k and S_{k+1} be the sets of tops in the defective cells of k and of $k+1$. They meet only in tops of 2-level 1 of Ω_{k+1}, so $b_1\geq|S_k|+|S_{k+1}|-6\geq 2$.

- $a_1\geq 1$. Then $K=3$, and (a_1,b_1) is $(2,2)$, $(1,2)$, $(1,3)$ or $(1,4)$. For $(2,2)$, $T_0=0$, every defective cell has one top, and the tops of Ω_k, Ω_{k+1}, Ω_{k+2} are at (ρ_k,ε_k), at $(\rho_k,-\varepsilon_k)=(\rho_{k+1},\varepsilon_{k+1})$ and at $(\rho_{k+1},-\varepsilon_{k+1})=(\rho_k,\varepsilon_k)$; the cells of $k+2$ are constant with $c=0$, so the tops of Ω_k relative to b_{k+3} are at (ρ_k,ε_k), that is at $(\rho_k-1,-\varepsilon_k)$ relative to b_k: impossible. For $(1,2)$, $|S_k\cup S_{k+1}|\geq 6$ forces equality everywhere: every top lies in $S_k\cup S_{k+1}$, every defective cell has one top, and so the top x of 2-level 1 of Ω_k is at (ρ_k,ε_k), those of Ω_{k+1} at $(\rho_k,-\varepsilon_k)=(\rho_{k+1},\varepsilon_{k+1})$, and the top z of 2-level 1 of Ω_{k+2} at $(\rho_{k+1},-\varepsilon_{k+1})=(\rho_k,\varepsilon_k)$, while Ω_{k+2} has no top of 2-level 0. So the cell of z is constant with $c=0$, and x relative to b_{k+3} is at (ρ_k,ε_k), that is at $(\rho_k-1,-\varepsilon_k)$ relative to b_k: impossible. For $(1,3)$ and $(1,4)$, $T_0\leq 1$, so Ω_k or Ω_{k+1} has no top of 2-level 0. In the first case the defective cells of k without the top of 2-level 1 of Ω_k are of type β, at least ℓ_k-1 of them; in the second the defective cells of $k+1$ without the top of 2-level 1 of Ω_{k+2} are of type α with tops of 2-level 1, at least $\ell_{k+1}-1$ of them. Both are too many.
- $a_1=0$. Only Ω_{k+1} has tops of 2-level 1. The cells of k have no top of 2-level 1, all other cells have no upper top, and so every constant cell is empty: every top lies in a defective cell of k, of type α, β or B', or of $k+1$, of type α or A. Let n_α, n_β, n_B and m_0, m_1, m_A count these cells, with m_0 and m_1 the cells of type α of $k+1$ with a top of 2-level 0 and 1. Each top of 2-level 1 of Ω_{k+1} is the upper top of a cell β or B' of k and the top of a cell α or A of $k+1$, so

$$b_1=n_\beta+n_B=m_1+m_A,\qquad T=\ell_k+n_B+m_0+m_A=\ell_{k+1}+n_\alpha+n_B+m_A\leq 6,$$

with $\ell_k=n_\alpha+n_\beta+n_B$ and $\ell_{k+1}=m_0+m_1+m_A$. Moreover $n_\alpha,n_\beta\leq(q-1)/2$ for the line L_k, and $m_0,m_1\leq(q'-1)/2$ for L_{k+1}, by Lemma 10.15(c)(iv). If $\ell_k=6$, then $n_B=m_0=m_A=0$ and $n_\alpha=n_\beta=3$, so $T\geq\ell_{k+1}+3>6$; if $\ell_k=5$, then $n_B=1$, $n_\alpha=n_\beta=2$, and again $T>6$; $\ell_{k+1}\in\{5,6\}$ fails in the same way. So $\ell_k=\ell_{k+1}=4$ and $5\parallel e'$. Now $n_\beta=4-n_\alpha-n_B\geq 2+m_A$ and $m_1\geq 2+n_B$, so $m_A=n_B=0$, $n_\beta=m_1=2$, $n_\alpha=m_0=2$ and $T=6$. Thus Ω_k has two tops of 2-level 0 at (ρ_k,ε_k), and Ω_{k+1} has two tops of 2-level 1, at $(\rho_k,-\varepsilon_k)$ and at $(\rho_{k+1},\varepsilon_{k+1})$, and two tops of 2-level 0 at $(\rho_{k+1},\varepsilon_{k+1})$. Put $\rho=\rho_k=\rho_{k+1}$ and $\varepsilon=\varepsilon_{k+1}=-\varepsilon_k$. The labels M_1 of the tops of 2-level 1 lie in the lines $2L_k$ and L_{k+1}, which share two points and so coincide; the tops of 2-level 0 of Ω_{k+1} have the labels $M_0=L_{k+1}-M_1$, and those of Ω_k the labels $L_k-\frac12 M_1=\frac12 M_0$. With $\beta'=\varepsilon b_{k+1}\kappa^\rho$, the elements $[1,\beta',\varepsilon\nu]_{1,I}$, $\nu\in L_{k+1}$, are the four elements of order $n=2\cdot 3^I e'$ in one coset $\Phi_5(x_1)$ by (C4). Those with $\nu\in M_1$ are the two tops x_1, x_2 of 2-level 1; for the other two, y_1 and y_2, the elements $2y_i=[\beta',\varepsilon\nu]_{0,I}$ are the tops of 2-level 0 of Ω_{k+1}, and $N/2-y_i=[-\beta'/2,-\varepsilon\nu/2]_{0,I}$, by (C1), are those of Ω_k. This is Lemma 10.18(b).

In every case B has a local chain, which proves the proposition. □

Corollary 10.20. *If a local configuration B at level N has no local chain, then the order of every element of B divides $72e'$.*

Proof. By Proposition 10.14 every element has 2-level at most 3, and then by Proposition 10.19 every element has 3-level at most 2. □

Declarations

Data and code availability. The finite verifications cited in the proofs, a formalisation in Lean 4 of the combinatorial steps, and independent cross-checks are archived at Zenodo, doi:10.5281/zenodo.00000000. Deep Bhattacharjee carried out the computations and the checks in Lean, Julia, C, C++, Python and shell scripts, with AI assistance from Claude (Anthropic). Both authors read and approved the final text.

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Formerly, Electro-Gravitational Space Propulsion Laboratory (EGSPL),
Bhubaneswar, Odisha 751030, India

Formerly, University of Calcutta, Kolkata, India
Email address: d.bhattacharjee@erl-forschung.de
Email address: itsdeep@live.com

Formerly, National Taiwan University, Taipei 106319, Taiwan
Email address: rai.bhattacharya.in@gmail.com